Cognitive Space Theory

Foundations: cellular sheaves, obstruction,
and the geometry of coherence

A theory under development,
with epistemic status attached to every claim

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How to read this book

What this is

This book develops *Cognitive Space Theory*: an account of structured meaning in which a cognitive state is a *section of a cellular sheaf* over a simplicial complex, and in which the ways a cognitive state can fail are *cohomological* — classified, localised and counted by the same algebra that produces them.

The theory is under development. It is presented here as mathematics, not as a specification: no programming language appears, no system is described, and no result depends on a particular implementation. Where a construction is a design intention rather than a theorem, it says so in the margin, in colour, on the line where the claim is made.

Why every claim carries a tag

A theory under development contains, at any moment, a mixture of proved statements, statements proved only under hypotheses one may not be able to meet, statements that have been computed but not proved, statements that are intentions, and statements that were believed and are now known to be false. Written in a single register these are indistinguishable, and a reader studying such a text learns falsehoods with the same confidence as theorems.

So every substantive claim here carries one of these:

[PROVED]	A complete proof is given here, from stated hypotheses.
[PROVED UNDER STATED CONDITIONS]	Proved, but only under hypotheses that may not be available.
[MEASURED]	A number produced by a procedure that can be re-run. Not a proof.
[IMPLEMENTED, UNPROVED]	Computable as stated; no proof of correctness given.
[PROPOSED]	A design intention. Not proved.
[CONTESTED BY AUDIT]	Disputed on stated grounds, in place.
[REFUTED]	Shown false, or tested and failed. Retained *with* the refutation.

Refuted material is kept rather than deleted. Knowing why a construction fails is most of what is needed in order not to rebuild it.

The three layers

Each central object is presented three times over, so that a single reading can be shallow or deep:

1. a boxed **Intuition** in plain language, with no symbols;
2. the formal **Definition**, **Construction** or **Proposition**, with proof;
3. a **Worked example** on an explicit small complex, carried through to numbers.

The worked examples are cumulative. A single running example — a four-vertex complex K_4, which is *derived* in Chapter 4 rather than postulated — is reused throughout, so that the same object is measured by every instrument the book builds. Where an exercise appears, it is because the step is short and doing it yourself is faster than reading it.

Agendas, and what an incomplete section looks like

Every section of Parts I and II ends with an **Agenda** box in green, containing three lists:

- *Settled* — what the section establishes, by proof or by citation, and which may be used downstream without further argument;
- *Open* — what it does *not* establish. These are the honest gaps, stated in the same place as the material that needs them;
- *Next steps* — the ordered work that would close the gaps.

An agenda whose *Open* list is non-empty is a section that is mathematically incomplete, and the incompleteness is printed rather than implied. Later parts of the book, which have not yet been brought up to the standard of Parts I and II, instead carry a blue **Work plan** box. The two are not the same:

Agenda (green)	Prose exists. It is correct as far as it goes, and the box says how far that is.
Work plan (blue)	Prose does *not* exist, or predates the foundations. The box is the instruction for writing it.

Reading order

Part I is short, has no prerequisites, and states the one hypothesis everything else is answerable to. Read it first.

Part II is the mathematical spine and is meant to be read in order: the base complex is *constructed* in Chapter 4 before it is used in Chapter 5, the sheaf is laid on it in Chapter 6, and the operator that measures everything appears in Chapter 7. Chapter 8 then corrects a type error that the first four chapters deliberately carry: the state is written as though the space it lives in were fixed, and it is not.

A reader who wants the shortest honest path to the point of the theory should read §1.1, §2.3 and Theorem 6.8. Those three say, respectively, why a complex is necessary at all, what is lost by using a graph instead, and what the higher cells buy in exact algebraic terms.

Contents

Part I

Orientation

Chapter 1

The Claim

1.1 The founding hypothesis

Cognitive Space Theory rests on one hypothesis, and the whole apparatus of this book is a consequence of taking it seriously.

> *Meaning admits no global chart.*

That is: there is no single coordinate system in which everything that can be meant is simultaneously and consistently expressible. Meaning is coordinatizable only *locally*, in patches, and the patches do not agree on their overlaps.

> **Intuition.** A map of the Earth is a sheet of paper with coordinates on it. No single sheet covers the globe without tearing or distortion, so cartographers use an *atlas*: many local sheets, plus rules for translating a position on one sheet into a position on another. The rules are the real content. The sheets are interchangeable; the translations are where the curvature of the Earth shows up.
>
> The hypothesis above says that meaning is like the globe and not like the sheet. If it were like the sheet — if one big coordinate system sufficed — then a cognitive state would be a vector, cognition would be linear algebra in a fixed basis, and this book would have nothing to say. Everything here exists because the translations between local frames fail to compose.

The hypothesis is not decoration and it is not unfalsifiable. It has a precise mathematical shadow, developed in Part II: if meaning admitted a global chart, then the complex would have one vertex, the sheaf would have no restriction maps, the coboundary would be zero, the Laplacian would be zero, and every state would be coherent. The theory would collapse to its own trivial case. *Structure exists exactly to the extent that local coordinatizations fail to agree.* Chapter 5, Example 5.2, computes that collapse explicitly, so that the reader can see the degenerate case and know what the non-degenerate one is departing from.

> **Claims and non-claims.** **[PROPOSED]** The founding hypothesis is a modelling commitment, not a theorem, and nothing in this book proves it. What the book does is make it *consequential*: every construction is derived from it, so that if the hypothesis is wrong, the constructions are not merely unmotivated but provably vacuous. That is the strongest

honest status a founding hypothesis can have.

1.2 What is being modelled

Three words are used throughout and each is used in a restricted, technical sense. They are declared here because their ordinary meanings would import claims the theory does not make.

region	A place where meaning is locally coordinatizable. Formally a vertex of the complex. *Not* an agent, a module, a subsystem, or a processor.
content	What is expressible within one region, in that region's own coordinates. Formally an element of a stalk. *Not* a belief; a region does not hold an opinion, because a region is not the kind of thing that can have one.
thought	A choice of content in every region that survives every comparison. Formally a global section. There is *one* of these at a time, expressed in many local coordinate systems — not one per region.

Claims and non-claims. The third row is the one that is easiest to get wrong, and getting it wrong changes the mathematics rather than merely the prose. If each region holds its own belief, then disagreement between regions is a normal and permanent feature to be negotiated, and the natural theory is a theory of consensus. That theory exists, it is good, and it is *not this one*: see Hansen–Ghrist's discourse sheaves [19], discussed in §3.4, where the vertices are genuinely separate agents.

Under the reading taken here there is one thought and many coordinate systems, so disagreement is not a social fact but a *defect of the atlas*: either the frames can be adjusted until they agree, or they cannot, and the obstruction to adjusting them is a topological invariant. That is what makes the dynamics of Part II onwards possible at all. A consensus theory has nothing to converge to except agreement; this theory has something to converge to *and* a classification of the cases where convergence is impossible.

1.3 The thesis, and what each clause forbids

> *A cognitive space is a cellular sheaf on a finite simplicial* 2*-complex. Its state is a section, which relaxes. Its restriction maps are the theory's only carrier of learned structure, and they mold. Its complex grows and prunes, and growth is triggered by an obstruction rather than by a rule.*

Every clause forbids something, and the forbidding is a live constraint on what may be written later.

clause	forbids
sheaf on a 2-**complex**	dropping to a graph, where the sheaf condition is vacuous (Theorem 6.8)
the state **relaxes**	treating the state as stored data rather than as a dynamical variable
restriction maps **mold**	freezing the maps at the identity, which provably empties the theory (Proposition 7.8)
grows and prunes	monotone architectures that only accumulate
growth from an **obstruction**	hand-written growth rules; the trigger must be a computed invariant

1.4 What Cognitive Space Theory is not

Stated as refusals, because each of these is a place where an adjacent and better-established framework would absorb the theory if the boundary were left implicit.

1. **Not a knowledge graph.** A knowledge graph is a labelled relation with no dynamics and no obstruction theory. The working test is sharp: *if a proposed feature would behave identically on a labelled graph, it does not belong here.*
2. **Not a geometric semantics in a single space.** The standard geometric account of concepts [14] places them as convex regions of one global quality space. Cognitive Space Theory is precisely the denial of that global space (§1.1). The two are not variants; they disagree on whether the global chart exists.
3. **Not a compositional semantics.** Categorical accounts of how meanings compose [10] are about *building* a meaning from parts. This theory is about meaning *failing* to assemble, and about what that failure is a measurement of. A composition calculus always composes; the object of study here is the residue when assembly does not go through.
4. **Not a model of brains.** Mechanisms are occasionally borrowed from neuroscience as *design constraints*. No claim is made to explain neural data, and nothing here should be evaluated against it.
5. **Not a thermodynamic or free-energy theory.** The cost functionals introduced later are this theory's own construction. Naming them after an established principle would import authority that has not been earned.
6. **Not a claim about consciousness, agency or life.** Where a construction has a suggestive name, the name is a label for a morphism and nothing further is asserted.

Agenda.

Settled.

S1. The founding hypothesis is stated in a form whose failure has a computable consequence (collapse to Example 5.2).

S2. The three primitive words are fixed, and the agent reading is explicitly excluded with its reason.

S3. The thesis is stated as a list of prohibitions, each pointing at the result that enforces it.

Open.

O1. No independent argument is given *for* the founding hypothesis. At present it is motivated by the fact that its negation trivialises the theory, which is a consistency argument, not evidence.

O2. "Region" is defined by what it is not. A positive criterion — given a domain, what determines how many regions there are and where their boundaries fall — is supplied only operationally, by the ingest construction of Chapter 4, and it inherits that construction's arbitrariness in the choice of witness set.

O3. The prohibition "growth from an obstruction, not a rule" is stated but not yet honoured: the obstruction that would serve as trigger is identified only as a candidate in §8.4.

Next steps, in order.

1. Find one phenomenon that a global-chart semantics cannot represent and a sheaf semantics can, stated so that the difference is a computation and not an interpretation. Contextuality (§6.2) is the leading candidate and would convert O1 from a consistency argument into an argument from evidence.
2. Give a positive criterion for region individuation, or accept explicitly that regions are an input to the theory and not an output, and say so here rather than leaving it to be inferred.
3. Close §8.4 to a theorem; if it closes, restate the fifth clause of the thesis as a result rather than a commitment.

Chapter 2

The Picture Before the Symbols

Nothing in this chapter is used later. It exists so that the definitions of Part II are recognised when they arrive, rather than merely accepted.

2.1 Charts, atlases, and the failure to agree

> **Intuition.** Suppose a body of knowledge is covered by overlapping regions, and inside each region you have a way of writing things down — a coordinate system private to that region. Two regions that overlap must be able to compare what they say about the overlap. Comparison requires a *translation*: a rule taking what is written in one region's coordinates into a form the overlap can read.
>
> Three things can then happen.
>
> - The translations agree everywhere. There is a single consistent thing being described, written down in many ways. This is a *thought*.
> - The translations disagree, but the disagreement can be removed by adjusting the translations themselves — the regions were saying the same thing in badly aligned frames. This is a *misalignment*, and it is repairable.
> - The translations disagree, and no adjustment of them removes the disagreement, because going around a loop of regions and coming back returns you somewhere else. This is a *hole*. It is not repairable by translation, only by changing the regions.
>
> The entire technical content of this book is: making those three cases precise, proving that they are exhaustive and mutually orthogonal, computing which one you are in, and saying what to do in each.

The three cases are not an informal taxonomy. They are the three summands of an orthogonal direct-sum decomposition of the space of pairwise comparisons (Theorem 11.5), they are computable by linear algebra, and the third is a topological invariant that no amount of local adjustment can change.

2.2 A single region cannot be wrong

Take the smallest possible cognitive space: one region, no overlaps. There are no comparisons to make, so no comparison can fail, so every content whatsoever is a thought. The space has no structure and no capacity for error.

This is worth stating early because it is the exact sense in which the theory is *about* the relations and not about the regions. Content lives in the regions; *everything else* lives between them. Example 5.2 carries this out in symbols and finds that every invariant the book defines is either zero or trivial on the one-region space.

2.3 The degenerate corner: what a neural network is, from here

The framework has a corner in which it reproduces machinery that already exists, and locating that corner exactly is the most direct way to say what the rest of the framework is for.

> **Intuition.** Suppose you make three simplifications at once:
>
> 1. allow only regions and overlaps of two — no three-way overlaps;
> 2. make every translation the identity, so that all regions share one coordinate system after all;
> 3. keep only the relaxation dynamics, and never adjust the translations or the regions.
>
> What remains is a graph with vectors on its nodes, diffusing toward agreement. That is a graph neural network, and Proposition 7.7 makes the identification exact for the linear case.

Each simplification has a provable cost, and the three costs are the reason the rest of the book exists:

simplification	**what it provably costs**
no three-way overlaps	the sheaf condition becomes vacuous, so nothing can ever obstruct and growth can never be derived (Theorem 6.8); and the three-way decomposition loses a summand, so misalignment and holes stop being distinguishable (Theorem 11.5)
identity translations	the only thought is exact consensus, and relaxation is plain averaging; diffusion drives everything to the global mean (Proposition 7.8)
no adjustment	the theory's only carrier of learned structure is frozen, and the sheaf becomes a decoration on a graph

The second row is not a criticism from outside: it is the published diagnosis of oversmoothing in graph neural networks, and the published repair is exactly to make the translations non-trivial [5]. The first row has an independent published counterpart in expressiveness: message passing that sees higher cells is strictly stronger than message passing that sees only edges [6].

> **Claims and non-claims.** This is the one place where the theory can be checked against an established literature rather than argued for. It should therefore be the first thing a sceptical reader is shown, and it is deliberately placed before any definition. If the identification in Proposition 7.7 were wrong, the theory would be in serious trouble; it is not wrong, and it is proved.

2.4 Reading, as a sequence of operations

Intuition. What happens when a body of knowledge absorbs a text? Not storage, and not appending. The stages, each of which becomes a named operation in Part II and later:

1. **Ingest.** The text arrives as a relation — these things occur together in this passage, those in that one — and a relation determines a complex canonically, with no parameters (Chapter 4).
2. **Anchor.** Some of the new material is identified with material already present. This is the only step in the whole pipeline containing a free choice, and it is where interpretation lives: two readers who anchor the same text differently have read different texts.
3. **Glue.** The two complexes are joined along what was anchored.
4. **Ferment.** The join has properties neither part had. Some disagreement is discharged by relaxation, some by adjusting translations — and some cannot be, because the act of joining created a hole that was not present in either piece (§8.4).
5. **Consolidate.** What survives is compressed and retained; what does not is pruned.

Step 4 is the interesting one and the reason this is called fermentation rather than insertion. *New obstructions are a property of the join, not of either part.* A text you already agreed with entirely teaches you nothing, and a text you cannot anchor at all teaches you nothing; the content is in the residue.

Agenda.

Settled.

- S1. The three-case taxonomy (thought / misalignment / hole) is stated informally and is discharged by Theorem 11.5.
- S2. The degenerate corner is located precisely and each simplification is paired with the result that prices it.
- S3. The reading pipeline is stated as five named stages, four of which are given formal definitions later.

Open.

- O1. Stage 5, consolidation, has no formal counterpart in Part II. It is treated in later parts whose foundations predate this one.
- O2. Stage 2, anchoring, is identified as the locus of interpretation but is given no criterion. Nothing here says which anchoring is *better*, only that the choice exists and matters.

Next steps, in order.

1. Define anchoring formally as a span with an isomorphism of sheaves on its apex (sketched in §8.4), and state the invariance question: which downstream quantities are independent of the anchoring chosen.
2. Bring consolidation into Part II or state explicitly that it is deferred, so that stage 5 is not silently missing.

Chapter 3

Sources, Position, and the Literature Graph

3.1 Naming discipline

This theory borrows vocabulary from several mature fields. Borrowed words arrive with expectations attached, and an unearned word is a claim made silently. The following rules are enforced throughout.

1. **A borrowed term is used only with its source meaning, or not at all.** Where a weaker notion is meant, a different word is coined.
2. **No term is used whose associated theorems are not being invoked.** This retires the word *stratified* from the description of the base complex; §5.3 explains what was meant, why the word was wrong, and where a genuine stratification does appear.
3. **Suggestive names are labels, never arguments.** If a construction is called reproduction, growth or memory, that name may not be used as a premise in any proof.

> **Claims and non-claims.** Terms retired from this book under rule 2, each with its replacement: *stratified* (of the base complex) $\to$ nothing; the two-scale structure it named is not assumed, and a genuine stratification appears instead on the configuration space, §8.2. *Belief* (of stalk content) $\to$ *content*, since "belief" presupposes a bearer. *Organ*, *agent*, *subsystem* (of vertices) $\to$ *region*, since vertices are not processors. *Free energy* (of cost functionals) $\to$ the functional's own name.

3.2 The five lineages

Cognitive Space Theory sits at the meeting of five bodies of work. For each, what is taken and what is deliberately not taken.

1. Cellular sheaves and their spectral theory. Cellular sheaves are due to Shepard [25] and were developed as a general applied framework by Curry [12]. The metric theory this book depends on — the sheaf Laplacian $L = \delta^*\delta$, its spectrum, its Cheeger-type inequalities — is Hansen and Ghrist [18]. *Taken:* the entire apparatus of Chapters 6 and 7, essentially unmodified. *Not taken:* their setting is a fixed complex with fixed stalks. Everything about growth and molding is outside it and must be built.

2. Data-to-complex constructions. Dowker's theorem on the homology of relations [13], its functorial and stable refinement [9], the nerve construction, and the persistence machinery with its stability theorems [11, 8, 7]. *Taken:* the ingest operator of Chapter 4, and the demand that ingest be *stable*. *Not taken:* persistence itself. This book measures a complex at one scale, not a filtration across scales; Mapper [27] is named and rejected, for the reason in §4.4.

3. Sheaf-theoretic contextuality. Abramsky and Brandenburger [2] show that non-locality and contextuality are exactly the non-existence of a global section of a presheaf of distributions over a cover of measurement contexts; the cohomological refinement is [1]. *Taken:* the identification of incoherence with a global-section obstruction, and — decisively — the observation that the *coefficient algebra* is a free parameter with consequences (§6.2). *Not taken:* any quantum-mechanical claim. The site is different and no physical interpretation is asserted.

4. Higher-order networks and sheaves on graphs in learning. Signal processing beyond pairwise interactions [4, 24]; sheaf neural networks [16]; neural sheaf diffusion [5]; the expressiveness separation for higher-cell message passing [6]. *Taken:* the diagnosis of oversmoothing as a statement about $\ker L$, which is the sharpest available argument that non-identity restriction maps are necessary rather than ornamental. *Not taken:* the learning setting. These are supervised models trained on labelled data; nothing here is.

5. Synchronization, fusion, and inference of structure. The connection Laplacian and vector diffusion maps [26]; $O(d)$ synchronization [3]; sheaves for heterogeneous sensor integration [22, 23]; learning sheaf Laplacians from observed signals [17]. *Taken:* this is the literature home of *molding* — adjusting restriction maps toward consistency is the synchronization problem, and Robinson's consistency radius is this book's discord in an applied setting. *Not taken:* their objectives are estimation against a ground truth. There is no ground truth here.

3.3 The literature graph

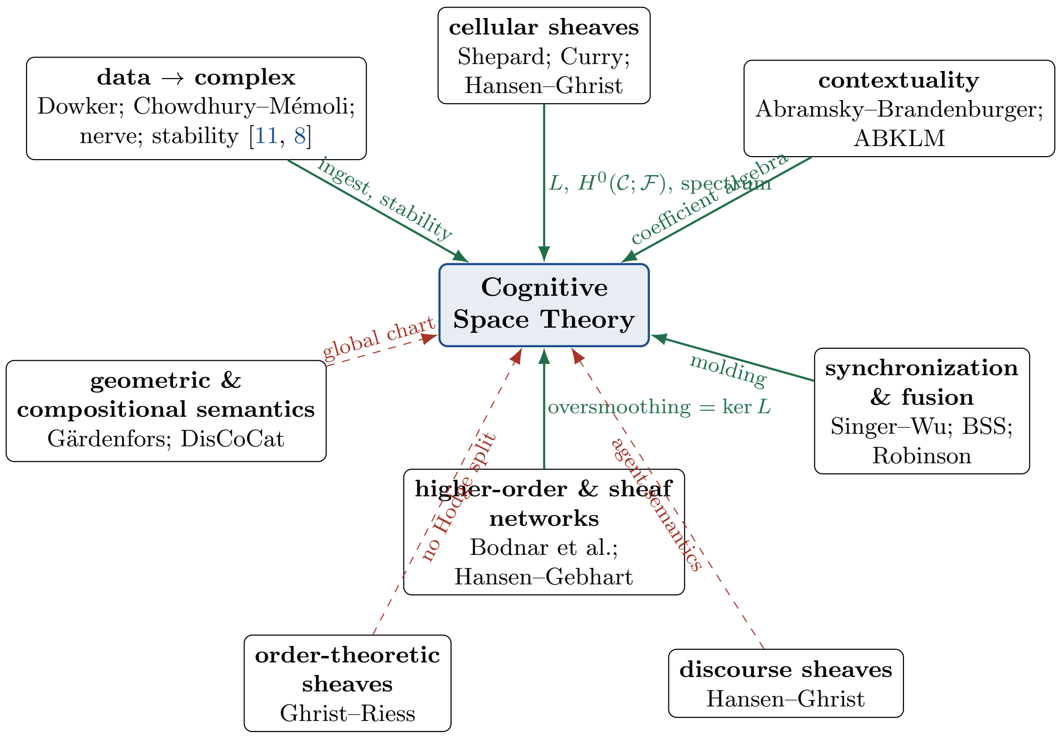

Figure 3.1: Provenance. Solid green: taken, with what is taken on the edge. Dashed red: a nearest neighbour whose central commitment is *rejected*, with the point of divergence on the edge. The dashed edges carry more information than the solid ones, because a theory is located by what it refuses.

3.4 Nearest neighbours, and exactly where they diverge

Three bodies of work are close enough that the difference must be stated precisely rather than gestured at.

Discourse sheaves [19]. The closest published relative: the same Laplacian, the same disagreement functional, on the same kind of object. The divergence is semantic and it is total. There, a vertex is an agent with private opinions and the restriction maps express how an agent's opinion appears in a shared discourse; disagreement is between people. Here, a vertex is a chart and disagreement is curvature. The mathematics is shared; the question asked of it is not.

Neural sheaf diffusion [5]. Shares the diagnosis of identity restriction maps and the remedy of learning non-trivial ones. Two divergences: their base is a graph, so the sheaf condition never binds (Theorem 6.8); and their restriction maps are learned by supervised gradient descent against labels, whereas molding here has no external target and must be driven by an intrinsic functional.

Sheaf contextuality [2]. Shares the central identification — incoherence is the failure of a global section — on a different site. The divergence is the coefficient algebra: their stalks carry distributions over a semiring, in which positivity generates the obstruction; this book's stalks are vector spaces over a field, chosen so that a metric and a three-way Hodge decomposition exist. §6.4 states what that choice costs, since it is a real cost and not a free upgrade.

Agenda.

Settled.

S1. Every construction used in Part II has a named lineage, with the take/leave boundary drawn explicitly.

S2. The three nearest neighbours are separated from this theory by a stated divergence rather than by tone.

S3. A naming discipline is in force, with the retired terms listed.

Open.

O1. No lineage is claimed for the dynamics. Molding has a home in the synchronization literature, but the two-phase separation of relaxation from consolidation does not have one here, and it is not yet established whether one exists.

O2. The claim in §3.4 that molding "must be driven by an intrinsic functional" names no functional. Until one is given and shown to have the required stationary points, this is a requirement rather than a design.

Next steps, in order.

1. Search the synchronization literature for an unsupervised objective whose stationary points are flat connections, and either adopt it with attribution or record that none was found.
2. Extend Figure 3.1 with a sixth cluster for the dynamics once O1 is resolved — or record the absence of one as a finding, since a construction with no lineage is either novel or naive and it matters which.

Part II

Foundations

Chapter 4

Where the Complex Comes From

4.1 The problem: $\mathcal{C}$ may not be assumed

Every account of sheaves on a complex begins with the complex already present. For a theory of cognition this is not available: if the base complex is postulated, then every structural claim the theory makes is a claim about a hand-drawn picture, and the vertices — the regions of §1.1 — are exactly as arbitrary as the person drawing them.

So this chapter comes first. It asks: given input of an arbitrary nature, what canonically determines a simplicial complex? The answer wanted must have three properties.

(P1) **Minimal input.** It should require as little structure on the input as possible — in particular no metric, since there is no reason to believe raw input carries one.

(P2) **No free parameters.** A construction with knobs makes every downstream invariant a function of the knobs.

(P3) **Stability.** A small perturbation of the input must produce a small perturbation of the complex. Without this, absorbing new material is not a controlled operation and no invariant computed after ingest means anything.

4.2 A relation determines a complex

> **Intuition.** The weakest possible thing you can say about a body of input is: *these things occur together, witnessed by that.* A passage of text witnesses the co-occurrence of the terms in it; an observation witnesses the co-occurrence of the features it exhibits. That is a relation between two sets — things on one side, witnesses on the other — and nothing more.
>
> It turns out a relation is already enough. Declare a collection of things to be jointly bound when *some single witness witnesses all of them at once.* This is Dowker's construction, and everything the base complex is required to be follows from it rather than being assumed.

Definition 4.1 (Dowker complexes of a relation)**.** Let X, Y be finite sets and $R \subseteq X \times Y$ a relation, written $x\,R\,y$. Define two abstract simplicial complexes:

$$\mathrm{D}_X(R) = \left\{\, \sigma \subseteq X \;:\; \sigma \neq \emptyset,\ \exists\, y \in Y \text{ with } x\,R\,y \text{ for all } x \in \sigma \,\right\},$$

$$\mathrm{D}_Y(R) = \{\, \tau \subseteq Y \ : \ \tau \neq \emptyset,\ \exists x \in X \text{ with } x \mathrel{R} y \text{ for all } y \in \tau \,\}.$$

Call $\mathrm{D}_X(R)$ the *ingest complex* of R and Y its *witness set.*

Proposition 4.2 (Downward closure is free)**.** **[PROVED]** *$\mathrm{D}_X(R)$ is an abstract simplicial complex: it is downward closed.*

Proof. Let $\sigma \in \mathrm{D}_X(R)$ with witness y, and let $\emptyset \neq \tau \subseteq \sigma$. Every $x \in \tau$ lies in σ, hence $x \mathrel{R} y$. So y witnesses τ and $\tau \in \mathrm{D}_X(R)$. □

This is worth pausing on. In the standard presentation, downward closure is a *modelling assumption* — one declares that every sub-collection of a jointly bound fact is itself recorded, and the reader is asked to accept it. Under Definition 4.1 it is a one-line consequence of what a witness is. The same holds for the other structural feature usually assumed:

Proposition 4.3 (n-ary binding is free)**.** **[PROVED]** *In $\mathrm{D}_X(R)$, a k-simplex records a single $(k{+}1)$-ary joint binding and is not determined by the pairwise bindings among its vertices. Explicitly, there exist relations R for which every pair $\{x_i, x_j\} \subseteq \sigma$ lies in $\mathrm{D}_X(R)$ while σ does not.*

Proof. Take $X = \{x_1, x_2, x_3\}$, $Y = \{y_{12}, y_{13}, y_{23}\}$, and let y_{ij} be related to exactly x_i and x_j. Each pair has a witness, so all three edges are present; no single y is related to all of x_1, x_2, x_3, so $\{x_1, x_2, x_3\} \notin \mathrm{D}_X(R)$. Adding a fourth witness y_{123} related to all three fills the triangle. The two relations have the same 1-skeleton and different complexes. □

Remark 4.4 (Why this replaces the usual argument)*.* The usual justification for simplicial complexes over graphs is that a fact depending on three hypotheses is one three-way fact rather than three pairwise ones. That argument is correct but weak: it justifies *higher-order* structure, and a hypergraph supplies that more cheaply and without downward closure. What it does not justify is the *simplicial* structure specifically. Propositions 4.2 and 4.3 together do: downward closure is not a price paid for the chain complex, it is what a witness relation already satisfies. The chain complex — and with it everything in Chapters 6 and 7 — is then available for free.

4.3 Dowker duality

Theorem 4.5 (Dowker, 1952)**.** **[PROVED]** *(Cited; not proved here.) For any relation $R \subseteq X \times Y$, the complexes $\mathrm{D}_X(R)$ and $\mathrm{D}_Y(R)$ are homotopy equivalent. In particular they have isomorphic homology and cohomology in every degree. [13]*

The content for this theory: *ingest does not privilege a side.* The complex built on the things and the complex built on the witnesses carry the same topology, so the invariants of Chapter 7 are invariants of the relation, not of the arbitrary decision to call one side "objects" and the other "contexts". A functorial and stable refinement, which is what (P3) requires, is in Chowdhury–Mémoli [9].

> **Claims and non-claims.** This is the point at which formal concept analysis is often proposed, since it also produces structure canonically from a binary relation via a Galois connection. It is not used here, and the reason is structural rather than one of taste: the output of that construction is a *lattice.* A lattice has no boundary operator, hence no cochain complex, hence no cohomology and no Laplacian — so none of Chapters 6–7 can

be run on it. The Dowker complex keeps the one property that made the relational input attractive (canonical, parameter-free, no metric required) and adds the homotopy-theoretic structure the rest of the book consumes.

4.4 The alternatives, and their prices

construction	input needed	parameters	stability
Dowker complex	a relation	none	yes [9]
nerve of a cover	a cover of a space	the cover	good cover $\Rightarrow \mathcal{N}(\mathcal{U}) \simeq X$
Vietoris–Rips / Čech	a metric	scale ε	yes [11, 8]
Mapper	metric, filter, cover, clustering	four	poor

The nerve construction is the conceptual archetype — it is where the word "region" comes from, and the nerve theorem is the statement that a good cover loses no topology — but it presupposes a space to cover, which ingest does not have. Vietoris–Rips requires a metric on the input and a scale; it is the right choice when the input genuinely carries a metric, and the persistence machinery exists precisely to remove the arbitrariness of ε by taking all scales at once. Mapper is named because it is the best-known data-to-complex pipeline and because it fails (P2) and (P3) badly enough that its output is not a measurement.

Claims and non-claims. Requirement (P3) is the one that is easy to skip and expensive to lose. The stability theorems [11, 8] are what make "absorb this text" a controlled operation: they bound the change in the computed invariants by the size of the change in the input. Without such a bound, a one-word change to a document could in principle alter b_1 arbitrarily, and no downstream quantity would be reproducible. Any future ingest operator proposed for this theory must come with a stability statement or be rejected.

4.5 The ingest operator, and the running example

Construction 4.6 (Ingest)**.** **[PROPOSED]** Input arrives as a finite relation $R \subseteq X \times Y$ between a set X of *things* and a set Y of *witnesses*. Ingest returns the complex $\mathcal{C} = \mathrm{D}_X(R)$, truncated to dimension 2 (§5.2). Assigning content to its cells is Chapter 6; joining it to an existing complex is §8.4.

Example 4.7 (The running example K_4, derived)**.** Let $X = \{\mathsf{a}, \mathsf{b}, \mathsf{c}, \mathsf{d}\}$ and $Y = \{y_1, y_2\}$ with

$$R = \big\{(\mathsf{a}, y_1), (\mathsf{b}, y_1), (\mathsf{c}, y_1), (\mathsf{c}, y_2), (\mathsf{d}, y_2)\big\}.$$

Read it concretely: two source passages, y_1 and y_2; the first exhibits $\mathsf{a}, \mathsf{b}, \mathsf{c}$ together, the second exhibits c, d together. Then

$$K_4 \;=\; \mathrm{D}_X(R) \;=\; \big\{\, \mathsf{a}, \mathsf{b}, \mathsf{c}, \mathsf{d},\ \mathsf{ab}, \mathsf{ac}, \mathsf{bc}, \mathsf{cd},\ \mathsf{abc} \,\big\},$$

a filled triangle $\{\mathsf{a}, \mathsf{b}, \mathsf{c}\}$ with a pendant edge $\{\mathsf{c}, \mathsf{d}\}$. The triangle is filled because a single witness y_1 carries all three at once; had the three terms merely occurred in pairs across three different passages, the triangle would be empty (Proposition 4.3). Every instrument built in this book is applied to this object.

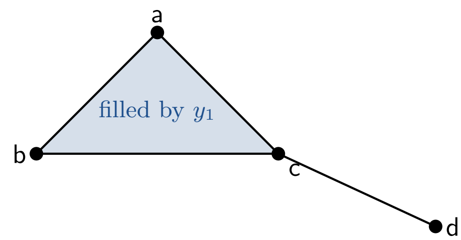

Figure 4.1: The running example $K_4 = \mathrm{D}_X(R)$ of Example 4.7. Nothing about it is chosen: it is what the relation R says.

Worked example. *Dowker duality, checked by hand.* The witness-side complex is $\mathrm{D}_Y(R) = \{y_1, y_2, \{y_1, y_2\}\}$: the pair $\{y_1, y_2\}$ is a simplex because c is related to both. So $\mathrm{D}_Y(R)$ is a single edge, which is contractible. And $\mathrm{D}_X(R)$ is a filled triangle with a whisker, also contractible. Both are contractible, hence homotopy equivalent, in agreement with Theorem 4.5. □

Note what the check would have caught: if the triangle abc had been left *unfilled* while its three edges were present, $\mathrm{D}_X(R)$ would have $b_1 = 1$ whereas $\mathrm{D}_Y(R)$ still has $b_1 = 0$, contradicting the theorem — which is another way of seeing that filling is forced by the relation and not optional.

Agenda.

Settled.

S1. Downward closure (Proposition 4.2) and non-reducibility of n-ary binding to pairwise binding (Proposition 4.3) are theorems about relations, not modelling assumptions.

S2. Ingest is side-independent, by Dowker's theorem.

S3. The running example is derived from a relation rather than drawn.

S4. Stability is stated as a *requirement* on any ingest operator, and the alternatives are ranked against it.

Open.

O1. Ingest satisfies (P1) and (P2), but the choice of the witness set Y is itself a modelling decision and is not canonical. The construction has no parameters *given* R; producing R from raw input is unaddressed.

O2. The stability of D_X is cited [9] for the persistent setting. The one-scale statement actually used here — how much K and its invariants move when a bounded number of pairs are added to or removed from R — is not stated and not proved.

O3. Truncation to dimension 2 (Construction 4.6) discards cells the relation genuinely produces. Its effect on b_1 and on $H^1(\mathcal{C}; \mathcal{F})$ is not analysed anywhere.

O4. Nothing here says how a witness set is *grown*. Ingest builds a complex from a relation; it does not say what happens to Y when new input arrives with new witnesses.

Next steps, in order.

1. State and prove a one-scale stability bound: for R, R' with $|R \triangle R'| \le k$, bound the change in b_0, b_1 and in $\lambda_2(L)$. This is the missing form of (P3) and everything downstream that claims to be a measurement depends on it.
2. Analyse 2-truncation. Either bound the error it induces in the invariants actually used, or drop the truncation and pay the combinatorial cost knowingly.

3. Give at least one concrete, fully specified R from a non-toy input, and compute K, b_0, b_1 on it. Until this exists, ingest is a definition rather than an operator.
4. Decide whether Y is part of the state. If witnesses persist, they belong in the theory; if they are consumed at ingest, say so and record that the complex retains no trace of what justified its cells.

Chapter 5

The Base Complex

5.1 Simplicial complexes

Definition 5.1 (Abstract simplicial complex)**.** Let V be a finite set. An *abstract simplicial complex* $\mathcal{C}$ on V is a family of nonempty subsets of V, called *simplices*, which is downward closed: if $\sigma \in \mathcal{C}$ and $\emptyset \neq \tau \subseteq \sigma$ then $\tau \in \mathcal{C}$. A simplex of cardinality $k+1$ is a *k-simplex*; write $\dim \sigma = k$. If $\tau \subseteq \sigma$ we write $\tau \trianglelefteq \sigma$ and call τ a *face* of σ. The set of k-simplices is $\mathcal{C}_k$, and $\dim \mathcal{C} = \max_\sigma \dim \sigma$.

0-simplices are vertices — the *regions* of §1.1; 1-simplices are edges, the places where two regions are compared; 2-simplices are filled triangles. By Proposition 4.2 the closure condition is automatic for any complex arising from ingest.

Example 5.2 (The one-region space)**.** **[PROVED]** Let $\mathcal{C} = \{\{v\}\}$, a single vertex. Then $\mathcal{C}_1 = \mathcal{C}_2 = \emptyset$, so with any sheaf whatsoever (Chapter 6):

$$C^1(\mathcal{C};\mathcal{F}) = 0, \qquad \delta = 0, \qquad L = 0, \qquad \omega(x) = 0 \text{ for all } x, \qquad H^0(\mathcal{C};\mathcal{F}) = \mathcal{F}(v).$$

Every content is a thought; nothing can fail; every invariant the book defines is zero or trivial.

This is the collapse promised in §1.1, and it is the precise sense in which the theory is a theory of the relations. Content lives in the stalks; *all* structure lives in the maps between them. If meaning admitted a global chart, Example 5.2 would be the whole theory.

5.2 Why dimension two, and why not one

The book works with complexes of dimension at most 2. Both bounds need justification, and they are of very different kinds.

Not dimension one. A graph is a 1-dimensional simplicial complex, so working on a graph is a strict special case, not a different choice. What is lost is stated exactly in Theorem 6.8 (the sheaf condition becomes vacuous, so nothing can obstruct) and Theorem 11.5 (the three-way decomposition loses a summand, so repairable misalignment and irreparable holes stop being distinguishable). These are the two load-bearing results of Part II and they are both statements that dimension 1 is degenerate.

Not dimension three or more. This bound is pragmatic and is not claimed to be principled. Cells of dimension 3 would make δ^2 non-vacuous in turn and would give H^2 a role. Nothing in this book needs them, the combinatorial cost of downward closure grows as 2^{k+1} per k-simplex, and no phenomenon has been identified that requires them.

> **Claims and non-claims. [CONTESTED BY AUDIT]** The asymmetry above should be uncomfortable. The argument against dimension 1 is a theorem; the argument against dimension 3 is that nothing has needed it yet. If a later construction turns out to require an obstruction living in H^2, the truncation is what will be blocking it, and the truncation was chosen for convenience. This is recorded here rather than discovered later.

5.3 On the word "stratified"

Earlier presentations of this material described $\mathcal{C}$ as *stratified*, meaning that it carried two nested scales — a coarse complex whose vertices were subsystems, and inside each, a fine complex whose vertices were concepts. That description is withdrawn, for two independent reasons.

First, the word was wrong. In mathematics a stratification is a decomposition into locally closed smooth pieces satisfying frontier and tangency conditions (Whitney), or the base data for constructible sheaves and the exit-path category. None of those theorems were being invoked, and none of them apply to a finite complex with no smooth structure. Under the naming discipline of §3.1, rule 2, the term is retired.

Second, and more seriously, the structure it named was not earned. Two nested scales were justified by the claim that the coarse vertices were genuinely separate subsystems. Under the reading of §1.1 vertices are charts, not subsystems, and that ground is gone. Rather than re-justify a two-scale structure on a new basis, this book takes the opposite position:

> *$\mathcal{C}$ has one scale. Multi-scale structure, where it exists, is to be* derived *from the sheaf — for instance by spectral partition of L — and not postulated.*

Remark 5.3 (Where a genuine stratification does appear). The word is not lost, only moved. There is a real stratification in this theory, but it is on the space of *configurations* $(\mathcal{C}, \mathcal{F})$ rather than on $\mathcal{C}$: the strata are indexed by the dimension vector, molding moves within a stratum, and growth crosses to an adjacent one. §8.2 develops this. It is a better home for the word because there the strata do real work — they are what makes the identity invariant locally constant.

5.4 The running example, combinatorially

For K_4 of Example 4.7, with vertex order $(\mathsf{a}, \mathsf{b}, \mathsf{c}, \mathsf{d})$, the degrees are $2, 2, 3, 1$ and the ordinary graph Laplacian of the 1-skeleton is

$$L_{K_4} = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}. \tag{5.1}$$

Proposition 5.4 (Spectrum of the running example). **[PROVED]** $\det(L_{K_4} - \lambda I) = \lambda(\lambda-1)(\lambda-3)(\lambda-4)$, *so* $\operatorname{spec} L_{K_4} = \{0, 1, 3, 4\}$.

Proof. Direct expansion gives $\lambda^4 - 8\lambda^3 + 19\lambda^2 - 12\lambda = \lambda(\lambda-1)(\lambda-3)(\lambda-4)$. The eigenvalue 0 is forced since every row sums to zero, so $L_{K_4}\mathbf{1} = 0$; its multiplicity is 1 because the 1-skeleton is connected (Proposition 7.8). $\square$

The integrality is an accident of this graph and should not be expected in general. It is why K_4 was chosen: every quantity in Part II can be carried through in exact arithmetic.

Agenda.

Settled.

S1. $\mathcal{C}$ is a single-scale finite complex of dimension ≤ 2; the two-scale "stratified" description is withdrawn with its reason.

S2. The degenerate one-region case is computed, and it is what the founding hypothesis denies.

S3. The lower dimension bound is justified by two theorems; the upper one is flagged as pragmatic.

Open.

O1. Multi-scale structure is asserted to be derivable by spectral partition. No derivation is given, no criterion says how many parts, and nothing shows the derived coarse complex is again a Dowker complex of anything.

O2. The dimension-2 truncation is unjustified except by convenience (recorded as **[CONTESTED BY AUDIT]**).

Next steps, in order.

1. Define the coarse-graining operator explicitly — a spectral or modular partition of $\mathcal{C}$ inducing a simplicial quotient — and state what it preserves. Until then, "derived, not postulated" is a promise.
2. Check whether the derived coarse complex admits a witness relation presenting it; if it does, coarse-graining and ingest are compatible and the theory has one notion of region, not two.
3. Identify one phenomenon requiring H^2, or record that none was found and close the dimension question.

Chapter 6

The Sheaf

> **Intuition.** On top of the shape we lay data. Each region carries a space of things expressible in that region's own coordinates. Each comparison site carries a space in which the neighbouring regions' contents are to be compared, and each region has a translation into it. A cellular sheaf is exactly this. A thought is then a choice of content in every region that survives every translation.

6.1 Two algebras, not one

Before any definition, a distinction that earlier presentations of this material collapsed and that costs real theorems if collapsed.

A stalk must do two jobs, and they are answerable to different algebra:

	coefficient algebra R	**scalar field** $\mathbb{K}$
carries	content: what can be expressed in a region	the metric: how far two contents are from agreeing
determines	what an obstruction *means*	whether L, relaxation and the Hodge decomposition exist
candidates	$\mathbb{B}$, $\mathbb{R}_{\geq 0}$, a lattice, a field	$\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$ only (Proposition 6.2)

Choosing $R = \mathbb{K}$ — taking content to be a vector over the same field that carries the metric — is the choice this book makes, but it is a choice with a price, and §6.4 states the price. Making it silently is the error to avoid.

6.2 The coefficient algebra: what content is

The general form of the question is settled in the contextuality literature [2], where a presheaf assigns to each context the distributions over its joint outcomes, valued in a commutative semiring R, and *contextuality is precisely the non-existence of a global section.* That is the same diagram as this book's, over a different site, and there the semiring is an explicit parameter with consequences:

R	content is	obstruction detected
$\mathbb{B} = (\{0,1\}, \vee, \wedge)$	possibility	logical (possibilistic) obstruction
$\mathbb{R}_{\geq 0}$	probability	probabilistic obstruction
$\mathbb{R}$, $\mathbb{C}$ (signed)	signed amplitude	*weakened or absent*: allowing negative weights admits global sections that positivity forbids
a lattice	propositions	order-theoretic obstruction, via a Tarski Laplacian [15]

Claims and non-claims. [CONTESTED BY AUDIT] The third row is the important one and it is stated here in the weakest form I can defend: passing from a positive semiring to a signed one can destroy obstructions that positivity created, because the constraint being violated was positivity and not linearity. The precise statement in the source literature — for which models, and in what generality, a signed global section always exists — must be checked against [2] and [1] before it is quoted as a theorem of this book. It is recorded now because the *direction* of the effect is what motivates §6.4, and that much is not in doubt.

6.3 The scalar field is forced

Definition 6.1 (Discord axiom)**. [PROPOSED]** The disagreement of a state is required to be a single element of $\mathbb{R}_{\geq 0}$, vanishing exactly when there is no disagreement, and minimisable.

This is a modelling axiom and it is where the real commitment is made — not in the classification below, which is only its consequence.

Proposition 6.2 (The field is forced to $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$)**. [PROVED]** *Suppose stalks are finite-dimensional modules over an associative division algebra $\mathbb{K}$ which is finite-dimensional over $\mathbb{R}$, equipped with an involution and a Hermitian form $\langle \cdot, \cdot \rangle$ for which $\langle x, x \rangle$ satisfies Definition 6.1. Then $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$.*

Proof. The hypotheses make $\mathbb{K}$ a finite-dimensional associative division algebra over $\mathbb{R}$. By Frobenius' theorem the only such algebras up to isomorphism are $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{H}$ [21]. □

Claims and non-claims. The proposition is one line and the hypotheses are everything. What has been shown is not that the field must be one of three, but that *Definition 6.1 costs exactly that much*: demand a single non-negative real measure of disagreement and the algebra is classified. Any future proposal to work over another field is a proposal to give up Definition 6.1, and should be stated that way.

Proposition 6.3 (The metric theory fails over finite fields)**. [PROVED]** *Let $\mathbb{F}_q$ have odd characteristic, let $n \geq 3$, and let B be a nondegenerate symmetric bilinear form on $\mathbb{F}_q^n$. Take $\mathcal{C}$ to be a single edge $e = \{u, v\}$ with $\mathcal{F}(u) = \mathcal{F}(v) = \mathcal{F}(e) = \mathbb{F}_q^n$ and identity restriction maps. Then there is a state $x \neq 0$ with $\omega(x) = 0$ but $\delta x \neq 0$.*

Proof. The quadratic form $Q(z) = B(z, z)$ in $n \geq 3$ variables over a finite field is isotropic: it has a nontrivial zero [20]. Pick $z \neq 0$ with $Q(z) = 0$ and set $x_u = 0$, $x_v = z$. Then $(\delta x)_e = z \neq 0$, so $\delta x \neq 0$, while $\omega(x) = B(z, z) = 0$. □

So over $\mathbb{F}_q$ the equivalence $\omega(x) = 0 \iff \delta x = 0$ — Proposition 7.5, on which every later result depends — is false. Cohomology survives over any field; the *metric* theory does not.

6.4 The trade-off, stated as a cost

	$\mathbb{K}$-vector stalks	**lattice stalks** [15]
Laplacian	$L = \delta^* \delta$	Tarski Laplacian
$H^0(\mathcal{C}; \mathcal{F})$	$\ker L$	fixed points
convergence	spectral, rate λ_2	monotone, by Tarski
three-way Hodge split	**yes**	no
positivity obstructions	**lost**	preserved

This book takes the left column. The reason is Theorem 11.5: the three-way decomposition is what distinguishes repairable misalignment from irreparable holes, and it is what makes the routing of §2.3 possible at all. The right column keeps a class of obstruction that the left column destroys.

Claims and non-claims. This is a real loss and it is placed here rather than in a limits chapter so that it cannot be forgotten. Cognitive Space Theory is a *metric* theory of coherence, and it purchases its dynamics by giving up possibilistic obstruction. Should a phenomenon be found that is purely possibilistic — incoherent in the $\mathbb{B}$ semiring but coherent over $\mathbb{K}$ — this theory will not see it, by construction and not by oversight.

Claims and non-claims. **[PROPOSED]** Between $\mathbb{R}$ and $\mathbb{C}$, this book's recommendation is $\mathbb{C}$, and the reasons are cheap: every result below holds verbatim with transpose replaced by conjugate transpose; restriction maps become unitary rather than orthogonal, and $U(d)$ is connected whereas $O(d)$ is not, so the reflection-parity obstruction to continuously adjusting maps disappears; phase permits asymmetric relations inside a self-adjoint operator, removing the need for a second directed structure alongside $\mathcal{C}$; and the contextuality literature this theory borrows from is complex-valued. The text below is written for a general $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ and specialises to $\mathbb{R}$ in worked examples only.

6.5 Which way do the restriction maps point?

Proposition 6.4 (The direction is forced). **[PROVED]** *Suppose we want a linear operator δ from vertex data to edge data whose kernel is exactly the set of assignments $x = (x_v)_{v \in V}$ that agree on every edge. Then the structure maps must go from faces to cofaces, $\mathcal{F}_{u \trianglelefteq e} : \mathcal{F}(u) \to \mathcal{F}(e)$.*

Proof. An assignment gives one vector $x_v \in \mathcal{F}(v)$ per vertex and nothing on edges. For the endpoint values x_u, x_v of $e = \{u, v\}$ to be compared they must be brought into a common space, and the only space available to e is $\mathcal{F}(e)$. Hence maps $\mathcal{F}(u) \to \mathcal{F}(e)$, $\mathcal{F}(v) \to \mathcal{F}(e)$ are required, and agreement on e reads $\mathcal{F}_{u \trianglelefteq e}(x_u) = \mathcal{F}_{v \trianglelefteq e}(x_v)$.

If instead the maps ran downward, $\rho_{e \to u} : \mathcal{F}(e) \to \mathcal{F}(u)$, then from vertex data no map into $\mathcal{F}(e)$ exists, so no operator $C^0(\mathcal{C}; \mathcal{F}) \to C^1(\mathcal{C}; \mathcal{F})$ is constructible from the structure maps and the condition "x agrees on e" is not expressible. Downward maps assemble instead into a boundary operator $\partial : C^1(\mathcal{C};\mathcal{F}) \to C^0(\mathcal{C};\mathcal{F})$ whose kernel is edge data summing to zero at vertices — a flow condition, not an agreement condition. That is a cellular *cosheaf*, and its invariants are homology, not the cohomology used here. □

6.6 The definition

Definition 6.5 (Cellular sheaf)**.** A *cellular sheaf* $\mathcal{F}$ of finite-dimensional $\mathbb{K}$-inner-product spaces on $\mathcal{C}$ assigns

(i) to each $\sigma \in \mathcal{C}$ a finite-dimensional inner-product space $\mathcal{F}(\sigma)$ over $\mathbb{K}$, the *stalk*;

(ii) to each face relation $\sigma \trianglelefteq \tau$ a linear *restriction map* $\mathcal{F}_{\sigma\trianglelefteq\tau} : \mathcal{F}(\sigma) \to \mathcal{F}(\tau)$,

subject to $\mathcal{F}_{\sigma\trianglelefteq\sigma} = \mathrm{Id}$ and, for every chain $\sigma \trianglelefteq \tau \trianglelefteq \upsilon$,

$$\mathcal{F}_{\tau\trianglelefteq\upsilon} \circ \mathcal{F}_{\sigma\trianglelefteq\tau} \;=\; \mathcal{F}_{\sigma\trianglelefteq\upsilon}. \tag{6.1}$$

Equivalently, $\mathcal{F}$ is a functor from the face poset of $\mathcal{C}$ to $\mathbf{Vect}_{\mathbb{K}}$.

Remark 6.6 (The inner product is where confidence lives). "Inner-product space" is not a technical convenience. The inner product on $\mathcal{F}(\sigma)$ is a free modelling parameter: rescaling it reweights that cell's contribution to the disagreement, and two sheaves with the same stalks and the same restriction maps but different stalk metrics have *different Laplacians and different coherent states*. Precision or confidence, wherever the theory needs it, is carried here and nowhere else.

Definition 6.7 (Cochains)**.** For $k \geq 0$ set $C^k(\mathcal{C};\mathcal{F}) = \bigoplus_{\sigma\in\mathcal{C}_k} \mathcal{F}(\sigma)$, the *k-cochains*, with the direct-sum inner product $\langle x, y\rangle = \sum_\sigma \langle x_\sigma, y_\sigma\rangle$.

6.7 What the 2-cells do: functoriality is $\delta^2 = 0$

Theorem 6.8 (Functoriality on a triangle is exactly $\delta^1\delta^0 = 0$)**.** **[PROVED]** *Let $\tau = \{u, v, w\} \in \mathcal{C}_2$ with edges uv, uw, vw. With the coboundary maps of Definition 7.1 and (7.2), the following are equivalent:*

(a) $(\delta^1\delta^0 x)_\tau = 0$ for every $x \in C^0(\mathcal{C};\mathcal{F})$;

(b) the composite $\mathcal{F}(p) \to \mathcal{F}(\tau)$ is independent of the intermediate edge, for each vertex $p \in \tau$ — that is, (6.1) holds on τ.

Consequently, if $\dim\mathcal{C} \leq 1$ the condition is vacuous and every *assignment of stalks and linear maps is a sheaf.*

Proof. Write $\rho_{p\trianglelefteq e}$ for vertex-to-edge and $\rho_{e\trianglelefteq\tau}$ for edge-to-triangle maps. Expanding,

$$(\delta^1\delta^0 x)_\tau = \rho_{vw\trianglelefteq\tau}(\delta x)_{vw} - \rho_{uw\trianglelefteq\tau}(\delta x)_{uw} + \rho_{uv\trianglelefteq\tau}(\delta x)_{uv},$$

and substituting $(\delta x)_{pq} = \rho_{q\trianglelefteq pq}x_q - \rho_{p\trianglelefteq pq}x_p$ and collecting the coefficient of each of x_u, x_v, x_w gives

$$\begin{aligned}(\delta^1\delta^0 x)_\tau = {}& \big[\rho_{uw\trianglelefteq\tau}\rho_{u\trianglelefteq uw} - \rho_{uv\trianglelefteq\tau}\rho_{u\trianglelefteq uv}\big]x_u \\ &+ \big[\rho_{uv\trianglelefteq\tau}\rho_{v\trianglelefteq uv} - \rho_{vw\trianglelefteq\tau}\rho_{v\trianglelefteq vw}\big]x_v \\ &+ \big[\rho_{vw\trianglelefteq\tau}\rho_{w\trianglelefteq vw} - \rho_{uw\trianglelefteq\tau}\rho_{w\trianglelefteq uw}\big]x_w.\end{aligned}$$

Since x_u, x_v, x_w range independently over their stalks, the expression vanishes identically if and only if each bracket vanishes, which is exactly (b). If $\dim \mathcal{C} \leq 1$ there are no chains $\sigma \trianglelefteq \tau \trianglelefteq v$ of length two, so (6.1) imposes no condition. □

This is the central structural fact of Part II, and it is what the choice of a 2-complex buys. Its consequences:

Corollary 6.9 (On a graph nothing can obstruct)**. [PROVED]** *If* $\dim \mathcal{C} \leq 1$ *the set of sheaves on* $\mathcal{C}$ *with fixed stalk dimensions is the* full *affine space* $\prod_{v \trianglelefteq e} \operatorname{Hom}(\mathcal{F}(v), \mathcal{F}(e))$*. Any continuous adjustment of restriction maps is therefore unobstructed: it can be carried out in any direction, from any starting point, without leaving the space of sheaves.*

Proof. Immediate from Theorem 6.8: with no constraint, every tuple of linear maps is a sheaf. □

Remark 6.10 (Why this is the whole argument for growth)*.* On a 2-complex the sheaves form the subvariety of $\prod \operatorname{Hom}$ cut out by the quadratic equations of Theorem 6.8(b). Adjusting restriction maps — molding — is motion *within* that variety. When the data demand a configuration outside it, no adjustment reaches the demand, and the only remaining move is to change $\mathcal{C}$ itself. That is the sense in which growth becomes *derivable* rather than triggered by a hand-written rule, and by Corollary 6.9 it is unavailable on a graph. The clause "growth from an obstruction" in the thesis of §1.4 rests entirely on this.

> **Claims and non-claims. [PROPOSED]** It is tempting to quantify: with all stalks of dimension d, the ambient space has $(2E + 3T)d^2$ parameters and the constraints supply at most $3Td^2$ equations, suggesting a subvariety of codimension $3Td^2$. The equations are *not* obviously independent, the variety is not obviously smooth, and no such count is asserted here. Establishing the true codimension — and whether the variety is connected — is exactly what would turn Remark 6.10 from a mechanism into a theorem with a stated obstruction class.

6.8 Flatness, holonomy, and what a filled triangle is

Proposition 6.11 (Invertible sheaves are local systems)**. [PROVED UNDER STATED CONDITIONS]** *Suppose every restriction map is invertible. Then* $\mathcal{F}$ *is a local system on* $\mathcal{C}$*: on each simplex the composite maps define parallel transport,* $H^0(\mathcal{C}; \mathcal{F})$ *is the space of parallel sections, and on a connected* $\mathcal{C}$ *the sheaf is determined up to isomorphism by the holonomy representation* $\pi_1(|\mathcal{C}|) \to GL(\mathcal{F}(v_0), \mathbb{K})$ *at a basepoint.*

Proof sketch. Invertibility makes the functor of Definition 6.5 factor through the fundamental groupoid of $|\mathcal{C}|$; (6.1) is the statement that transport is independent of the path within a simplex. The identification of $H^0(\mathcal{C}; \mathcal{F})$ with parallel sections is then Proposition 7.5. Full details are standard and are not reproduced; see [12]. □

> **Intuition.** Read together with Theorem 6.8: transport around the boundary of a *filled* triangle is required to be trivial. A filled triangle is a place where the connection is forced to be flat. An *unfilled* cycle carries holonomy, and that holonomy is a curvature the system can measure and cannot remove by re-choosing frames. So the three structural commitments —

> higher cells, non-identity maps, obstruction-driven growth — are one geometric statement: *fill where flatness is asserted; measure where it is not; grow where the measurement will not go away.*

6.9 The potency ladder, and why the identity is fatal

Restriction maps admit three levels of expressiveness:

level	maps	what it can express
0	identity	"the same thing, verbatim" — no knowledge in the wiring
1	unitary / orthogonal	"the same content in a different frame" — reversible, lossless
2	general linear	"agreeing only on a subspace" — lossy, and the loss is itself a measurement of the connection

Level 0 is not a simplification but a deletion, and Proposition 7.8 makes that precise once the Laplacian is available.

Agenda.

Settled.

S1. Content algebra and metric field are separated, and the metric field is classified by Proposition 6.2 given Definition 6.1.

S2. Finite fields are excluded by an explicit counterexample (Proposition 6.3).

S3. The cost of the metric choice — loss of possibilistic obstruction — is stated where the choice is made.

S4. The direction of restriction maps is forced (Proposition 6.4).

S5. Theorem 6.8: functoriality on 2-cells is exactly $\delta^2 = 0$, and it is vacuous on graphs.

Open.

O1. The recommendation $\mathbb{K} = \mathbb{C}$ is argued but not adopted: the book's worked examples remain real, and no result below has been re-derived in the complex case even though all are expected to hold verbatim.

O2. The signed-semiring claim in §6.2 is **[CONTESTED BY AUDIT]** and must be verified against its sources before use.

O3. Remark 6.10 describes a mechanism, not a theorem. No obstruction class is exhibited; nothing yet *computes* that a required configuration lies outside the sheaf variety.

O4. Nothing here says how stalk dimensions are chosen, or what fixes the stalk metrics of Remark 6.6.

O5. Proposition 6.11 is proved only in sketch, and only under invertibility, which level 2 of the potency ladder abandons.

Next steps, in order.

1. Settle $\mathbb{K}$. If $\mathbb{C}$, restate Chapters 6 and 7 with conjugate transposes and re-run the worked example; if $\mathbb{R}$, record why the arguments for $\mathbb{C}$ were declined.
2. Verify the signed-semiring statement against [2, 1] and either promote it to **[PROVED]** with a precise hypothesis or withdraw it.
3. Determine the codimension and connectedness of the sheaf variety on a 2-complex. This is the single highest-value open problem in Part II: it converts growth from a mechanism into a derivation.

4. Give a criterion for stalk dimension and stalk metric, or state that both are inputs.
5. Write out Proposition 6.11 in full, and state what survives when restriction maps are merely injective, or general.

Chapter 7

Coboundary, Laplacian, and the Three Channels

7.1 The coboundary

Definition 7.1 (Coboundary). Fix an arbitrary orientation of each simplex. The degree-zero coboundary $\delta^0 : C^0(\mathcal{C};\mathcal{F}) \to C^1(\mathcal{C};\mathcal{F})$ acts edgewise on $e = (u, v)$ by

$$(\delta^0 x)_e \;=\; \mathcal{F}_{v\trianglelefteq e}(x_v) \;-\; \mathcal{F}_{u\trianglelefteq e}(x_u) \;\in\; \mathcal{F}(e), \tag{7.1}$$

and the degree-one coboundary $\delta^1 : C^1(\mathcal{C};\mathcal{F}) \to C^2(\mathcal{C};\mathcal{F})$ acts on $\tau = (u, v, w)$ by

$$(\delta^1 y)_\tau \;=\; \mathcal{F}_{vw\trianglelefteq\tau}(y_{vw}) - \mathcal{F}_{uw\trianglelefteq\tau}(y_{uw}) + \mathcal{F}_{uv\trianglelefteq\tau}(y_{uv}). \tag{7.2}$$

A 0-cochain with $\delta^0 x = 0$ is a *global section*. We write δ for δ^0 where no ambiguity arises.

Lemma 7.2 (Orientation-independence). **[PROVED]** *Reversing the orientation of a simplex replaces the corresponding component of δx by its negative. Consequently $\ker\delta$, $\|\delta x\|^2$ and $\delta^*\delta$ are independent of the orientations chosen.*

Proof. Reversal exchanges the roles of the endpoints in (7.1), negating that entry. Writing $\delta' = D\delta$ with D blockwise diagonal with entries ± 1, we get $\delta'^*\delta' = \delta^* D^* D\delta = \delta^*\delta$ since $D^*D = I$; vanishing and squared norms are likewise unaffected. □

Remark 7.3. This licenses "fix an arbitrary orientation", and it is worth checking rather than assuming, because the analogous statement fails for the *signed* per-edge quantity $(\delta x)_e$, which is defined only up to sign. Any diagnostic reporting $(\delta x)_e$ rather than $\|(\delta x)_e\|^2$ is reporting an orientation-dependent quantity and is not well defined.

7.2 The Laplacian and discord

Definition 7.4 (Sheaf Laplacian and discord). The degree-zero sheaf Laplacian is $L = \delta^*\delta : C^0(\mathcal{C};\mathcal{F}) \to C^0(\mathcal{C};\mathcal{F})$, and the *discord* of a state x is

$$\omega(x) = \langle x, Lx\rangle = \|\delta x\|^2 = \sum_{e=\{u,v\}} \left\|\mathcal{F}_{v\trianglelefteq e}(x_v) - \mathcal{F}_{u\trianglelefteq e}(x_u)\right\|^2. \tag{7.3}$$

Proposition 7.5 (Basic properties). **[PROVED]** *L is self-adjoint and positive semidefinite, and*

$$\omega(x) = 0 \iff \delta x = 0 \iff x \in \ker L,$$

with $\ker \delta = H^0(\mathcal{C}; \mathcal{F})$.

Proof. Self-adjointness: $(\delta^*\delta)^* = \delta^*\delta$. Positive semidefiniteness: $\langle x, Lx\rangle = \|\delta x\|^2 \geq 0$. If $Lx = 0$ then $\|\delta x\|^2 = 0$, so $\delta x = 0$ because $\|\cdot\|$ is a norm — this is the step that fails over a finite field (Proposition 6.3); conversely $\delta x = 0$ gives $Lx = 0$. Finally the cochain complex begins in degree 0, so $H^0(\mathcal{C}; \mathcal{F}) = \ker \delta^0 / \operatorname{im} \delta^{-1} = \ker \delta$. □

> **Intuition.** ω is one number: the total squared disagreement across all comparison sites. It vanishes exactly when there is a single coherent thought. The achievement of the construction is that "a coherent thought" has become a precise object — a vector in the kernel of a self-adjoint operator — rather than a mood.

7.3 The three channels

Theorem 7.6 (Hodge decomposition). **[PROVED]** *Let $\mathcal{F}$ be a cellular sheaf on a complex of dimension ≤ 2 over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, so that $\delta^1\delta^0 = 0$ (Theorem 6.8). Then*

$$C^1(\mathcal{C}; \mathcal{F}) \;=\; \underbrace{\operatorname{im}\delta^0}_{\textit{gradient}} \oplus \underbrace{\mathcal{H}^1}_{\textit{harmonic}} \oplus \underbrace{\operatorname{im}(\delta^1)^*}_{\textit{curl}}, \qquad \mathcal{H}^1 := \ker \delta^1 \cap \ker(\delta^0)^*, \tag{7.4}$$

the sum is orthogonal, and $\mathcal{H}^1 \cong H^1(\mathcal{C}; \mathcal{F})$. If $\dim \mathcal{C} \leq 1$ then $\delta^1 = 0$ and the third summand is zero.

Proof. Orthogonal complements give $C^1(\mathcal{C}; \mathcal{F}) = \operatorname{im}\delta^0 \oplus \ker(\delta^0)^*$ and $C^1(\mathcal{C}; \mathcal{F}) = \ker \delta^1 \oplus \operatorname{im}(\delta^1)^*$. Since $\delta^1\delta^0 = 0$ we have $\operatorname{im}\delta^0 \subseteq \ker\delta^1$, so intersecting the first decomposition with $\ker\delta^1$ gives $\ker\delta^1 = \operatorname{im}\delta^0 \oplus (\ker\delta^1 \cap \ker(\delta^0)^*) = \operatorname{im}\delta^0 \oplus \mathcal{H}^1$. Substituting into the second decomposition yields (11.1), and orthogonality is inherited from the two complements. Finally $H^1(\mathcal{C}; \mathcal{F}) = \ker\delta^1 / \operatorname{im}\delta^0 \cong \mathcal{H}^1$ by the first identity. If $\dim \mathcal{C} \leq 1$ then $C^2(\mathcal{C}; \mathcal{F}) = 0$, so $\delta^1 = 0$ and $\operatorname{im}(\delta^1)^* = 0$. □

> **Intuition.** This is the taxonomy of §2 made exact. A disagreement pattern $y \in C^1(\mathcal{C}; \mathcal{F})$ decomposes uniquely and orthogonally into three parts, and each has a different repair:
>
> - **gradient**: $y = \delta x$ for some state. The disagreement is an artefact of the current state and is removed by *relaxing the state*.
> - **curl**: in $\operatorname{im}(\delta^1)^*$. Circulation that the filled triangles see. Removable by *adjusting the restriction maps*.
> - **harmonic**: in $\mathcal{H}^1 \cong H^1(\mathcal{C}; \mathcal{F})$. Neither. It is a topological invariant of the complex, and the only way to remove it is to *change the complex*.
>
> On a graph the curl channel is empty, so misalignment and holes are indistinguishable and there is nothing for molding to discharge. That is the first row of the table in §2.3, now proved.

7.4 The degenerate corner, proved

Proposition 7.7 (Linear feedforward networks are path sheaves)**. [PROVED]** *Let* $W_i \in \mathbb{K}^{n_{i+1}\times n_i}$ *for* $i = 0, \dots, L-1$*. Let* $\mathcal{C}$ *be the path complex on vertices* $v_0, \dots, v_L$ *with edges* $e_i = \{v_i, v_{i+1}\}$ *and no* 2*-cells, and define a sheaf by* $\mathcal{F}(v_i) = \mathbb{K}^{n_i}$*,* $\mathcal{F}(e_i) = \mathbb{K}^{n_{i+1}}$*,*

$$\mathcal{F}_{v_i \trianglelefteq e_i} = W_i, \qquad \mathcal{F}_{v_{i+1} \trianglelefteq e_i} = \mathrm{Id}.$$

Then for $x \in C^0(\mathcal{C}; \mathcal{F})$*,*

$$\omega(x) = \sum_{i=0}^{L-1} \|x_{i+1} - W_i x_i\|^2,$$

so $x \in H^0(\mathcal{C}; \mathcal{F})$ *if and only if* $x_{i+1} = W_i x_i$ *for all* i *— that is, if and only if* x *is the trace of a forward pass. Moreover the map* $H^0(\mathcal{C}; \mathcal{F}) \to \mathbb{K}^{n_0}$*,* $x \mapsto x_0$*, is a linear isomorphism, so* $\dim H^0(\mathcal{C}; \mathcal{F}) = n_0$*.*

Proof. Substituting the stated maps into (7.1) gives $(\delta x)_{e_i} = x_{i+1} - W_i x_i$, and (7.3) gives the displayed discord. By Proposition 7.5, $x \in H^0(\mathcal{C}; \mathcal{F})$ iff every term vanishes, which is the recursion. The evaluation map is linear; it is injective because the recursion determines $x_1, \dots, x_L$ from x_0, and surjective because any x_0 extends by the recursion. The sheaf axioms are vacuous by Theorem 6.8 since $\dim \mathcal{C} = 1$. □

So the set of valid activation traces of a linear network is literally $H^0(\mathcal{C}; \mathcal{F})$ of a path-shaped cellular sheaf, and the forward pass is the unique global section extending a given input.

> **Claims and non-claims.** The nonlinearity is not absorbed. A pointwise σ makes the restriction maps nonlinear and leaves $\mathbf{Vect}_{\mathbb{K}}$ entirely. Two routes exist — sheaves valued in lattices with a Tarski Laplacian [15], or enlarging the stalks to graphs of the nonlinearity — and neither is developed in this book. Proposition 7.7 is exact for the linear case and is not claimed beyond it.

Proposition 7.8 (Identity restriction maps empty the theory)**. [PROVED]** *Suppose every stalk is* $\mathbb{K}^d$ *and every restriction map is the identity. Then*

$$L = L_{\mathcal{C}} \otimes I_d,$$

where $L_{\mathcal{C}}$ *is the ordinary graph Laplacian of the* 1*-skeleton. Hence* $\operatorname{spec} L$ *is* $\operatorname{spec} L_{\mathcal{C}}$ *with each eigenvalue repeated* d *times;* $\dim\ker L = d \cdot b_0$*; and on a connected* $\mathcal{C}$*,*

$$H^0(\mathcal{C}; \mathcal{F}) = \{x : x_u = x_v \ \forall u, v\}, \qquad e^{-tL}x \xrightarrow[t\to\infty]{} \text{the constant assignment at } \tfrac{1}{|V|}\textstyle\sum_v x_v.$$

Proof. With identity restrictions (7.3) reads $\omega(x) = \sum_{\{u,v\}\in E} \|x_u - x_v\|^2$, the Dirichlet form of $L_{\mathcal{C}}$ applied coordinatewise in each of the d components, so $L = L_{\mathcal{C}} \otimes I_d$. The spectrum of a Kronecker product with the identity is that of the factor with multiplicity d. The kernel of $L_{\mathcal{C}}$ is spanned by indicator vectors of connected components, of which there are b_0; tensoring with $\mathbb{K}^d$ multiplies dimension by d. On a connected complex the kernel is spanned by $\mathbf{1}$, so $H^0(\mathcal{C}; \mathcal{F})$ is the diagonal, and $e^{-tL} \to$ the orthogonal projection onto $\ker L$, which is the mean. □

Claims and non-claims. This is the sharpest single argument in the book, because it is a theorem and not a preference. With identity maps: the only coherent thought is exact consensus; relaxation is plain averaging; and diffusion provably drives every region to the global mean regardless of where it started. That last statement is the phenomenon known as oversmoothing, and the published repair is exactly to make the restriction maps non-trivial [5]. Everything a sheaf can express that a graph cannot — that two regions agree *after* a change of frame, or agree only on a subspace — lives in non-identity restriction maps. Freezing them at the identity is not a cheap approximation; it is a proof that the sheaf is decoration.

7.5 The running example, computed

Worked example. Take K_4 (Example 4.7) over $\mathbb{K} = \mathbb{R}$ with identity restrictions and $d = 2$, and the state

$$x_{\mathsf{a}} = \begin{pmatrix}1\\0\end{pmatrix}, \quad x_{\mathsf{b}} = \begin{pmatrix}1\\0\end{pmatrix}, \quad x_{\mathsf{c}} = \begin{pmatrix}0\\1\end{pmatrix}, \quad x_{\mathsf{d}} = \begin{pmatrix}0\\1\end{pmatrix}.$$

Discord, edge by edge, using $\omega_e = \|x_u - x_v\|^2$:

$$\begin{array}{lll} \{\mathsf{a},\mathsf{b}\}: & \|(0,0)\|^2 & = 0, \\ \{\mathsf{b},\mathsf{c}\}: & \|(1,-1)\|^2 & = 2, \\ \{\mathsf{a},\mathsf{c}\}: & \|(1,-1)\|^2 & = 2, \\ \{\mathsf{c},\mathsf{d}\}: & \|(0,0)\|^2 & = 0, \end{array} \qquad \Longrightarrow \qquad \omega(x) = 4.$$

Localisation. The two guilty edges tie at $\omega_e = 2$, so the arg max is not unique: a diagnostic returning a single "worst edge" here returns an artefact of tie-breaking order, not a measurement. The honest return is the tied set $\{\{\mathsf{b},\mathsf{c}\},\{\mathsf{a},\mathsf{c}\}\}$, which reads back correctly as "c is the region at odds with the rest".

Kernel projection. The 1-skeleton is connected, so by Proposition 7.8 the kernel is the diagonal and the projection is the constant assignment at the mean $\bar{x} = \frac{1}{4}\big((1,0) + (1,0) + (0,1) + (0,1)\big) = (\frac{1}{2}, \frac{1}{2})$. Hence

$$\|x_0\|^2 = 4 \cdot \left(\tfrac{1}{4} + \tfrac{1}{4}\right) = 2, \qquad x_\perp = x - x_0 = \left(\pm\tfrac{1}{2}, \mp\tfrac{1}{2}\right)_v, \qquad \|x_\perp\|^2 = 4 \cdot \tfrac{1}{2} = 2,$$

and Pythagoras checks: $\|x\|^2 = 2 + 2 = 4$. □

Exercise 7.9. Recompute $\omega(x)$ after replacing one restriction map by a non-identity: let $\mathcal{F}_{\mathsf{c} \trianglelefteq e}$ be rotation by 90° on the two edges at c inside the triangle, and the identity elsewhere. Does ω rise or fall — and does the resulting assignment satisfy (6.1) on the filled triangle $\{\mathsf{a},\mathsf{b},\mathsf{c}\}$? The second half is the important half, and Theorem 6.8 says exactly what to check.

Agenda.

Settled.

S1. L is self-adjoint PSD and $H^0(\mathcal{C};\mathcal{F}) = \ker L$ (Proposition 7.5): a coherent thought is a matrix kernel.

S2. Theorem 11.5: three orthogonal channels, with the third existing only above dimension 1.

S3. Proposition 7.7: linear feedforward networks are exactly $H^0(\mathcal{C};\mathcal{F})$ of a path sheaf.

S4. Proposition 7.8: identity maps force consensus and averaging, and the published oversmoothing diagnosis is this statement.

Open.

O1. The routing law — relax on gradient, mold on curl, grow on harmonic — is stated as intuition. No dynamical system is defined, no convergence is proved, and nothing shows the three repairs do not interfere.

O2. $\dim H^0(\mathcal{C};\mathcal{F}) \geq 1$ is *not* automatic once restriction maps are non-identity. A sheaf with $H^0(\mathcal{C};\mathcal{F}) = 0$ admits no coherent thought but the zero one, and no condition is given that prevents molding from arriving there.

O3. Neither the harmonic component nor the curl component is computed anywhere, including on the running example, because with identity maps on K_4 both are zero. The book has therefore never exhibited a nonzero element of the channel its architecture depends on.

O4. Nothing here treats the normalised Laplacian, and the convergence rate λ_2 is named without being bounded for any family of interest.

Next steps, in order.

1. Compute a nonzero harmonic class. Build the smallest sheaf on K_4 with non-identity maps for which $\mathcal{H}^1 \neq 0$ and print the class. Until this exists, the three-channel decomposition is a theorem with no worked instance, which is the weakest state a load-bearing result can be in.
2. State the routing law as a dynamical system and prove it decreases ω; identify its fixed points.
3. Find a condition on the restriction maps guaranteeing $\dim H^0(\mathcal{C};\mathcal{F}) \geq 1$, and check whether molding preserves it. If it does not, molding can destroy the possibility of coherent thought and must be constrained.
4. Bound $\lambda_2(L)$ for sheaves arising from ingest, so that "relaxation converges" has a rate.

Chapter 8

Configuration, Time, and Gluing

The four preceding chapters carry a deliberate type error, and this chapter corrects it.

8.1 The error in “the current state”

It is natural to define a cognitive space as a triple $(\mathcal{C}, \mathcal{F}, s)$ with s a distinguished 0-cochain called the current state. The word *current* presupposes a time parameter that the triple does not contain, and adding one does not fix the problem, because the space s lives in is itself time-dependent: $C^0(\mathcal{C}; \mathcal{F}) = \bigoplus_v \mathcal{F}(v)$ changes whenever $\mathcal{C}$ grows or a stalk dimension changes. A trajectory of s is therefore *not* a curve in a fixed vector space, and writing $s(t) \in C^0(\mathcal{C}; \mathcal{F})$ asserts something false.

8.2 Configuration space, and the stratification

Construction 8.1 (The configuration space)**.** **[PROPOSED]** Let $\mathfrak{B}$ be the space whose points are pairs $(\mathcal{C}, \mathcal{F})$ — a finite complex of dimension ≤ 2 together with a cellular sheaf on it. Over each point sits the vector space $C^0(\mathcal{C}; \mathcal{F})$, and these assemble into a bundle $p : \mathcal{E} \to \mathfrak{B}$. A *history* is a path γ in $\mathfrak{B}$ together with a section of $\gamma^*\mathcal{E}$.

Definition 8.2 (The dimension vector)**.** For a configuration $(\mathcal{C}, \mathcal{F})$ put $\mathrm{K}(\mathcal{C}, \mathcal{F}) = \big(\dim \mathcal{F}(\sigma)\big)_{\sigma \in \mathcal{C}}$.

Construction 8.3 (Stratification by K)**.** **[PROPOSED]** Decompose $\mathfrak{B}$ by the value of K. Within a stratum the complex and all stalk dimensions are fixed and only the restriction maps vary, so a stratum is the sheaf variety of §6.8 — the subvariety of $\prod \mathrm{Hom}$ cut out by Theorem 6.8(b). Then:

- **molding** is motion within a stratum, and K is constant along it *by the type of the motion*, not by a separate argument;
- **growth** is a transition to an adjacent stratum, and it changes K by an amount determined by which transition was taken.

This is where the word retired in §5.3 belongs. The strata here do work: they are why a dimension-vector invariant is *locally constant*, which is the standard reason invariants are discrete.

Claims and non-claims. **[PROPOSED]** Construction 8.3 is a proposal with the right shape and no proofs. $\mathfrak{B}$ has been given no topology; "adjacent" has no definition; the strata are not known to be smooth, connected, or even nonempty for a given K; and the sheaf variety's structure is exactly the open problem recorded in Chapter 6. What is claimed is only that this is the right *place* for the two-phase separation to live, and that placing it here would convert three separately-argued claims into one structural fact.

8.3 Continuous time

Definition 8.4 (Sheaf heat flow)**.** Within a fixed stratum, the relaxation of the state is the flow

$$\frac{ds}{dt} = -Ls, \qquad s(t) = e^{-tL}s(0). \tag{8.1}$$

Proposition 8.5 (Relaxation converges to a thought)**.** **[PROVED]** *$e^{-tL}s(0) \to P_{\ker L}s(0)$ as $t \to \infty$, where $P_{\ker L}$ is the orthogonal projection onto $H^0(\mathcal{C};\mathcal{F})$, and the rate is governed by the smallest nonzero eigenvalue λ_2 of L: $\left\|e^{-tL}s - P_{\ker L}s\right\| \le e^{-\lambda_2 t}\left\|s - P_{\ker L}s\right\|$.*

Proof. L is self-adjoint PSD (Proposition 7.5), so it has an orthonormal eigenbasis with eigenvalues $0 = \lambda_1 \le \lambda_2 \le \dots$. In that basis e^{-tL} acts by $e^{-t\lambda_i}$, which is 1 on $\ker L$ and tends to 0 elsewhere, with the slowest decay $e^{-\lambda_2 t}$ on $(\ker L)^\perp$. □

The discrete update $s_{t+1} = s_t - \eta L s_t$ is the explicit Euler scheme for (8.1); the continuous form is the primary object and the discrete one an approximation to it, not the other way round.

Claims and non-claims. Two things are being suppressed and should not be.

First, if $H^0(\mathcal{C};\mathcal{F}) = 0$ then (8.1) converges to 0: the only coherent state is silence and relaxation destroys the state. This is the open item O2 of Chapter 7 appearing as a dynamical failure rather than an algebraic curiosity.

Second, taking the state to be a *point* of $C^0(\mathcal{C};\mathcal{F})$ rather than a *distribution* on it is a modelling choice with an alternative: a density evolving by the Fokker–Planck equation associated with (8.1) has as its stationary measure the Gaussian with precision L, a Gaussian Markov random field. That version carries uncertainty natively. It is not developed here, and the choice is recorded as a choice.

8.4 Gluing, and a candidate for the growth address

Chapter 4 produces a complex from input; this section joins it to what is already present, and asks what the join creates.

Construction 8.6 (Anchor and glue)**.** **[PROPOSED]** Let $\mathcal{C}$ be the present complex and $\mathcal{C}'$ the ingest complex of new input. An *anchoring* is a span $\mathcal{C} \hookleftarrow A \hookrightarrow \mathcal{C}'$ of subcomplex inclusions together with an isomorphism of the two restricted sheaves on A. The glued complex is the pushout $\mathcal{C} \cup_A \mathcal{C}'$, carrying the sheaf determined by the two restrictions and the isomorphism.

The anchoring is the only free choice in the pipeline of §2.4, and it is where interpretation lives: two anchorings of the same input into the same complex are two readings of the same text.

Proposition 8.7 (Mayer–Vietoris for cellular sheaves)**.** **[PROVED]** *Let $\mathcal{C} = A \cup B$ with A, B subcomplexes and let $\mathcal{F}$ be a sheaf on $\mathcal{C}$. Then the sequence of cochain complexes*

$$0 \to C^\bullet(\mathcal{C};\mathcal{F}) \to C^\bullet(A;\mathcal{F}|_A) \oplus C^\bullet(B;\mathcal{F}|_B) \to C^\bullet(A \cap B;\mathcal{F}|_{A\cap B}) \to 0$$

given by restriction and difference is exact, and hence induces a long exact sequence

$$\cdots \to H^0(\mathcal{C};\mathcal{F}) \to H^0(A) \oplus H^0(B) \xrightarrow{-} H^0(A \cap B) \xrightarrow{\partial} H^1(\mathcal{C};\mathcal{F}) \to \cdots$$

Proof. Cochains are direct sums over cells, and the cells of $A \cup B$ are the union of those of A and B, with the cells of $A \cap B$ counted in both. Exactness is then inclusion–exclusion cellwise: a cochain on $\mathcal{C}$ is precisely a pair of cochains on A and B agreeing on $A \cap B$ (exactness in the middle and injectivity on the left), and every cochain on $A \cap B$ extends to A by zero elsewhere (surjectivity on the right). The maps commute with δ because restriction does. The long exact sequence is the standard consequence. □

Intuition. Read the connecting map ∂. An element of $H^0(A \cap B)$ is a coherent thought *on the overlap*. If it is not the difference of coherent thoughts on the two sides, then ∂ sends it to a nonzero class in $H^1(\mathcal{C})$: the act of gluing has created a hole that neither piece had. That is what it means for reading to change you — *the new obstruction is a property of the join, not of either part.* A text you already agreed with entirely contributes nothing to the image of ∂; a text you cannot anchor at all has empty $A \cap B$ and also contributes nothing.

Claims and non-claims. **[PROPOSED]** This is offered as a *candidate* for the growth address and not as a solution. What recommends it is that its type is right: the theory needs a 1-dimensional obstruction that is measured from data rather than manufactured from the current state, and a class in the image of ∂ is by construction not a coboundary. What is missing is substantial. The map ∂ depends on the anchoring of Construction 8.6, and nothing says how it varies with that choice; no procedure turns a class in H^1 into a specific cell to add; and the relation between this class and the harmonic component of Theorem 11.5 — which is where the routing law expects to find the address — is stated nowhere.

8.5 The definition, restated

Definition 8.8 (Cognitive space)**.** A *cognitive space* over $\mathbb{K}$ is a pair $\mathcal{M} = (\mathcal{C}, \mathcal{F})$ consisting of a finite abstract simplicial complex $\mathcal{C}$ of dimension at most 2 (Definition 5.1) and a cellular sheaf $\mathcal{F}$ of finite-dimensional $\mathbb{K}$-inner-product spaces on $\mathcal{C}$ (Definition 6.5). A *state* of $\mathcal{M}$ is an element $s \in C^0(\mathcal{C};\mathcal{F})$; it is *coherent* when $\delta s = 0$. A *history* is a path in the configuration space $\mathfrak{B}$ (Construction 8.1) together with a section of the pulled-back cochain bundle along it.

The state is not part of the space. That is the whole content of the correction: $\mathcal{M}$ is the structure, s is a point of a vector space attached to it, and the two move on different footings — s by Proposition 8.5 within a fixed configuration, $\mathcal{M}$ by molding within a stratum and growth across strata.

Agenda.

Settled.

S1. The state is separated from the space, and the type error in "current state" is identified and corrected.

S2. Relaxation has a continuous form with proved convergence and rate (Proposition 8.5).

S3. Mayer–Vietoris for cellular sheaves is proved (Proposition 8.7) and supplies a candidate of the correct type for the growth address.

Open.

O1. $\mathfrak{B}$ has no topology, so Construction 8.1 is a bookkeeping device rather than a space and "path in $\mathfrak{B}$" is not yet meaningful.

O2. Construction 8.3 is entirely unproved: no smoothness, no connectivity, no definition of stratum adjacency.

O3. The dependence of ∂ on the anchoring is unexamined, and no procedure converts a class in H^1 into a cell to add.

O4. The relation between the Mayer–Vietoris class and the harmonic component of Theorem 11.5 is not established, although the routing law requires them to be the same object.

O5. The Fokker–Planck alternative to a point-valued state is recorded and not developed.

Next steps, in order.

1. Topologise $\mathfrak{B}$. The natural candidate within a stratum is the subspace topology from $\prod \mathrm{Hom}$; between strata, define adjacency by elementary growth moves and check that the resulting relation is a partial order.
2. Prove or disprove that the image of ∂ has nonzero harmonic part. This is the load-bearing question: if the classes created by gluing are harmonic, the routing law and the ingest pipeline are the same mechanism, and if they are not, one of the two is wrong.
3. Determine how ∂ varies with the anchoring, and identify which downstream quantities are anchoring-independent. Anything that is not invariant is a property of the reader, not of the text, and must be labelled as such.
4. Define the growth move: given a harmonic class, which cell is added. Until this exists, "growth is derivable" remains a claim about where the trigger comes from and says nothing about what the trigger does.

Part III

Coherence

Chapter 9

The Coherence Score

Work plan. ¡RECAST¿

Goal. Keep the decomposition $\rho = \alpha \cdot \tilde{\rho}$ intact — it is the single highest-value result in the V0 record — while correcting its *status*: in V2 ρ is a diagnostic and a Lyapunov function, never an objective.

Raw material. This chapter's existing text; `MATH_TOOLBOX.md` I.2 (the exact decomposition, the Courant–Fischer window, the measured refutation of the "dead dynamic range" worry), I.3 (the weighted Anderson–Morley bound and its stated hypothesis); `TATIANA_V1_IDENTITY_MAP.md` §8 DON'T 2.

Steps.

1. **Keep the whole decomposition.** $\alpha = \left\|x^{\perp}\right\|^2 / \|X\|_F^2$ (dissent fraction) and $\tilde{\rho}$ (mode index) as a genuine Rayleigh quotient on $(\ker L)^{\perp}$, hence $\tilde{\rho} \in [\lambda_2(L_K)/B,\ \lambda_{\max}(L_K)/B]$ by Courant–Fischer — both endpoints computable from the graph *before any data.* **¡KEEP¿**, **[PROVED]**.
2. **Get the kernel right, in the text, where it can be checked.** $\ker L$ is the *per-component* constants, not the global diagonal. Two earlier write-ups got this wrong, and when $b_0 > 1$ the error scores a between-component separation as dissent when it is in the kernel. State the correct version and note the trap.
3. **Carry the Anderson–Morley hypothesis explicitly.** The bound requires $\|\mathcal{F}_{v \trianglelefteq e}\| \leq 1$; orthogonal restriction maps suffice, general learned maps do not. An early proof silently assumed identity maps. Any theorem statement must carry the norm bound. **[PROVED UNDER STATED CONDITIONS]**, and say under which conditions.
4. **Retract the novelty framing.** An earlier draft called the weighted bound "the one genuinely new theorem." It is standard spectral graph theory. Correct this in place rather than quietly dropping it.
5. **Add the V2 status paragraph — this is the recast.** State plainly: *do not minimise ρ.* Perfect coherence is death — the flow drives the state to its harmonic projection and generically to zero. ρ diagnoses and certifies descent; it is never the thing being optimised. The typed diagnosis survives and is the point: high α/low $\tilde{\rho}$ is a structural split, low α/high $\tilde{\rho}$ is one bad edge.
6. **Keep §9.5** (the measured refutation of the dead-dynamic-range worry) as written — it is a clean example of a speculative objection being killed by measurement, and $\alpha = (n-1)(1-\bar{c})/n$ is exact.

Done when. The decomposition is stated with its hypotheses, and no reader could come away thinking ρ is an objective function.

This chapter derives the two numbers the engine computes on every tick, and then shows that one of them is secretly two numbers with different meanings.

9.1 Why a normaliser is needed at all

Intuition. The discord ω has no natural scale. A bigger complex has more edges and therefore a bigger ω for no interesting reason, and a state scaled by a factor of 10 has a discord scaled by 100. What is wanted is a dimensionless thermometer in $[0, 1]$: 0 for perfect agreement, 1 for the maximum disagreement the shape permits. The subtlety is getting it *without* an eigendecomposition, because the score is computed on every tick and an eigensolver is not free.

The natural normaliser is $\lambda_{\max}(L)\,\|x\|^2$, since $\omega(x) = x^\top Lx \le \lambda_{\max}(L)\,\|x\|^2$ by the Rayleigh bound. But computing $\lambda_{\max}$ is exactly what we are trying to avoid. The way out is a classical degree bound.

Theorem 9.1 (Anderson–Morley). **[PROVED]** *Let G be a finite simple graph with combinatorial Laplacian L_G. Then*

$$\lambda_{\max}(L_G) \;\le\; \max_{\{u,v\}\in E}\bigl(\deg u + \deg v\bigr) \;=:\; B. \tag{9.1}$$

Proof. Write $L_G = D - A$ and let $Q = D + A$ be the signless Laplacian. Because $Q = NN^\top$ for N the unsigned incidence matrix, Q is positive semidefinite, and $\lambda_{\max}(L_G) \le \lambda_{\max}(Q)$: passing to the bipartite double cover exchanges L_G and Q while preserving the spectral radius, and L_G is a principal restriction of that cover's signless Laplacian.

It therefore suffices to bound $\lambda_{\max}(Q)$. Let y be a unit eigenvector of Q for $\lambda_{\max}$, let u maximise $|y_u|$, and let v be a neighbour of u maximising $|y_v|$. From $\lambda_{\max} y_u = \deg(u)y_u + \sum_{w\sim u} y_w$ we get

$$(\lambda_{\max} - \deg u)\,|y_u| \;\le\; \deg(u)\,|y_v|\,,$$

and symmetrically $(\lambda_{\max} - \deg v)\,|y_v| \le \deg(v)\,|y_u|$. Multiplying the two inequalities and cancelling $|y_u|\,|y_v| > 0$ gives $(\lambda_{\max} - \deg u)(\lambda_{\max} - \deg v) \le \deg u \deg v$, i.e. $\lambda_{\max}^2 - \lambda_{\max}(\deg u + \deg v) \le 0$, whence $\lambda_{\max} \le \deg u + \deg v \le B$. □

Remark 9.2 (The bound is loose, and the looseness costs something). (9.1) is tight only on special graphs — it is exact on complete bipartite graphs. On the running example K_4 it gives $B = 5$ while $\lambda_{\max} = 4$, a 25% over-estimate. Over-bounding $\lambda_{\max}$ compresses ρ toward zero, which is one of three contributors to the dead dynamic range documented in Section 9.5. Finding a tighter eigensolver-free bound is Open Problem 26.4.

9.2 The score

Definition 9.3 (Coherence score). Let $X \in \mathbb{R}^{|V|\times d}$ stack the active stalk vectors as rows, with Frobenius norm $\|X\|_F$. In the identity-restriction regime the *coherence score* is

$$\boxed{\rho \;=\; \frac{\omega}{B\,\|X\|_F^2}} \tag{9.2}$$

When $E = \emptyset$ or $\|X\|_F = 0$, the score is *undefined* and returns $\perp$ — never 0 and never 1.

Proposition 9.4 ($\rho \in [0, 1]$). **[PROVED UNDER STATED CONDITIONS]** *With identity restriction maps, $0 \le \rho \le 1$.*

Proof. Lower bound: $\omega \geq 0$ and $B\,\|X\|_F^2 > 0$ whenever the score is defined. Upper bound: by Proposition 7.8, $L = L_K \otimes I_d$, so $\lambda_{\max}(L) = \lambda_{\max}(L_K) \leq B$ by Theorem 9.1. Hence $\omega = x^\top L x \leq \lambda_{\max}(L)\,\|x\|^2 \leq B\,\|X\|_F^2$, using $\|x\| = \|X\|_F$ under the row-stacking identification. □

Claims and non-claims. The hypothesis in Proposition 9.4 is not cosmetic. With *non-identity* restriction maps the Kronecker factorisation fails, $\lambda_{\max}(L)$ is no longer the spectrum of any $|V| \times |V|$ matrix, and B need not bound it. The correct general hypothesis is an operator-norm condition: if every $\|\mathcal{F}_{v \trianglelefteq e}\| \leq 1$ then a weighted version of Theorem 9.1 still applies. Without it the score can exceed 1. The recorded design decision is that this case is *reported*, not silently clamped — a value $\rho > 1$ is evidence that the normalisation hypothesis has been violated, and truncating it destroys the only evidence there is.

Remark 9.5 (Why $\perp$ is not 0). An empty edge set means no comparison was made. Returning 0 would report perfect agreement, which is a measurement; returning $\perp$ reports that no measurement exists. The distinction propagates: the controller of Chapter 22 carries a third mode precisely so that "no measurement" does not default to "all is well".

9.3 The score answers two questions at once

Intuition. The numerator ω is *blind* to consensus, because consensus is exactly $\ker L$ and the kernel contributes nothing to the quadratic form. The denominator $\|X\|_F^2$ counts consensus in full. So ρ is disagreement divided by (disagreement plus consensus), tangled together with a second and quite different quantity saying which *shape* the disagreement takes. Untangling them costs one orthogonal projection.

Proposition 9.6 (Exact factorisation)**.** **[PROVED UNDER STATED CONDITIONS]** *Assume identity restrictions. Let $x = x_0 \oplus x_\perp$ be the orthogonal decomposition with x_0 the projection onto* $\ker L$*, and suppose $x_\perp \neq 0$. Then $\omega(x) = x_\perp^\top L x_\perp$, $\|X\|_F^2 = \|x_0\|^2 + \|x_\perp\|^2$, and*

$$\boxed{\rho \;=\; \underbrace{\frac{\|x_\perp\|^2}{\|X\|_F^2}}_{\alpha} \cdot \underbrace{\frac{\omega}{B\,\|x_\perp\|^2}}_{\tilde{\rho}}} \tag{9.3}$$

where α, the dissent fraction*, says how much of the state is not consensus, and $\tilde{\rho}$, the* mode index*, says which mode the disagreement occupies.*

Proof. Since $Lx_0 = 0$ and L is symmetric, the cross terms in $x^\top L x = (x_0 + x_\perp)^\top L (x_0 + x_\perp)$ vanish, because $x_0^\top L x_\perp = (Lx_0)^\top x_\perp = 0$. The norm identity is Pythagoras. The factorisation is multiplication and division of (9.2) by $\|x_\perp\|^2$. □

Remark 9.7 (The kernel is per-component, not the global mean). For a connected 1-skeleton, $\ker L$ is the diagonal and $x_0 = \mathbf{1} \otimes \bar{x}$ with $\bar{x}$ the global mean. For $b_0 > 1$ this is *false*: by Proposition 7.8 the kernel consists of assignments constant on *each component*, so x_0 subtracts the **per-component** mean. Using the global mean when $b_0 > 1$ scores between-component separation as dissent, when in fact it lies *inside* the kernel and is not disagreement at all. This is a correctness condition, not an optimisation.

Proposition 9.8 (The achievable window)**.** **[PROVED UNDER STATED CONDITIONS]** *$\tilde{\rho} = R_L(x_\perp)/B$ is a Rayleigh quotient of L restricted to $(\ker L)^\perp$, so by Courant–Fischer*

$$\tilde{\rho} \in \Big[\ \lambda^+_{\min}(L_K)/B, \quad \lambda_{\max}(L_K)/B\ \Big], \tag{9.4}$$

where $\lambda^+_{\min}$ is the smallest nonzero *eigenvalue — the Fiedler value when K is connected. Both endpoints are graph invariants, computable from the 1-skeleton before any data is seen, and needing recomputation only when the topology changes.*

Proof. $x_\perp$ ranges over $(\ker L)^\perp$, an invariant subspace on which L is positive definite with spectrum the nonzero eigenvalues. The Rayleigh quotient of a symmetric operator on an invariant subspace attains exactly the closed interval between the extreme eigenvalues there. Dividing by the constant B gives (9.4). □

Claims and non-claims. Four points, in order of importance.

(i) (9.3) resolves the non-monotonicity of Remark 22.2: adding a well-aligned organ inflates $\|X\|_F^2$ and barely moves $\|x_\perp\|^2$, so α dilutes while $\tilde{\rho}$ is untouched. The controller should gate on $\tilde{\rho}$ and report α alongside.

(ii) (9.4) is **normalisation, not derivation**. Setting a threshold ε as a quantile q of the window still requires choosing q. What changes is that ε is expressed in units the graph supplies and adapts automatically to rewiring. It is a magic number made topology-relative, not a magic number eliminated, and it must not be sold as more than that.

(iii) The split is exact *only* for identity restrictions. With non-identity maps, $\ker L$ is not the per-component constants and no $|V| \times |V|$ matrix carries the spectrum of L. The correct behaviour is then to report α, $\tilde{\rho}$ and the window as $\perp$ rather than approximate them. ρ itself is unaffected.

(iv) When $x_\perp = 0$ exactly, $\alpha = 0$ and $\tilde{\rho} = \perp$: perfect consensus has no mode. Returning 0 for $\tilde{\rho}$ would assert "the most global mode possible", a measurement that was not made.

9.4 The running example, continued

Worked example. Continue Example 4.7 with the state of Chapter 7, where we found $\omega = 4$, $\|X\|_F^2 = 4$ and $\|x_\perp\|^2 = 2$.

The degrees are $2, 2, 3, 1$, so

$$B = \max\{2{+}2,\ 2{+}3,\ 2{+}3,\ 3{+}1\} = 5.$$

Hence

$$\rho = \frac{4}{5 \cdot 4} = \frac{1}{5}, \qquad \alpha = \frac{2}{4} = \frac{1}{2}, \qquad \tilde{\rho} = \frac{4}{5 \cdot 2} = \frac{2}{5},$$

and the factorisation checks: $\alpha \cdot \tilde{\rho} = \frac{1}{2} \cdot \frac{2}{5} = \frac{1}{5} = \rho$.

By Proposition 5.4 the nonzero spectrum is $\{1, 3, 4\}$, so

$$\text{window} = \left[\tfrac{1}{5},\ \tfrac{4}{5}\right] = [0.2,\ 0.8],$$

and $\tilde{\rho} = 0.4$ sits at window position $(0.4 - 0.2)/(0.8 - 0.2) = \frac{1}{3}$.

Reading the pair. $\alpha = \frac{1}{2}$ is large: half the state's energy is not consensus. $\tilde{\rho}$ at $\frac{1}{3}$ of its window is low. By the typed reading of Remark 22.3, large α with small $\tilde{\rho}$ is a *two-camp split* — which is exactly how the state was built. The single number $\rho = 0.2$ reads as "mild disagreement" and misses this completely. □

Exercise 9.9. Add a fifth organ P attached only to S, with $x_{\mathsf{P}} = x_{\mathsf{S}} = (1, 0)$: a perfectly aligned new organ contributing no disagreement. Recompute B, ρ, α and $\tilde{\rho}$. Which changed, and does the change match the dilution claim in item (i) above?

9.5 The dead dynamic range

Claims and non-claims. [REFUTED] (measured 2026-07-28). It was claimed that the compressed dynamic range of ρ is caused by encoder anisotropy acting through α. The claim does not survive. The deployed encoder returns unit-normalised vectors, for which

$$\alpha = \frac{(n-1)(1-\bar{c})}{n}, \tag{9.5}$$

with $\bar{c}$ the mean pairwise cosine. At the observed $\bar{c} \approx 0.449$ and $n = 7$ this gives $\alpha \approx 0.46$, which is not small. The compression is produced *jointly* by α, by $\tilde{\rho}$ sitting mid-window, and by B over-bounding $\lambda_{\max}$. On a seven-organ complex the measurement was $\tilde{\rho} \approx 0.47$ inside a window $[0.146, 0.854]$: the mode index has healthy dynamic range that ρ was multiplying away.

Exercise 9.10. Derive (9.5). Assume $\|x_v\| = 1$ for all v, take x_0 to be the constant assignment at the mean, and expand $\|x_\perp\|^2 = \sum_v \|x_v - \bar{x}\|^2$ in terms of pairwise inner products. The identity is three lines, and it is why the refutation is airtight rather than merely suggestive.

9.6 The precision-weighted generalisation

Construction 9.11 (Weighted discord and normaliser)**. [IMPLEMENTED, UNPROVED]** Give each edge a *precision* $\pi_e > 0$ and set

$$\omega_\pi(x) = \sum_{e=\{u,v\}} \pi_e \left\| \mathcal{F}_{v \trianglelefteq e}(x_v) - \mathcal{F}_{u \trianglelefteq e}(x_u) \right\|^2, \tag{9.6}$$

the Dirichlet form of $L_\pi = \delta^\top P \delta$ with $P = \operatorname{diag}(\pi_e)$. Define the precision-weighted degree and the weighted normaliser

$$d_\pi(v) = \sum_{e \ni v} \pi_e, \qquad B_\pi = \max_{e=\{u,v\}} \big(d_\pi(u) + d_\pi(v)\big),$$

and replace B by B_π throughout.

Proposition 9.12 (The weighted bound)**. [PROVED]** *In the identity-restriction regime, $\lambda_{\max}(L_\pi) \le B_\pi$.*

Proof. For rational weights, the weighted graph Laplacian is the unweighted Laplacian of the multigraph in which edge e is replaced by π_e parallel copies; Theorem 9.1 applied there gives the bound with degrees replaced by weighted degrees. The general case follows because both sides are continuous in the weights and the rationals are dense. □

Claims and non-claims. This is where organ-level confidence legitimately enters the architecture. It does *not* enter as a stalk variance: the dimensional analysis of Chapter 10 shows a confidence-derived stalk floor is d-extensive and swamps everything else at $d = 384$. As an edge precision, entering as $s_e = -\ln c_e/d$, its trace contribution is $O(1)$ and it is dimensionally harmless. The same number is load-bearing in one place and destructive in the other, and the entire difference is where in the sheaf it is attached.

Part IV

Semantic Geometry

Chapter 10

Semantic Geometry

Work plan. ¡KEEP¿

Goal. Keep the geometry and make its dispatch rule — which metric for which job — the chapter's organising principle rather than a remark.

Raw material. This chapter's existing text; `MATH_TOOLBOX.md` IV.1 (the estimation/comparison dispatch), IV.2 (Cone–Bures: kept as a metric, its closeness claim refuted), IV.3 (what a concept is), I.6 (the variance floor — the largest measured downstream effect in the V0 record).

Steps.

1. **Lead with the dispatch rule.** ESTIMATION uses information/Fisher–Rao geometry, where a theorem exists (precision- weighted fusion is a genuine MLE); COMPARISON uses Bures–Wasserstein transport geometry, where a modelling choice exists. They genuinely disagree — by Agueh–Carlier the Wasserstein barycenter of Gaussians has the *plain* weighted mean, not the precision-weighted one. Never blend comparison into estimation.
2. **Own the seam.** π_v fuses in information geometry but its output is immediately compared in transport geometry by ω; π_v is *not* the transport barycenter of its own fine complex. This is a paragraph the book owes a reader, not a bug. Write it.
3. **Record the rejected repair.** Training a blend weight between the two geometries is **[REFUTED]**, on four grounds worth keeping because they generalise: it replaces a choice-with-a-reason by a fitted-constant-with- none; no training signal exists; one scalar wants a sweep, not an optimizer; and the two uses are asymmetric, so blending damages a proven MLE for a fitted number.
4. **Upgrade what a concept *is*.** A Gaussian stalk's level sets are ellipsoids, hence convex, so the SPD stalk already *is* a convex conceptual region in Gärdenfors' sense — concepts have extent, and treating one as a point discards exactly the structure this geometry builds. State this as the chapter's conclusion.
5. **Keep the variance-floor fix and its numbers.** The correct invariant is "floor carries trace $O(1)$" and not "shared epsilon"; getting this wrong once cost a 47× slowdown, and fixing it moved the epistemic term from 229.76 to 0 and made the transport term non-inert. **[MEASURED]**. It is the best worked example in the book of a hardcoded constant silently flattening a whole geometry.
6. **Keep Cone–Bures with its refutation attached.** It survives as a genuine metric (proved, measured, bounded, and it deletes a hand-set threshold). Its "tracks true Hellinger–Kantorovich to under 1%" claim does *not* survive real text. Keep both facts adjacent; report the diagnostic ratio as telemetry and never as a claim.

Done when. A reader can state which metric to use for a given job and why, and can explain the seam between them without treating it as an oversight.

Part I built the shape and Part II measured disagreement across it. This part supplies the geometry *inside* an organ: what a concept is, how many concepts collapse into the single vector a stalk holds, and how far apart two uncertain concepts are.

10.1 A concept is a Gaussian

Intuition. A concept is not a point. It is a point plus an admission of how uncertain that point is, and — more subtly — of the *directions* in which it is uncertain. A concept may be sharply located along one axis of meaning and vague along another. A Gaussian with a structured covariance is the smallest object that records all three.

Definition 10.1 (Concept)**.** A *concept* is a Gaussian $\mathcal{N}(\mu_i, \Sigma_i)$ on $\mathbb{R}^d$ with low-rank-plus-isotropic covariance

$$\Sigma_i = U_i U_i^\top + D_i I_d, \qquad U_i \in \mathbb{R}^{d \times k_i},\ D_i > 0. \tag{10.1}$$

Here μ_i is the embedding, $U_i U_i^\top$ is anisotropic learned spread, and D_i is an isotropic noise floor — the epistemic humility of the concept.

The rank k_i is kept small (in practice $k_i \leq 8$). This is not a performance compromise: it is what makes every computation below reduce to a $k \times k$ problem rather than a $d \times d$ one, and it is the reason the geometry is affordable at all.

10.2 The coarse-graining π_v

Intuition. An organ holds many concepts but exposes one vector to the coarse complex. Which vector? Not the plain average: a near-certain concept should dominate a vague one. The right notion of weight is *precision* — inverse variance — and the resulting formula is exactly how independent noisy measurements of a single quantity are optimally combined.

Construction 10.2 (The coarse-graining map)**.** For an organ v with active concepts $\{\mathcal{N}(\mu_i, \Sigma_i)\}_{i=1}^n$, the coarse stalk value is

$$\boxed{\pi_v \;=\; \Big(\sum_{i=1}^n \Sigma_i^{-1}\Big)^{-1} \Big(\sum_{i=1}^n \Sigma_i^{-1} \mu_i\Big)} \tag{10.2}$$

If v has no active concept then $\pi_v = \perp$, and v is excluded from the computation of ρ rather than contributing a zero vector.

Proposition 10.3 (π_v is the maximum-likelihood fusion)**.** **[PROVED]** *Suppose $\theta \in \mathbb{R}^d$ is a latent vector and each concept is an independent observation $\mu_i = \theta + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \Sigma_i)$. Then the maximum-likelihood estimate of θ is (10.2).*

Proof. The log-likelihood is $\ell(\theta) = -\frac{1}{2}\sum_i (\mu_i - \theta)^\top \Sigma_i^{-1}(\mu_i - \theta) + \text{const}$. Its gradient is $\nabla_\theta \ell = \sum_i \Sigma_i^{-1}(\mu_i - \theta)$, and setting it to zero gives $\big(\sum_i \Sigma_i^{-1}\big)\theta = \sum_i \Sigma_i^{-1}\mu_i$. The matrix $\sum_i \Sigma_i^{-1}$ is positive definite (each Σ_i^{-1} is, since $D_i > 0$), hence invertible, and ℓ is strictly concave, so the stationary point is the unique maximum. □

Remark 10.4 (Three properties worth naming). The fusion (10.2) is *order-independent* (both sums are commutative), *non-destructive* (every concept contributes to both the precision sum and the mean; none is discarded), and *n-ary* rather than iterated-pairwise. These are not incidental: an order-dependent or lossy summary would make the coarse measurement depend on the sequence in which an organ happened to acquire its concepts.

Two computational paths, both exact

- **Isotropic fast path.** When all U_i are empty, $\Sigma_i^{-1} = D_i^{-1} I$ and (10.2) collapses to the scalar precision-weighted average
$$\pi_v = \frac{\sum_i D_i^{-1}\mu_i}{\sum_i D_i^{-1}}, \tag{10.3}$$
computable in $O(nd)$ with no matrix inversion.
- **General path.** With $U_i \neq \varnothing$, invert each Σ_i by the Sherman–Morrison–Woodbury identity
$$\Sigma_i^{-1} = D_i^{-1} I - D_i^{-1} U_i \big(I_{k_i} + D_i^{-1} U_i^\top U_i\big)^{-1} U_i^\top D_i^{-1}, \tag{10.4}$$
accumulate into a dense information matrix, and solve once, at $O(ndk + d^3)$.

Lemma 10.5 (Woodbury for this covariance)**.** **[PROVED]** (10.4) *holds for* $\Sigma = UU^\top + DI$ *with* $D > 0$*.*

Proof. The Woodbury identity states $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$. Take $A = DI$, $C = I_k$, $V = U^\top$. Then $A^{-1} = D^{-1}I$ and $C^{-1} + VA^{-1}U = I_k + D^{-1}U^\top U$, which is positive definite and hence invertible. Substituting gives (10.4). The cost is dominated by the $k \times k$ inverse, not by anything of size d. □

Remark 10.6 (The floor is not optional). A lower bound $D_i \geq 10^{-6}$ is imposed. Without it, a concept reported as near-certain produces $1/D_i \approx 10^{16}$ and the information matrix loses all contribution from every other concept to rounding. The floor pins π_v strongly but finitely. Numerically the general path was checked against a dense $\big(\sum_i \Sigma_i^{-1}\big)^{-1}$ solve and agreed to 2.2×10^{-16}; equal precisions reduce to the plain mean, and a 9:1 precision ratio pulls π_v to the confident concept, as required.

Remark 10.7 (This map is the two-level glue). After execution, each organ's coarse stalk is *overwritten* by its own π_v. That is what makes the coarse ρ and the fine geometry one measurement at two resolutions rather than two unrelated numbers. It also means π_v is lossy in a load-bearing way: whatever the fine complex knew that does not survive (10.2) is invisible to every coarse diagnostic.

10.3 Distance between uncertain concepts

> **Intuition.** How far apart are two concepts? Comparing centres alone throws away the spreads. The 2-Wasserstein — optimal-transport — distance between Gaussians accounts for both, and is a genuine metric rather than a heuristic score.

Definition 10.8 (2-Wasserstein between Gaussians)**.**

$$W_2^2\big(\mathcal{N}(\mu_1,\Sigma_1),\mathcal{N}(\mu_2,\Sigma_2)\big) = \|\mu_1-\mu_2\|^2 + \operatorname{Tr}\Sigma_1 + \operatorname{Tr}\Sigma_2 - 2\operatorname{Tr}\Big[\big(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2}\big)^{1/2}\Big]. \tag{10.5}$$

The covariance part alone is the squared Bures–Wasserstein distance d^2_{BW} on $\mathrm{SPD}(d)$.

Proposition 10.9 (Isotropic reduction to $O(k^3)$)**. [PROVED]** *If $\Sigma_1 = D_1 I$ is isotropic and $\Sigma_2 = U_2U_2^\top + D_2 I$ with $U_2 \in \mathbb{R}^{d\times k}$, then, writing $\{\sigma_j^2\}_{j=1}^k$ for the eigenvalues of the $k\times k$ matrix $U_2^\top U_2$,*

$$W_2^2 = \|\mu_1-\mu_2\|^2 + \big(\operatorname{Tr}\Sigma_1 + \operatorname{Tr}\Sigma_2\big) - 2\sqrt{D_1}\Big[\sum_{j=1}^{k}\sqrt{\sigma_j^2 + D_2} + (d-k)\sqrt{D_2}\Big]. \tag{10.6}$$

Proof. Since $\Sigma_1^{1/2} = \sqrt{D_1}I$, we get $\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2} = D_1\Sigma_2$ and hence $\big(D_1\Sigma_2\big)^{1/2} = \sqrt{D_1}\,\Sigma_2^{1/2}$, so the cross term is $2\sqrt{D_1}\operatorname{Tr}\Sigma_2^{1/2}$. The eigenvalues of $\Sigma_2 = U_2U_2^\top + D_2I$ are $\sigma_j^2 + D_2$ on $\operatorname{im} U_2$ and D_2 with multiplicity $d-k$ on $\ker U_2^\top$, because the nonzero eigenvalues of $U_2U_2^\top$ coincide with those of $U_2^\top U_2$. Summing $\sqrt{\lambda}$ over the spectrum gives (10.6). Only a $k\times k$ eigendecomposition is needed. ☐

This is exact for isotropic Σ_1, not a diagonal approximation.

10.4 The two terms are not commensurable

Construction 10.10 (Reporting the split)**.** W_2^2 is a sum of two quantities measuring different things, and they are returned separately rather than added:

$$W_2^2 = \underbrace{\|\mu_1-\mu_2\|^2}_{\text{semantic: } \textit{what} \text{ they say}} + \underbrace{\operatorname{Tr}\Sigma_1 + \operatorname{Tr}\Sigma_2 - 2\operatorname{Tr}\big[(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2})^{1/2}\big]}_{\text{epistemic: } \textit{how sure} \text{ they are}}. \tag{10.7}$$

Proposition 10.11 (Degeneration in the isotropic regime)**. [PROVED]** *If both concepts are isotropic, $\Sigma_i = D_iI$ with U_i empty, then*

$$W_2^2 = \|\mu_1-\mu_2\|^2 + d\big(\sqrt{D_1}-\sqrt{D_2}\big)^2. \tag{10.8}$$

In particular $D_1 = D_2$ implies $W_2^2 = \|\mu_1-\mu_2\|^2$ exactly.

Proof. Set $k=0$ in Proposition 10.9. Then $\operatorname{Tr}\Sigma_1 + \operatorname{Tr}\Sigma_2 = d(D_1+D_2)$ and the cross term is $2\sqrt{D_1}\cdot d\sqrt{D_2}$, so the covariance part equals $d\big(D_1+D_2-2\sqrt{D_1D_2}\big) = d(\sqrt{D_1}-\sqrt{D_2})^2$. ☐

Claims and non-claims. Proposition 10.11 recorded two uncomfortable facts. Both were true when written; both were subsequently repaired. The diagnosis is retained because it is what forced the repair and because the structural lesson survives it.

(i) The machinery was inert. Every grown concept carried the same prior $D = 1.0$ with empty U, so $D_1 = D_2$ and W_2^2 was *exactly* squared Euclidean distance — an optimal-transport apparatus with a Wasserstein-shaped proof attached and nothing to do. After the repair, D is the shrinkage floor of Remark 10.15 and varies with effective sample size and with rank, so $D_1 \neq D_2$ in general. On two stalks with equal means and ranks 0 and 1, the epistemic term moves from 0 to 0.206: the geometry is live.

(ii) It would have become actively harmful the moment calibration was fixed. The se-

mantic term is *d-intensive* — bounded by 4 for unit-normalised embeddings, whatever d is — while the epistemic term is *d-extensive*. At $d = 384$, a prior $D_1 = 1.0$ against a confident concept with $D_2 = -\ln 0.95$ gave an epistemic term of roughly 230 against a semantic term of 2: about 115 to 1, driven entirely by a self-reported confidence that the architecture elsewhere declines to trust. Merge, split and consolidation would then have been governed by hallucinated confidences. This is a latent defect that switches on precisely when the system is *improved*, which is the worst kind.

The lesson survives both fixes and is why the terms are reported apart: summing a d-intensive and a d-extensive quantity and thresholding the sum is a dimensional error, not a modelling preference.

Exercise 10.12. Verify the 115:1 figure. Take $d = 384$, $D_1 = 1.0$, $D_2 = -\ln 0.95$, both isotropic, and evaluate (10.8). Then repeat with the shrinkage floor $D = O(1/d)$ of Remark 10.15 and confirm the epistemic term drops to order 1. The point of doing this by hand is that the failure is visible in the exponents, before any numbers are substituted.

10.5 Fusion and its entropy

Definition 10.13 (Precision-weighted fusion)**.** The fusion of two concepts is the information-form product

$$\Lambda_i = \Sigma_i^{-1}, \qquad \Sigma_\star = (\Lambda_1 + \Lambda_2)^{-1}, \qquad \mu_\star = \Sigma_\star(\Lambda_1\mu_1 + \Lambda_2\mu_2), \tag{10.9}$$

which is (10.2) for $n = 2$. Full covariances are fused via Woodbury (10.4), never element-wise.

Proposition 10.14 (Fused entropy by the matrix determinant lemma)**.** **[PROVED]** *The differential entropy $h = \frac{1}{2}\ln\big((2\pi e)^d \det \Sigma_\star\big)$ is computable from a $k \times k$ determinant: for $\Sigma = UU^\top + DI$,*

$$\det \Sigma = D^d \det\big(I_k + D^{-1}U^\top U\big). \tag{10.10}$$

Proof. The matrix determinant lemma gives $\det(A + UV^\top) = \det A \cdot \det(I + V^\top A^{-1}U)$. With $A = DI$ and $V = U$ we have $\det A = D^d$ and $V^\top A^{-1}U = D^{-1}U^\top U$. □

Remark 10.15 (Where confidence actually goes). **[IMPLEMENTED, UNPROVED]** An earlier design set the noise floor from the generating model's self-reported confidence via $D = -\ln c$, with a named prior $D = 1.0$ when c was unavailable. Both were removed. The floor is now a shared shrinkage floor

$$D = \varepsilon + \frac{\kappa}{d\,(n_{\text{eff}} + \kappa)} = O(1/d),$$

independent of any reported confidence. The scaling is the whole point: D feeds a d-extensive Bures term, so a floor of order 1 contributes a trace of order d and swamps a semantic term bounded by 4. At $d = 384$ the two retired forms give isotropic traces of about 19.7 and 384 respectively; the shrinkage floor gives 0.88.

Confidence is *relocated, not discarded*: its home is the edge precision of Construction 9.11, entering as $s_e = -\ln c_e/d$ so that its trace contribution is $O(1)$. There it is dimensionally harmless and load-bearing; as an isotropic stalk variance it was neither. Against the deployed model, token log-probabilities are unavailable, so c is genuinely unknown and the engine warns once rather than fabricating a value. This is why ρ, and not self-reported confidence, is the primary signal.

Claims and non-claims. The pattern in this chapter is worth extracting, because it recurs. Each of the three defects — inert geometry, dimensional mismatch, fabricated confidence — was invisible to every test that checked whether the code *ran*, and visible immediately to a check of whether the quantities being combined had the same units. Dimensional analysis is the cheapest correctness tool available here and it was applied late.

Part V

Cohomology and the Growth Address

Chapter 11

Cohomology and the Diagnostic Hierarchy

Work plan. ¡RECAST¿

Goal. This is the chapter V2 changes most in *role.* In V0 the cohomology was a diagnostic hierarchy. In V2 the Hodge decomposition is the routing law of the wake phase and the harmonic component is the growth address — so this chapter carries the architecture, not commentary on it.

Raw material. This chapter's existing text (§"the hierarchy is vacuous as currently fed" is the key surviving content); `MATH_TOOLBOX.md` III.1 (Proposition 8.2 and the reframe it forced), III.2 (measuring a genuine η: four runs, one retraction), III.6 (the b_1-swamping bug and the triangle rule); `TATIANA_V1_IDENTITY_MAP.md` §4.1, §9.

Steps.

1. **Open with Proposition 8.2, not with the hierarchy.** $[\delta x] = 0$ in $H^1(\mathcal{C}; \mathcal{F})$ *unconditionally* — a coboundary is by definition not a gluing failure. This holds for every state, every sheaf, identity restrictions or not. It is trivial to prove and it is the deepest correct criticism the project received: the engine's only 1-cochain is always exact, so growth has no address, structurally, no matter how the state is computed. **[PROVED]**. Everything else in the chapter is downstream of it.
2. **State what it kills.** The original growth intuition (read the address off $H^1(\mathcal{C}; \mathcal{F})$ of a state the system already computes) is dead permanently. So is the "residuals of learned restriction maps" repair, which is **[REFUTED]** for the identical reason — those residuals are still coboundaries of the learned maps, still exact.
3. **Give the fix as the reframe it is.** Growth needs a genuinely *measured* 1-cochain η that is not the coboundary of any single global state, plus the Hodge split; the harmonic component is the address. State this as the pivot of the whole architecture.
4. **Flag the V2-specific hole loudly.** In V0, η came from organs emitting pairwise judgements that disagreed. *V2 has no organs.* Where a genuinely non-exact η comes from for a single kernel is **open** and is the most important unanswered question in the book. Do not paper over it; give it its own section and forward-reference Chapter 26.
5. **Keep the elicitation lesson.** A symmetric $\{u, v, \text{agreement}\}$ schema *cannot* be a 1-cochain — antisymmetrising a symmetric table gives $\eta \equiv 0$ identically. Any future source of η, organ-based or not, must be antisymmetric by construction. Two earlier designs failed informatively before one worked.
6. **Keep the dimensionality trap.** A 1-dimensional harmonic subspace can report *that* an obstruction exists but never *where* — at $b_1 = 1$ every harmonic component is a multiple of one

fixed vector, so the "consistent growth address" it reports is an artifact. Any configuration used for a real address needs $b_1 \geq 2$. This nearly became a reported finding and must stay visible.

7. **Keep the triangle rule and the swamping bug.** Record a 2-simplex exactly when its three concepts co-fire; the earlier claim that "no triangles $\Rightarrow b_1 = 0$" was backwards — no triangles *maximises* b_1, and inserting each assembly as a clique manufactures $(n-1)(n-2)/2$ artifact cycles that swamp the real address. Also record that $b_1 = E - V + b_0 - F$ is *wrong* once triangles share edges; compute b_1 from ranks, always.
8. **Correct the headline statistic.** On a sparse graph a high "holes" fraction is the *null expectation* ($\mathbb{E} = b_1/|E|$), not evidence — measured twelve sigma below a null in one run. The usable output is per-cycle *localisation* of harmonic mass, which is what Construction 6 actually consumes. Replace the fraction with localisation wherever the old text reports it.

Done when. A reader can prove Proposition 8.2 unaided, explain why it makes the growth address the central problem, and state what would have to be true of a source of η for it to work — including that V2 does not yet have one.

11.1 Two pathologies that must not be confused

Definition 11.1 (Sheaf cohomology in low degrees)**.** With coboundaries $\delta^k : C^k(\mathcal{C};\mathcal{F}) \to C^{k+1}(\mathcal{C};\mathcal{F})$,

$$H^0(\mathcal{C};\mathcal{F}) = \ker \delta^0 \quad \text{(global sections)}, \qquad H^1(\mathcal{C};\mathcal{F}) = \ker \delta^1 / \operatorname{im} \delta^0 \quad \text{(gluing failure)}.$$

Intuition. A *structural hole* — $b_1 \neq 0$, a Betti number of the bare shape — means a *binding is missing*: no theorem yet ties some loop together. A *gluing failure* — $H^1(\mathcal{C};\mathcal{F}) \neq 0$ of the sheaf — means the data are *contradictory*: the organs cannot be reconciled. The first calls for growth; the second calls for repair. Treating a contradiction as a hole makes you add structure when you should resolve a conflict, and treating a hole as a contradiction makes you argue when you should build.

Remark 11.2 (b_1 is earned here)*.* b_1 is a defensible diagnostic *only* because simplices were defined to mean genuine n-ary binding (Remark 4.4). A filled 2-simplex really asserts a joint relation, so an unfilled 1-cycle really is a missing joint relation. Most applications of topological data analysis to knowledge graphs skip this justification and compute b_1 of a complex whose 2-cells mean nothing in particular, at which point the number is an artefact of the construction rather than a fact about the knowledge.

11.2 The hierarchy is vacuous as currently fed

This is the most important negative result in the book, and it is worth stating as a proposition rather than as a caveat.

Proposition 11.3 (Exactness of the emitted cochain)**. [PROVED]** *Suppose the* 1*-cochain submitted for cohomological analysis is obtained as* $y = \delta^0 x$ *for some global state* $x \in C^0(\mathcal{C};\mathcal{F})$*. Then* $[y] = 0$ *in* $H^1(\mathcal{C};\mathcal{F})$*, for every* x*, every sheaf* $\mathcal{F}$*, and every choice of restriction maps.*

Proof. $H^1(\mathcal{C};\mathcal{F}) = \ker \delta^1 / \operatorname{im} \delta^0$ and $y \in \operatorname{im} \delta^0$ by construction, so y represents the zero class. (That $y \in \ker \delta^1$ is automatic from $\delta^1 \delta^0 = 0$, but is not even needed: membership in the image is already the conclusion.) ∎

Claims and non-claims. [REFUTED] The consequence is not subtle. The engine computes its 1-cochain by applying δ to the current state — that is what ω is built from. So the class it then measures in $H^1(\mathcal{C};\mathcal{F})$ is identically zero, and the growth mechanism that was supposed to be addressed by a nonzero class *has no address*. Every reported $H^1(\mathcal{C};\mathcal{F})$ diagnostic in that pipeline was measuring the definition of a coboundary, not the state of the system.

The repair requires a source of 1-cochains that is *not* of the form δx. The natural candidate is pairwise judgements formed independently: an organ pair that renders a verdict on their shared edge without either of them consulting a global state. That is new engineering — organs that emit pairwise judgements, storage for 2-simplices, and an actual δ^1 — and it does not exist yet.

One proposed repair in the sources is recorded by the audit as *half circular*, and the circularity must be located precisely before it is adopted. See Open Problem 26.2.

Remark 11.4 (The negative result is publishable). Proposition 11.3 is not merely an internal correction. It identifies a failure mode that any sheaf-based diagnostic over a derived cochain will share, and the derivation is three lines. It belongs in the paper on cohomological obstruction as a stated negative result, not in a footnote.

11.3 The Hodge decomposition

Theorem 11.5 (Hodge decomposition for cellular sheaves)**. [PROVED]** *Let* $L_1 = \delta^0(\delta^0)^\top + (\delta^1)^\top\delta^1$ *be the degree-one Laplacian. Then*

$$C^1(\mathcal{C};\mathcal{F}) = \operatorname{im}\delta^0 \;\oplus\; \ker L_1 \;\oplus\; \operatorname{im}(\delta^1)^\top, \tag{11.1}$$

an orthogonal decomposition, and $\ker L_1 \cong H^1(\mathcal{C};\mathcal{F})$*.*

Proof. Orthogonality of $\operatorname{im}\delta^0$ and $\operatorname{im}(\delta^1)^\top$ follows from $\delta^1\delta^0 = 0$: for $y = \delta^0 a$ and $z = (\delta^1)^\top b$, $\langle y, z\rangle = \langle \delta^0 a, (\delta^1)^\top b\rangle = \langle \delta^1\delta^0 a, b\rangle = 0$. For $w \in \ker L_1$ we have $0 = \langle L_1 w, w\rangle = \|(\delta^0)^\top w\|^2 + \|\delta^1 w\|^2$, so $w \in \ker(\delta^0)^\top \cap \ker\delta^1$, which is the orthogonal complement of $\operatorname{im}\delta^0 \oplus \operatorname{im}(\delta^1)^\top$. Counting dimensions gives (11.1). Finally each class in $H^1(\mathcal{C};\mathcal{F}) = \ker\delta^1/\operatorname{im}\delta^0$ has a unique representative orthogonal to $\operatorname{im}\delta^0$, and that representative lies in $\ker L_1$. □

Intuition. The decomposition sorts every 1-cochain into three kinds of thing: the part that is the disagreement of some global assignment ($\operatorname{im}\delta^0$), the part that is a genuine irreducible contradiction ($\ker L_1$, the harmonic part), and the part that is a local circulation with no bearing on gluing. Proposition 11.3 says that as currently fed, the engine only ever produces cochains of the first kind.

Worked example. On the running example K_4 with the trivial sheaf ($\mathcal{F}(\sigma) = \mathbb{R}$): $\dim C^0(\mathcal{C};\mathcal{F}) = 4$, $\dim C^1(\mathcal{C};\mathcal{F}) = 4$ and the complex has one filled triangle so $\dim C^2(\mathcal{C};\mathcal{F}) = 1$. Since the 1-skeleton is connected, $b_0 = 1$ and $\operatorname{rank}\delta^0 = 4 - 1 = 3$. The only cycle in the 1-skeleton is the triangle $\{\mathsf{S},\mathsf{M},\mathsf{R}\}$, and it is filled, so $b_1 = 0$. Consistently: $\dim\ker\delta^1 = \dim C^1(\mathcal{C};\mathcal{F}) - \operatorname{rank}\delta^1 = 4 - 1 = 3 = \operatorname{rank}\delta^0$, so $H^1(\mathcal{C};\mathcal{F}) = 0$.

Now delete the 2-simplex, leaving the same 1-skeleton unfilled. Then $C^2(\mathcal{C};\mathcal{F}) = 0$,

$\ker \delta^1 = C^1(\mathcal{C}; \mathcal{F})$ has dimension 4, and $H^1(\mathcal{C}; \mathcal{F}) \cong \mathbb{R}^{4-3} = \mathbb{R}$, matching $b_1 = 1$: one unfilled loop, one class. This is the sense in which coning (Chapter 19) kills a class — it adds precisely the 2-cells that make the cycle a boundary. □

11.4 The uncertainty stack

Intuition. Coherence is not correctness. Five instruments, ordered by cost, answer five different questions, and they are not interchangeable. The most common error in systems of this kind is to let the cheapest instrument speak for all five.

Instrument	Question	Cost	When
ρ	Are my parts agreeing?	effectively free	every tick
$\hat{\rho}$ vs ρ	Did I *know* whether I would agree?	free (k-NN)	once history exists
$b_0,\ b_1$	What is structurally missing?	medium	on conflict
perturbation	Am I stable to paraphrase?	k model calls	high stakes only
VerifyOp	Am I actually *right*?	medium	promotion gate

Table 11.1: The five instruments and their distinct questions.

Remark 11.6 (Perturbation is a variance, not a derivative). Stability is measured as paraphrase-perturbation variance: run k semantically equivalent paraphrases and measure the spread of outcomes. This replaces an earlier proposal to compute a Fréchet derivative $\|\partial F/\partial x\|$ of the whole pipeline. That derivative does not exist: F factors through discrete token sampling, a DAG generator, and database lookups, none of which is differentiable. Earlier drafts also posited an infinite-dimensional Banach state space for the same purpose; it was removed for the same reason. The replacement costs k model calls and must be rationed.

Remark 11.7 (Expected precision and the feedback edge). $\hat{\rho}$ is a similarity-weighted k-nearest-neighbour prediction of ρ from a log of past (query embedding, ρ) pairs. The *signed* gap $\hat{\rho} - \rho$ separates three states that a single number cannot: *competence* (predicted agreement, observed agreement), *known ignorance* (predicted disagreement, observed disagreement — "confidently, I do not know this"), and *hallucination risk* (predicted agreement, observed disagreement). A VerifyOp failure updates the store, so self-knowledge improves with use. This is the only instrument in the stack whose accuracy is supposed to increase over time, and it is therefore the one that carries the architectural bet of Chapter 25.

Part VI

Identity

Chapter 12

Identity: What Survives Mutation

¡NEW¿ *This chapter does not exist in the V0 book. It is the centre of V2 and everything after it depends on it.*

Work plan. ¡NEW¿

Goal. Make "TATIANA stays itself while it changes" into a number that can be printed and a theorem that can fail. Nothing in this chapter may rest on the word "identity" being intuitively clear.

Raw material. `DOCS/TATIANA_V1_IDENTITY_MAP.md` §3 (the three levels), §4 (which phase conserves what); `MATH_TOOLBOX.md` II.2 (the $\mathbb{K}/W$ split and the consolidation formula), III.3 (Construction 6 fixes the growth increment); `TATIANA-V2/MATH/phase0_invariant_dimension.md` (the first measurement, already run — see step 5).

Steps.

1. **State the split first.** Identity is carried by the crystallized store $\mathbb{K}$, not by the working complex W; W may be destroyed at the end of every session. Give the reader the reason this is the right place for identity to live *before* any formula: a memory that is destroyed nightly cannot be what makes you you.
2. **Level 0 — the exact invariant.** Define $\mathrm{K}(\mathbb{K}) = (\dim \mathcal{F}_{\mathbb{K}}(\sigma))_\sigma$, the dimension vector, a K_0 class. Prove that the consolidation rule $R^{\mathbb{K}} \leftarrow \Pi_{O(d)}(R^{\mathbb{K}} + \gamma(\nu)(R^W - R^{\mathbb{K}}))$ conserves it *exactly*. The proof is short and must be given in full: a retraction onto $O(d)$ cannot change a dimension, molding cannot, state relaxation cannot.
3. **The one thing that moves it.** Growth, and only growth, and by a *derived* amount: $\dim \mathcal{F}(w) = \dim H^0(\mathcal{C};\mathcal{F})(\gamma;\mathcal{F}|_\gamma)$ from Construction 6 (Chapter 19). Nothing is hand-set. State the theorem shape explicitly, because it is the shape V2 is built to have: *two of three timescales conserve the invariant exactly; the third changes it computably.*
4. **Level 1 — the projective invariant.** Level 0 is coarse: it cannot distinguish two stores with the same dimensions and different maps. State the finer claim, that relative structure is preserved while absolute magnitude is not, $\mathcal{F}_{\mathbb{K}} \sim \lambda \mathcal{F}_{\mathbb{K}}$. Then make the observation that earns it: $\Pi_{O(d)}$ is the polar retraction $R \mapsto UV^\mathsf{T}$, which discards scale and keeps direction — written into the design months earlier as a gauge fixing, and only later recognised as the identity-preservation mechanism. Say plainly that the coincidence is the reason to trust it, and that this is **[CONTESTED BY AUDIT]** until step 5's measurement is extended to cover it directly.
5. **Report the measurement that exists.** Phase 0 has been run. Reproduce its result here in full, including its controls: on a 3-cycle and on a $b_1 = 0$ path, pre-existing $\mathrm{K}(\mathbb{K})$ is untouched across 500 wake/sleep ticks; the single growth event's size is $k_w = 1$ confirmed by two independent routes (holonomy $\dim\ker(H - \mathrm{Id})$ and cochain-space $\dim\ker(\delta^0(\delta^0)^\mathsf{T})$); $Q(t)$ moves while dimensions stay pinned. Tag **[MEASURED]**.

6. **State what the measurement does not cover, in the same breath.** Sleep-phase pruning is unimplemented, so the "may decrease" half of the invariant table is untested and the kill-test is *survived, not passed*. No 2-cells; $\gamma(\nu)$ constant; post-growth dynamics unwired; dense rather than Householder maps; synthetic input only. This list is not an appendix item — it goes beside the result.
7. **Level 2 — declare it out of scope and say why.** The full isomorphism class of $\mathcal{F}_{\mathbb{K}}$: a cellular sheaf on a graph is a representation of the incidence quiver, and anything richer than a single cycle is of wild representation type. Krull–Schmidt makes the decomposition into indecomposables unique but not computable. V2 reports levels 0 and 1 and does not chase level 2.

Done when. A reader can state what $\mathrm{K}(\mathbb{K})$ is, prove unaided that consolidation conserves it, say exactly how much growth changes it and why that number is not a choice, and list what the existing measurement does not cover.

Work plan. ¡NEW¿

Goal. §: The open question this chapter raises and cannot yet close — whether molding can erode the obstruction that growth depends on.

Raw material. Raised by the Phase 0 experiment; see `TATIANA-V2/MATH/phase0_invariant_dimension.md` §4 and the logbook entry for 2026-08-14.

Steps.

1. State the concern precisely. Molding changes restriction maps; the growth address is a harmonic class determined *by* those maps. In the experiment as run, molding is bounded to one nudge per tick before the same tick's stall check, so the question never arises. Unbounded, it might.
2. Say why it matters: if slow molding can dissolve a harmonic obstruction, the three-layer handoff of Chapter 13 is not a handoff but a race, and "growth fires when molding stalls" would be a statement about the schedule rather than about the geometry.
3. Do not answer it here. Record it as an open problem with a stated experiment (vary the molding budget per tick; measure whether the harmonic component's norm decays to below the stall tolerance without growth ever firing) and cross-reference Chapter 26.

Done when. The question is stated sharply enough that someone else could run the experiment that settles it.

Part VII

The Two-Phase Law

Chapter 13

The Two-Phase Law

¡NEW¿ *This chapter replaces the role a single global objective would have played. There is no such objective, and the reason is structural.*

Work plan. ¡NEW¿

Goal. Establish that TATIANA alternates between two phases with different objectives, that this is forced rather than chosen, and that cost lives in exactly one of them.

Raw material. `TATIANA_V1_IDENTITY_MAP.md` §4 (the table), §4.1 (wake, the Hodge routing law), §4.2 (sleep), §6 (the maintenance cost); `MATH_TOOLBOX.md` I.5 (predictive coding, $\dot{x} = -Lx$ is sheaf diffusion), II.2 ($\gamma(\nu)$, $\gamma_0 \ll 1$).

Steps.

1. **Open with the table**, since it is the whole chapter in one view: what drives each phase ($q_t \neq 0$ vs $q_t = 0$), what moves (s fast, $\mathcal{F}$ slow, $\mathcal{C}$ grows / $\mathcal{F}$ renormalizes, $\mathcal{C}$ prunes), the objective (discharge discord / pay down maintenance cost), and what happens to $\mathrm{K}(\mathbb{K})$ (conserved exactly / may decrease).
2. **Argue both phases are forced.** The model must be able to run with empty input; that requirement alone establishes an offline phase exists. Name it rather than deriving it from biology, and only then note the neuroscience that agrees.
3. **Wake — the Hodge routing law.** On the 2-complex, $C^1(\mathcal{C};\mathcal{F}) = \operatorname{im}\delta^0 \oplus H^1(\mathcal{C};\mathcal{F}) \oplus \operatorname{im}(\delta^1)^*$, orthogonally. State and prove the routing claim: *each summand is dischargeable by exactly one layer.* $\operatorname{im}\delta^0$ yields to changing s (the fast flow $\dot{s} = -L_{\mathcal{F}}s + \text{drive}$, provably, since $L_{\mathcal{F}}$ is PSD); $H^1(\mathcal{C};\mathcal{F})$ does not — it is holonomy of $\mathcal{F}$ around a cycle, and only molding or growth touches it; $\operatorname{im}(\delta^1)^*$ flags a bad 2-cell. Include the Lyapunov argument in full.
4. **The handoff is a stall, not a threshold.** This is the chapter's sharpest point and must not be blurred. When the constrained descent has converged and discord remains, it is *proven* that no further motion at that level helps. That is the growth trigger, derived from the geometry. Contrast explicitly with a rule of the form $\rho > c$, and say why the latter would be a hand-set number.
5. **Name the one free parameter.** The numerical tolerance on "converged" is an **[PROPOSED]** engineering choice. State it in the open. Do not hide it inside a λ.
6. **Sleep — renormalize, then down-select.** Apply $\Pi_{O(d)}$ globally: scale drops, relative structure survives (Chapter 12, level 1). Then consult the maintenance cost and prune structure that does not pay for itself.
7. **State where energy lives and why it lives nowhere else.** One term, one phase. Growth adds during wake; pruning removes during sleep; they never bid against each other inside a single tick. Give this its own paragraph — it is the entire answer to the objection that the design needs

a multi-term objective with weights.

8. **The maintenance cost.** $\mathcal{U}(\mathbb{K}) = L(\text{structure}) + L(\text{observations} \mid \text{structure})$. The description-length form is already derived (Chapter 18); what is new here is only its *placement.* Then name the one genuine free parameter of the entire design: the exchange rate between the cost of a deformation and the cost of a new cell, since MDL needs both in the same units. State it in the open and record that if it acquires siblings, the design has regressed.

Done when. A reader can say what moves in each phase, prove that the fast flow cannot discharge a harmonic component, explain why the growth trigger is not a threshold, and name the single exchange rate together with the reason a second one would be evidence of failure.

Work plan. ¡RETIRE¿

Goal. §: Record the refused alternative, so it is not proposed again.

Raw material. The predictive-coding discussion in `DOCS/Phase I --- Understand the Geometry o.txt` (second half); refusal recorded in `TATIANA_V1_IDENTITY_MAP.md` §8 DON'T 1 and §13.

Steps.

1. State the proposal as it was made and at its strongest: a single cognitive energy functional $F = E_{\text{pred}} + \lambda_C E_C + \lambda_{\mathcal{F}} E_{\mathcal{F}} + \lambda_s E_s + \lambda_I E_I + \lambda_K E_K$, with dynamics choosing whichever admissible transformation most decreases it. It is a coherent design and its appeal should be visible.
2. Give the refusal and its reason: six weights are six hand-set numbers trading against each other every tick, which is exactly the failure mode the discipline rule against hand-set constants exists to prevent. The two-phase separation achieves the same coordination with one exchange rate.
3. Record the standing consequence: *do not add a* λ. A second exchange rate means the two-phase separation has failed and should be re-derived, not patched.
4. Note honestly what the refused proposal got right, since parts of it survive under other names: prediction error precision-weighted is ω (Chapter 9); the compression term is the MDL law (Chapter 18); the stability term is $\gamma_0 \ll 1$'s anti-interference property. What is rejected is the *blending*, not the ingredients.

Done when. Someone rereading the predictive-coding notes can see why they were not adopted wholesale, and which of their parts already live elsewhere in the book.

Part VIII

Dynamics

Chapter 14

Growth Operators and the Algebra $\mathfrak{D}$

Work plan. ¡RECAST¿

Goal. Keep the typed structural operators. Rebuild the algebra $\mathfrak{D}$ on *measured* relations rather than asserted ones, and be honest that in V2 this chapter is mostly a programme rather than a result.

Raw material. This chapter's existing text; `MATH_TOOLBOX.md` I.1 (typed ops replacing Pachner moves), I.7 (the curvature controller — derived, unit-tested, never wired); `DOCS/Phase I --- Understand the Geometry o.txt` (the phased programme: invariants, then primitives, then interaction laws, then the algebra).

Steps.

1. **Keep the typed operators** bind, collapse, split, merge, cone, each logged to a mutation history. Keep the reason they replaced Pachner moves: a knowledge graph is not a manifold, so the preservation guarantee does not transfer. **¡KEEP¿**.
2. **Add the invariants step *before* the primitives.** The V0 text hunts for primitive operations first. The correct order — and the one the source programme itself corrects to — is invariants first: "primitive" is only meaningful relative to what must not move. Each operator must now declare its effect on $\mathrm{K}(\mathbb{K})$ (Chapter 12). Make that declaration part of the definition of an operator, not a later audit.
3. **Derive the relations; do not invent them.** The commutators are to be *measured*: apply candidate primitives in both orders on small explicit complexes and compute the difference. State clearly that $\mathfrak{D} = T(V)/I$ is a target, and that I is empty until such measurements exist. An algebra asserted without its relations is bookkeeping.
4. **Report the independence-relation test honestly.** If the independence relation is empty, $\mathfrak{D}$ is the free algebra and the whole apparatus counts directed acyclic graphs. Record this as the falsifier for this chapter, and record that it has not been run for V2's operator set.
5. **Keep the curvature controller with its warning.** "Run the flow, skip the surgery" is right, and Forman's weights are *not* free — they are fixed by whichever Laplacian the system actually prints ω and ρ from, so using a Hebbian coupling as the Forman weight computes the curvature of a *different* operator: a consistent-looking number about the wrong thing. Keep the flagged prediction that a low-precision edge may drift positive and be folded *because we are unsure of it*, which is backwards; check the sign behaviour before trusting the controller.
6. **Keep the floor derivation as a template.** The exact multiplicative flow can never sever an edge; what severs is the forward-Euler discretisation flipping sign — so the old weight floor was patching a numerical artefact that had been misdiagnosed as a safety mechanism. The replacement is exponential integration plus a control barrier, which turns out to be exactly the

object neuroscience calls weight-dependent plasticity. Keep also the floating-point trap: bare exponential integration underflows to zero and severs silently; only the barrier survives floating point.

7. **State the implementation status plainly.** No line of the curvature controller has ever moved a real edge weight. **[PROPOSED]**.

Done when. Every operator declares its effect on the invariant, and the chapter distinguishes what is proved from what is a programme awaiting measurement.

Intuition. Learning *is* change of shape. The complex binds organs that work together, prunes idle ones, and — when a procedure repeatedly succeeds — promotes it to a first-class operator. Four axes of growth, one of which grows the very set of moves the system can make.

14.1 The structural operators

Definition 14.1 (Base moves)**.** The base moves on $\mathcal{C}$ are the typed operators

$$\{\mathsf{bind},\ \mathsf{collapse},\ \mathsf{split},\ \mathsf{merge}\},$$

each logged to an immutable mutation history.

Remark 14.2 ("Pachner move" was discarded, correctly)*.* Earlier drafts called these Pachner moves. A knowledge complex is not a manifold, so the invariance guarantees that make Pachner moves useful — that any two triangulations of a PL manifold are connected by a finite sequence of them — simply do not transfer. Borrowing the name would have imported a theorem that does not apply.

Construction 14.3 (The four growth axes)**.** Evolution proceeds along

$$\underbrace{\delta\mathcal{C}_{\text{fine}}}_{\text{concepts}},\quad \underbrace{\delta\mathcal{C}_{\text{coarse}}}_{\text{coalitions}},\quad \underbrace{\delta\mathcal{F}}_{\text{stalk and restriction data}},\quad \underbrace{\delta\mathfrak{D}}_{\text{new operators}}.$$

The fourth axis is the essential one: the operator algebra is time-dependent, $\mathfrak{D}(t) \to \mathfrak{D}(t+1)$.

Definition 14.4 (Hebbian coupling and decay)**.** For a coalition σ with weight $w(\sigma, t) \in [0, 1]$: co-activation increments w; idle steps apply $w(\sigma, t+1) = (1-\gamma)w(\sigma, t)$ for a decay rate $\gamma \in (0, 1)$; a coalition falling below a bind threshold *collapses*. A pair past the bind threshold is a 1-simplex of K.

Proposition 14.5 (Idle coalitions collapse in finite time)**.** **[PROVED]** *If σ is idle from time t_0 and the bind threshold is $\theta > 0$, then σ collapses at the first $t \geq t_0 + \log_{1/(1-\gamma)}\big(w(\sigma, t_0)/\theta\big)$.*

Proof. Iterating the decay gives $w(\sigma, t_0 + m) = (1-\gamma)^m w(\sigma, t_0)$, which falls below θ as soon as $m \ln(1-\gamma) < \ln\big(\theta/w(\sigma, t_0)\big)$; since $\ln(1-\gamma) < 0$ this is $m > \log_{1/(1-\gamma)}(w(\sigma, t_0)/\theta)$. □

Construction 14.6 (Consolidation and microneurons)**.** Over the weight filtration, persistent homology reads which coalitions and concept-cycles *survive*; survivors are promoted into permanent structure. A *microneuron* is a promoted sub-tree: when a composite DAG repeatedly resolves conflict *and* passes verification, it is named and thereafter emitted by the planner as a single atomic generator of $\mathfrak{D}$. This is $\delta\mathfrak{D}$ made concrete. The promotion gate is VerifyOp (Chapter 21): a composite becomes a generator only if it survives the external oracle.

Claims and non-claims. The requirement that promotion pass VerifyOp is doing real work. Without it, “repeatedly resolved conflict” promotes whatever procedure most reliably makes the organs agree — which, in a system whose only cheap signal is agreement, selects for procedures that manufacture consensus rather than truth. Coherence is a control signal, not a fitness function, and the gate is what keeps that distinction enforced.

Chapter 15

The Fine Dynamics

Intuition. Inside an organ, reasoning is a *flow*. New material perturbs the internal state and a gradient descent pulls it back toward internal coherence, subject to hard constraints (axioms that must hold) and soft constraints (inferred facts, bendable).

Construction 15.1 (Fine state and deductive flow)**.** The fine state of an organ is a configuration $x \in (\mathbb{R}^d)^m$ over its m active concepts, with a scalar *conflict functional* $\Phi(x) \geq 0$ measuring internal inconsistency — the fine-level analogue of ω. Reasoning integrates

$$\dot{x} = -\nabla\Phi(x) \tag{15.1}$$

by a fixed-step fourth-order Runge–Kutta scheme. The reported δ*-strain* of a step is the achieved change $\Delta\Phi$. Two constraint types shape Φ: *rigid* constraints (user-stated axioms) act as hard orthogonal projections onto an affine subspace via a projector P; *fluid* constraints (inferred facts) act as soft quadratic penalties.

Proposition 15.2 (The flow is a descent)**. [PROVED]** *Along any solution of* (15.1), $\frac{d}{dt}\Phi(x(t)) = -\|\nabla\Phi(x(t))\|^2 \leq 0$, *with equality exactly at critical points. If* Φ *is bounded below — which it is, by* $\Phi \geq 0$ *— then* $\Phi(x(t))$ *converges and* $\nabla\Phi(x(t)) \to 0$ *along a subsequence.*

Proof. Chain rule: $\frac{d}{dt}\Phi = \langle\nabla\Phi, \dot{x}\rangle = -\langle\nabla\Phi, \nabla\Phi\rangle$. Monotone and bounded below implies convergent; integrating $\int_0^\infty \|\nabla\Phi\|^2\, dt = \Phi(x(0)) - \lim\Phi < \infty$ forces $\liminf\|\nabla\Phi\| = 0$. $\square$

Remark 15.3 (Observed sign behaviour). On a live run, injecting a new concept raised Φ by a structural jump (δ-strain $= +0.202$) and the subsequent flow pulled it back down (δ-strain $= -0.081$): a structural expansion followed by a deductive relaxation, which is the intended dynamics. Note that the expansion is *not* a descent step — it is an exogenous change of the functional itself, and only the relaxation is governed by Proposition 15.2.

Claims and non-claims. The closed form of Φ is an implementation-level choice — a penalised quadratic on the fine complex — and is deliberately not over-specified here. What is mathematically committed is (15.1) (a gradient flow on a nonnegative conflict functional) and the rigid/fluid split as hard projection versus soft penalty. Convergence to a *global* minimum is not claimed and would be false for a non-convex Φ.

Chapter 16

Scheduling: Antichains, and Why "Operad" Is Not Earned

Intuition. A reasoning act is a small program: a directed acyclic graph of operators, some depending on others. Independent operators can run at once, and the scheduler slices the DAG into layers of mutually independent work.

Definition 16.1 (Antichain foliation)**.** A reasoning act is a DAG $T = (N, \prec)$ of operator nodes. The scheduler foliates T into the antichain layers of $\prec$: layer 0 is the set of minimal nodes, and layer $\ell + 1$ is the set of nodes all of whose predecessors lie in layers $\leq \ell$. Each layer is executed in parallel. Correctness follows because nodes in one antichain share no $\prec$-dependency. Node *support* sets index which fine-complex vertices an operator reads and writes, so slices with disjoint support are safe to co-execute.

Remark 16.2 (The intended structure is a trace monoid). What Definition 16.1 describes is antichain layering of a DAG. An operad requires multi-input composition together with equivariance, associativity and unit axioms, none of which are stated or used anywhere. The structure the design is reaching for is a *trace monoid* $\mathbb{M}(\Sigma, I)$ in the sense of Mazurkiewicz: the free monoid on the operator alphabet modulo $ab = ba$ for pairs in an *independence relation* I, whose canonical Foata normal form is exactly the antichain layering the scheduler computes. That identification would be a *sharpening* rather than a downgrade, since it supplies normal forms, a Hilbert series, and a canonical form for deduplication. The terminology should be corrected regardless of whether the identification is completed.

Claims and non-claims. **[REFUTED]** The trace-monoid identification **does not currently hold**, and the reason is three separate defects, measured 2026-07-29 against the deployed operator set.

(i) The relation is not symmetric. In the scheduler, admitting *any* empty-support node — including a read-only one — marks the whole slice as a global mutation and blocks every later node. So 8 of the 15 operator pairs can share a slice in one order and not in the other. An independence relation is symmetric by definition; a non-symmetric relation defines no trace monoid, hence no Foata normal form and no Hilbert series. This is very likely an unintended scheduling behaviour and is worth fixing on its own merits.

(ii) The support sets are placeholders. Every support accessor returns a hardcoded

literal — $\varnothing$, $\{0\}$ or $\{1\}$ — with three of the six operators returning the same $\{0\}$, and a comment standing in for an analysis of what the operator actually touches. Any invariant computed from these describes six constants, not the system.

(iii) The convention for $\varnothing$ is inverted. In the engine an empty support means GLOBAL: touches everything, must run isolated. Read as a set, $\varnothing$ is disjoint from everything and would mean independent of everything. The two readings give capacities differing by a factor of 1.8 ($\kappa = 3.00$ versus $\kappa = 5.45$, against $\kappa = 6$ for the free monoid), so the quantity is not well defined until the convention is fixed.

Under the reading the engine actually implements, the algebra is *nearly free* — about 9% below the free monoid — because three of six operators share one support. The order of repair is: fix the symmetry defect; derive real read/write sets, including the shared state the current model ignores entirely (the store, the obstruction counter Ω, the mutation log, and the coarse stalks that π_v overwrites after every execution); only then recompute I. Operators that all mutate Ω do not commute whatever their vertex supports say. Until that is done, no capacity, Koszulity, or resolution claim about $\mathfrak{D}$ should appear anywhere.

Chapter 17

The Lift–Project Adjunction

Work plan. ¡KEEP¿

Goal. Keep the adjunction — it is standard, genuinely proved, and now does double duty as the memory architecture and as the candidate reproduction morphism.

Raw material. This chapter's existing text; `MATH_TOOLBOX.md` II.2 (the $\mathbb{K}/W$ two-complex model, the consolidation rule, the measured $Q(t)$ trajectory), II.1 (what was tried and abandoned for memory identity first).

Steps.

1. **Keep the theorem.** For a full subposet inclusion ι: ι^* restricts (instantiate, store → cache), $\iota_!$ is extension by zero (write back only what was touched), ι_* fills untouched cells by limits, and the (co)units are isomorphisms so $\iota^*\iota_! \cong \mathrm{Id}$. **[PROVED]**, standard.
2. **State the write-back choice and its reason.** $\iota_!$, never ι_*: the cache must assert *nothing* about untouched cells, which is the sheaf-theoretic form of "no measurement is not a measurement of zero."
3. **Present consolidation as the anti-interference mechanism.** $\gamma_0 \ll 1$ is what lets new learning coexist with old memory without overwriting it, and the session's state s_W is *never* written back — content and wiring persist, the episode does not.
4. **Report that $\gamma_0 \ll 1$ was forced twice, independently.** Once by the anti-interference argument, and once by measurement: no restriction-map parameterisation is affordable to learn from a single tick (the cheapest is over-parameterised threefold, a full orthogonal map by over five hundred). Two independent routes to one conclusion is worth stating as such.
5. **Keep $Q(t)$ as an instrument.** Mean $\|R - \mathrm{Id}\|_F^2$; constant sheaf $\iff$ $Q = 0$; measured moving monotonically off zero across four ticks, and *exactly* zero when nothing is verified. "$Q(t)$ leaving zero is not a proxy for consolidation; it is the definition." **[MEASURED]**.
6. **Keep the two bugs the tests caught**, since both would have rotted silently: a constructor that pre-filled every edge with identity so learned maps were dropped on rebuild, and a single Householder reflection landing in the wrong component of $O(d)$ so consolidation correctly refused to interpolate and $Q(t)$ would have stayed zero with every test passing.
7. **Add the forward reference to reproduction.** This same adjoint pair is Chapter 23's candidate morphism. Note it here.
8. **Keep the abandoned alternatives with their reasons.** Formal Concept Analysis **[REFUTED]** as substrate (data-determined; its canonicity comes precisely from discarding the path-dependence that makes a memory a memory — memory must *grow*, not be recomputed), and the K_0/Jordan–Hölder schema **[REFUTED]** (both candidate categories fail for opposite reasons). Both were dissolved by the growth-history identity this chapter provides.

Done when. The adjunction is proved, the write-back choice is justified by the $\perp$-invariant, and its second role in reproduction is signposted.

Intuition. Two worlds: free-form natural language, and structured reasoning DAGs. One map *lifts* language into a validated DAG; a partner map *projects* a DAG back into language. Meaning that carries no reasoning content collapses under the lift — it lands in the kernel.

Construction 17.1 (The boundary morphisms)**.** Let $\mathcal{L}$ send a natural-language request to a validated reasoning DAG, and $\mathcal{M}$ send a DAG to a natural-language rendering. The lift is guarded by a validator rejecting malformed DAGs at the boundary: duplicate identifiers, dangling children, cycles by depth-first colouring, absence of a terminal output node, and disconnection. The downstream engine therefore only ever receives well-formed programs.

Remark 17.2 (The kernel behaves as intended). Semantically void input maps into $\ker \mathcal{L}$: a joke embedded in a mathematical prompt produced zero DAG nodes, exactly as a kernel element should. This is an observation on one input, not a theorem.

Claims and non-claims. $\mathcal{L} \dashv \mathcal{M}$ is the intended categorical *framing* — lift left adjoint to project — and it organises the design usefully. The unit and counit are not constructed here and the triangle identities are not verified, so "adjunction" should be read as a structuring principle plus two concrete morphisms with an observed kernel behaviour, and not as a proved theorem. Stating this is part of the honesty invariant rather than an apology for it.

Part IX

The Later Constructions

Chapter 18

The MDL Growth Law

Work plan. ¡RECAST¿

Goal. Keep Construction 7 and give it its V2 placement: this is the *maintenance cost* of the sleep phase, and it is the only place cost lives in the entire architecture.

Raw material. This chapter's existing text; `MATH_TOOLBOX.md` III.4 (Construction 7, the per-direction code, the measurement); `DOCS/DERIVATION_MDL_GROWTH_LAW.md`; `TATIANA_V1_IDENTITY_MAP.md` §6 (placement, and the one free parameter).

Steps.

1. **Keep the superseded law as a worked failure.** The original had a numerator overcounting by a factor of k (fixed by holonomy propagation — only the first leg needs describing, the rest are determined) and, fatally, a denominator that did not depend on concept size at all, making a zero-dimensional concept always optimal: "attach always, with a concept that carries nothing." Keep this; it shows exactly how a plausible law can invert its own purpose.
2. **State Construction 7's repair.** A per-direction code: fix a direction in the limit, take its k readings around a traversal, and ask whether they agree. The saving per traversal follows from the intraclass correlation, and the law becomes a threshold on recurrence count that is never met when the saving is non-positive.
3. **Report the measurements.** Within 0.042 bits/traversal of exact quantised-Gaussian entropy; over-prices its own reference code so the threshold is *conservative* — it under-attaches, never over-attaches; and a shuffled-readings control drops the saving from $+5.289$ to -0.437 bits, so the saving measures the recurrence and not the coder. **[MEASURED]**.
4. **Keep the restored consequences.** Size-dependence returns per-direction; the subspace-selection rule (rank by reliability, keep a prefix) is now a *theorem* rather than a proposal; and a reliability floor exists independent of frequency.
5. **Keep the open self-reference problem.** Taking bits-per-real as $\frac{1}{2}\log n$ is self-referential but resolvable by iteration — *except* that it degenerates exactly where the growth law lives: at $n = 1$ the model is free and over-generation returns. Do not adopt it as derived-therefore-safe. State the alternatives.
6. **Keep the stated narrowing.** The law prices what was there, not which succession fired — a strict under-estimate, making it conservative by construction rather than by luck.
7. **Add the V2 placement paragraph.** This cost is consulted during *sleep* only. It never bids against growth inside a single tick. Name the exchange rate between deformation cost and new-cell cost as the one genuine free parameter of the design, and record that acquiring a second one is evidence the two-phase separation has failed.

Done when. The law is stated with its measurements and its open self-reference problem, and its placement in the sleep phase is explicit.

The question this chapter answers: *when is a new concept worth inventing?* Earlier designs answered it with a threshold someone chose. This one derives a threshold instead.

18.1 Two-part description length

Intuition. A memory is a compression of experience. Minimum description length makes that literal: the total cost of an explanation is what it takes to write down the model plus what it takes to write down the data *given* the model. Adding a concept always makes the first term worse. It is worth doing exactly when it makes the second term better by more.

Definition 18.1 (Two-part MDL)**.**

$$L_{\text{total}} = \underbrace{L(\text{model})}_{\text{the memory}} + \underbrace{L(\text{data} \mid \text{model})}_{\text{the experience}}.$$

The *model* is the complex $\mathcal{C}$ together with the sheaf $\mathcal{F}$ — what the system *is*. The *data* is the observed record: the sequence of assemblies and the succession relation read off it — what the system has *seen*.

Neither term alone is the criterion. Minimising the model term alone forbids all growth; minimising the data term alone is achieved by memorising everything. The criterion is the sum.

18.2 The model term

Proposition 18.2 (Cost of coning a cycle)**.** **[PROVED]** *Let γ be a 1-cycle of length k, and cone it with a new vertex w carrying a stalk of dimension d_w. Writing b for bits per real number, the added model cost is*

$$\Delta L(\mathit{model}) = b\, d_w\Big(1 + \sum_{e\in\gamma} d_e\Big). \tag{18.1}$$

Proof. Three kinds of new object are described. The vertex with its stalk costs $d_w \cdot b$ bits. Each of the k restriction maps out of $\mathcal{F}(w)$ is a $d_e \times d_w$ matrix, costing $d_w d_e b$ bits, summed over $e \in \gamma$. The k new triangles cost *nothing*: they are determined by the cone construction, not described separately. Summing gives (18.1). □

Remark 18.3 (Two consequences, both load-bearing)*.* First, the cost is *linear in d_w*. The dimension of the new stalk is the dominant free quantity, so whatever fixes d_w fixes most of the cost — which is what Section 19.6 is for. Second, *triangles are free.* They are implied by the cone, so the k new 2-cells cost nothing to describe. This is a real asymmetry, not a bookkeeping convenience, and it is why coning is cheap relative to what it buys.

18.3 The data term and the law

Proposition 18.4 (Data-term saving)**.** **[PROVED]** *Let the cycle γ occur n times in the record. Without the concept, each traversal is coded as k separate transitions, each costing its full surprisal; with the concept it is a single named event. Then*

$$\Delta L(\mathit{data} \mid \mathit{model}) = -n\big(c_{old} - c_{new}\big), \tag{18.2}$$

where c_{old} is the cost of coding one traversal the long way and c_{new} the cost of naming it.

Theorem 18.5 (The growth law). **[PROVED]** *Attach the concept if and only if* $\Delta L_{total} < 0$, *that is*

$$\boxed{n \; > \; \frac{b\, d_w \big(1 + \sum_{e \in \gamma} d_e\big)}{c_{old} - c_{new}}} \tag{18.3}$$

Proof. Add (18.1) and (18.2) and require the sum negative; rearrange, using $c_{\text{old}} > c_{\text{new}}$. □

> **Intuition.** Read what this is: a *recurrence threshold that is computed, not chosen.* It is the ratio of what a concept costs to store against what it saves per use — the number of times a pattern must repeat before it is worth naming.

Three consequences, two of which were not asked for:

1. n is a number the engine can print, rather than a parameter a human sets.
2. **It orders the holes.** When several harmonic classes compete, fill them in decreasing $|\Delta L_{\text{total}}|$. This removes a decision that had been silently delegated to a human.
3. **It predicts non-attachment.** A one-off inconsistency is *noise to be tolerated*, not a concept to be invented. The earlier threshold-based law had no way to express this, which is exactly how it would have over-generated.

Chapter 19

The Coned Concept

Work plan. **¡KEEP¿**

Goal. Keep Construction 6 essentially as written — it is the most-verified construction in the record — and add the one upgrade V2 needs: its output dimension is what moves the invariant, so this chapter and Chapter 12 must agree exactly.

Raw material. This chapter's existing text; `MATH_TOOLBOX.md` III.3 (derivation, two self-corrections, the measurements, provenance); `TATIANA-V2/MATH/phase0_invariant_dimension.md` (the dimension confirmed two independent ways).

Steps.

1. **Keep the derivation and its punchline.** Consistency on the new edges forces the value at w to *generate* the section over the whole cycle, and substituting into the old edges gives verbatim the categorical cone condition. The cone condition was not imposed — it fell out of demanding the topological cone do its job. **[PROVED]**.
2. **Keep the identification.** $\mathcal{F}(w) := \lim D_\gamma \cong H^0(\mathcal{C};\mathcal{F})(\gamma;\mathcal{F}|_\gamma)$: the new concept is *not* a summary of the cycle's members but their *compatible part.* This sentence is the chapter.
3. **Keep both self-corrections visible.** First, an earlier draft argued the limit is "the smallest object admitting the required maps" — backwards; every cone factors uniquely through the limit, so it is the *largest* non-redundant choice, a ceiling and not a floor. Second, the "refusal branch" (nothing consistent, so the construction declines) is logically real but *structurally unreachable*: a cycle carries harmonic mass if and only if it has sections.
4. **Keep the upgrade from preference to necessity.** The coned complex is a cochain complex *if and only if* $\mathcal{F}(w)$ is a cone over the diagram. Construction 6 is therefore not the best choice among several — it is the only choice for which the coned object is a sheaf at all. The universal property was never a preference; it was the existence condition. **[PROVED]**, and measured: an arbitrary $\mathcal{F}(w)$ gives 1.756, the limit gives 4.4×10^{-16}.
5. **Keep the parity observation.** Reachability of the degenerate case is governed by the parity of d, since every element of $SO(\text{odd})$ fixes an axis. This is a fact about this architecture specifically, because its restriction maps are orthogonal by construction.
6. **Keep the provenance paragraph, including the direction.** The universal-property route to a new concept is Goguen's, not this project's — and the variance is *opposite*: his blends are colimits, a sheaf's restriction maps force a limit. "Copying him would have gotten it backwards." What is genuinely original is that the sheaf's own restriction maps supply the diagram, so nothing new is introduced beyond what the memory already held.
7. **Add the invariant link.** $\dim \mathcal{F}(w) = \dim H^0(\mathcal{C};\mathcal{F})(\gamma;\mathcal{F}|_\gamma)$ is exactly the amount by which $\mathrm{K}(\mathbb{K})$ changes, known *before* the growth decision, from the same computation that produced the address. Cross-reference Chapter 12 and report that both routes to this number have been

computed and agree. **[MEASURED]**.

8. **Record what is still not derivable.** The label for the new concept is not derivable from a diagram. Name this as the one place genuine novelty enters rather than being computed, and connect it to Chapter 24.

Done when. The construction is proved to be forced rather than chosen, and its output dimension is stated identically here and in Chapter 12.

This is the central construction of the later work: the stalk of a newly-grown concept, *derived* rather than chosen.

19.1 The topological cone

Construction 19.1 (Coning a cycle)**.** Let $\gamma = (v_1, e_1, v_2, \dots, v_k, e_k, v_1)$ be a 1-cycle carrying harmonic mass, with $e_i = \{v_i, v_{i+1}\}$ (indices mod k). Attach a vertex w, edges $f_i = \{w, v_i\}$, and triangles $t_i = \{w, v_i, v_{i+1}\}$.

Proposition 19.2 (The cone kills the cycle)**. [PROVED]** *With $\partial t_i = e_i - f_{i+1} + f_i$,*

$$\partial\Big(\sum_i t_i\Big) = \sum_i e_i + \sum_i (f_i - f_{i+1}) = \gamma,$$

since the f terms telescope to zero. Hence γ becomes a boundary.

19.2 The stalks, chosen minimally

Construction 19.3 (New edge stalks)**.** Set

$$\mathcal{F}(f_i) := \mathcal{F}(v_i), \qquad \mathcal{F}_{v_i \trianglelefteq f_i} := \mathrm{Id}.$$

Claims and non-claims. This is an *engineering choice*, and a deliberately empty one. Any distortion introduced here would be a free parameter, and whatever the construction then achieved could be attributed to a cleverly chosen edge stalk rather than to the concept. All the content is forced into $\mathcal{F}(w)$ and the maps out of it, which is where it can be reasoned about.

The only remaining unknowns are $\mathcal{F}(w)$ and the legs $p_i := \mathcal{F}_{w \trianglelefteq f_i} : \mathcal{F}(w) \to \mathcal{F}(v_i)$.

19.3 The cone condition is forced, not imposed

Proposition 19.4 (Derivation of the cone condition)**. [PROVED]** *Write $r_i^- := \mathcal{F}_{v_i \trianglelefteq e_i}$ and $r_i^+ := \mathcal{F}_{v_{i+1} \trianglelefteq e_i}$. For a 0-cochain x:*

(i) δx vanishes on the new edges if and only if $x_{v_i} = p_i(x_w)$ for every i;

(ii) given (i), δx vanishes on the old edges for every *$x_w \in \mathcal{F}(w)$ if and only if*

$$\boxed{r_i^+ \circ p_{i+1} = r_i^- \circ p_i \quad \textit{for all } i} \tag{19.1}$$

Proof. (i) On the new edge f_i, using Construction 19.3,

$$(\delta x)_{f_i} = \mathcal{F}_{w \trianglelefteq f_i} x_w - \mathcal{F}_{v_i \trianglelefteq f_i} x_{v_i} = p_i(x_w) - x_{v_i},$$

which vanishes exactly when $x_{v_i} = p_i(x_w)$.

(ii) Substituting $x_{v_i} = p_i(x_w)$ into the old edge e_i,

$$(\delta x)_{e_i} = r_i^+ p_{i+1}(x_w) - r_i^- p_i(x_w) = \big(r_i^+ p_{i+1} - r_i^- p_i\big)(x_w).$$

A linear map vanishes on all of $\mathcal{F}(w)$ if and only if it is the zero map, which is (19.1). $\square$

> **Intuition.** Part (i) says something worth pausing on: **the value at w generates the section over the whole cycle**. That is what it means for a concept to explain a loop — the loop stops being k independent facts and becomes one.

Condition (19.1) is verbatim the definition of a *cone* over the diagram

$$D_\gamma : \quad \mathcal{F}(v_1) \xrightarrow{r_1^-} \mathcal{F}(e_1) \xleftarrow{r_1^+} \mathcal{F}(v_2) \xrightarrow{r_2^-} \mathcal{F}(e_2) \longleftarrow \cdots$$

with apex $\mathcal{F}(w)$ and legs p_i. The categorical cone condition was not imposed on the construction; it fell out of demanding that the topological cone do its job.

19.4 The construction

Construction 19.5 (The coned concept)**.**

$$\boxed{\mathcal{F}(w) := \varprojlim D_\gamma, \qquad p_i := \text{the limit's projections.}}$$

Proposition 19.6 (Concrete description)**.** **[PROVED]** *For finite-dimensional stalks,*

$$\varprojlim D_\gamma = \Big\{(x_1, \ldots, x_k) \in \bigoplus_i \mathcal{F}(v_i) \;\Big|\; r_i^- x_i = r_i^+ x_{i+1} \ \forall i\Big\} \;\cong\; H^0\big(\gamma; \mathcal{F}|_\gamma\big).$$

Proof. The limit of a finite diagram of vector spaces is the subspace of the product on which all the diagram's maps agree, which is the displayed equaliser. Comparing with Definition 7.1 restricted to the subcomplex γ, the agreement conditions are exactly $\delta x = 0$, so the space is $\ker \delta|_\gamma = H^0(\gamma; \mathcal{F}|_\gamma)$. $\square$

> **Intuition.** In words: the new concept *is* everything that was consistent around the loop but could not be glued. It does not summarise the cycle's members; it *is* the compatible part of them.

Three properties, each of which had been written down separately as a requirement:

Property	**Why it matters**
canonical — unique up to unique isomorphism	no hand-made choice enters the construction
path-dependent — depends on D_γ, i.e. on which cycle history produced	precisely the property whose absence caused an earlier formal-concept-analysis approach to be rejected
degenerate case	if $H^0(\gamma; \mathcal{F}\vert_\gamma) = 0$ the limit is the zero space — but see Proposition 19.7

19.5 The refusal branch is unreachable

Proposition 19.7 (Harmonic mass implies sections)**.** **[PROVED]** *On a k-cycle with vertex and edge stalks all of dimension n, and with no 2-cells,*

$$\dim H^1(\gamma; \mathcal{F}|_\gamma) = \dim H^0(\gamma; \mathcal{F}|_\gamma).$$

Consequently the cycle carries harmonic mass if and only if it has nonzero sections.

Proof. For a 1-dimensional complex the Euler characteristic satisfies $\chi = \dim C^0(\mathcal{C}; \mathcal{F}) - \dim C^1(\mathcal{C}; \mathcal{F}) = \dim H^0 - \dim H^1$. On a k-cycle with all stalks of dimension n there are k vertices and k edges, so $\dim C^0(\mathcal{C}; \mathcal{F}) = \dim C^1(\mathcal{C}; \mathcal{F}) = kn$ and $\chi = 0$, giving the equality. With no 2-cells the harmonic space *is* H^1. □

> **Claims and non-claims.** An earlier draft claimed the degenerate case is meaningful — "nothing was consistent around that loop, so the construction refuses rather than inventing". True, and it never happens. By Proposition 19.7, $H^0 = 0$ implies no harmonic mass, which implies no growth address, which implies the growth law never fires on that cycle at all. Confirmed numerically: the degenerate case reports that the harmonic space is already empty before coning, so there is nothing to kill.
>
> In general the gap is $\dim H^1 - \dim H^0 = k(n_e - n_v)$, so the branch *could* become reachable if edge stalks are ever given a different dimension from vertex stalks. They are not, currently.

19.6 The limit is a ceiling, not a floor

> **Claims and non-claims. [REFUTED]** An earlier draft justified Construction 19.5 by asserting that the limit is "the smallest object admitting the required maps". That is backwards, and the error is worth keeping visible.

Proposition 19.8 (Universality bounds from above)**.** **[PROVED]** *Every cone over D_γ factors uniquely through the limit. If a candidate apex $\mathcal{F}(w)'$ has jointly injective legs — i.e. carries no data that no restriction map ever sees — then that factoring map is injective, so*

$$\dim \mathcal{F}(w)' \;\leq\; \dim \varprojlim D_\gamma.$$

Proof. The universal property of the limit gives a unique $\phi : \mathcal{F}(w)' \to \varprojlim D_\gamma$ commuting with the legs. If $\phi(z) = 0$ then every leg $p_i'(z) = 0$; joint injectivity of the legs then forces $z = 0$, so ϕ is injective and the dimension inequality follows. □

So the limit is the *largest non-redundant* choice, not the smallest — a ceiling on what a new concept can usefully carry. The two determinations are therefore genuinely separate, and both are derived:

1. **Consistency fixes the shape.** $\mathcal{F}(w)$ must be a cone over D_γ; the limit is the universal one.
2. **MDL fixes the size.** By (18.1) the cost is linear in d_w, so the optimum is a *subspace* of the limit: keep the directions that pay for themselves, discard the rest.

Chapter 20

Retrieval: Rank, Do Not Threshold

Work plan. ¡RECAST¿

Goal. Keep the empirical result, which is large and well-measured, but change what the chapter is *for*: V2's own diagnosis is that retrieval cannot live in a metric at all, so this chapter becomes the evidence for that conclusion rather than a description of a component.

Raw material. This chapter's existing text; `MATH_TOOLBOX.md` V.1–V.2 (the bug, the fix, the self-audit), VII.1 (the six measurements that together say retrieval needs a generative process).

Steps.

1. **Keep the bug and its magnitude.** A relevance threshold compared against squared Bures–Wasserstein distance, with an unbounded scan and no limit, admitted 0.31% of ground-truth dependencies; at the shipped default, literally zero of three thousand ticks retrieved a pair. **[MEASURED]**.
2. **Keep all three root causes**, since each is instructive: it was a near-duplicate filter rather than a relevance filter; embeddings sit on a large common-mode pedestal that any absolute threshold mostly measures; and the epistemic term is dimension-extensive while the semantic term is dimension-intensive, so summing them lets the former dominate by two orders of magnitude regardless of meaning.
3. **Keep the fix and the honest framing of its size.** Rank, do not threshold; rank on cosine, not squared distance (stored means are not unit-norm, so squared-distance ranking silently penalises general concepts). Worth 0.31% $\rightarrow$ 46.6%. State the lesson explicitly: the whole design argument was about a four-point effect while the actual fix was worth 140$\times$.
4. **Keep the self-audit.** Two claims about the fix were themselves overstated and walked back — an optimum read off a flat sweep, and a parameter called "derived" that was an unmeasured budget coincidence. "Two independent constraints landing on the same number is exactly the tidiness motivated reasoning produces." Keep this sentence.
5. **Keep the refuted alternative.** Diffusion/heat-kernel retrieval, twelve configurations, none beating raw cosine, degrading monotonically with diffusion time — the pedestal makes the walk near-uniform so it amplifies hubness rather than removing it. **[REFUTED]**.
6. **Add the V2 recast.** State the conclusion this chapter is now evidence for: retrieval cannot live in the metric alone, cannot live in a single relation (citation and embedding similarity agree under half the time), and cannot live in the corpus. What is missing is a process that decides what to retrieve *about* — which in V2 is the wake phase's drive, not a lookup. Reframe accordingly and forward-reference Chapter 13.
7. **Carry the standing warning.** A simulator produces the topology its rules imply. Moving from "assume a structure and test for it" to "generate the structure" is a stronger design and *also* a way to hide assumptions inside an update rule. No claim about emergent structure without a

validation battery built *before* the generator.

Done when. The chapter reads as the argument that retrieval needs a process rather than a metric, with the measurements as evidence rather than as the point.

Intuition. Retrieval decides which stored concepts an organ sees. A threshold on distance answers "which concepts are close enough?", which sounds right and is not: it makes the number of results depend on how the embedding space happens to be scaled, and it silently returns nothing at all when the query lands in a sparse region.

Construction 20.1 (Bounded rank query)**.** Replace the ε-ball query with a bounded rank query returning the top k stored concepts, ranked by *cosine similarity* to the query mean, using a bounded max-heap of size k. Zero-norm vectors are dropped rather than scored, since a zero vector has no direction.

Proposition 20.2 (Why cosine and not squared distance)**.** **[PROVED]** *Expanding,*

$$\|\mu_q - \mu_i\|^2 = \|\mu_q\|^2 - 2\,\langle \mu_q, \mu_i \rangle + \|\mu_i\|^2 .$$

For a fixed query, $\|\mu_q\|^2$ is constant and drops out of any ranking. The term $\|\mu_i\|^2$ does not*. Hence squared distance reproduces the cosine ranking if and only if every stored mean is unit-norm.*

Claims and non-claims. Stored means are *not* unit-norm. By Construction 10.2, π_v is a weighted centroid of unit vectors, so its norm lies below 1 and falls further the more spread the concepts it summarises. Ranking by squared distance would therefore apply a per-concept penalty *proportional to how broad a concept is*, pushing exactly the general concepts down the list for a reason unrelated to the query.

There is also an evidential argument, which is decisive on its own: the measured recall figure against which the retrieval fix is accepted was obtained on a *cosine* ranking. Ranking by squared distance is a different ordering carrying an unmeasured bias, which would have made the acceptance criterion meaningless.

Remark 20.3 (Cost)*.* Scoring needs only μ; the low-rank factor U, the floor D and the stored reasoning text are not needed unless the row enters the heap. Reading μ, scoring, and deserialising the rest only on entry drops the per-row cost from $O(dk^2 + k^3)$ to $O(d)$ for the large majority of rows.

Claims and non-claims. k is an *engineering choice* ($k = 24$), not a derived quantity. What changed is that it is now constrained on both sides — too small starves the organ, too large defeats the point of retrieval — rather than being an unexamined constant. Saying so is the difference between a parameter and a magic number.

Part X

Verification and Control

Chapter 21

Verification: the External Oracle

Work plan. ¡RECAST¿

Goal. Keep the trichotomy, which is a real epistemic fix. Retitle and re-found the chapter: in V2 there is no "external oracle" to appeal to, and the verification lift's architecture is an open problem, not a given.

Raw material. This chapter's existing text; `MATH_TOOLBOX.md` I.8 (the trichotomy and the promotion gate), I.5 (the derivation of $\gamma(\emptyset)$ for unchecked ticks); Chapter 24 of this book, which now owns the open question.

Steps.

1. **Retitle.** "The External Oracle" presumes V0's architecture, where verification was one organ among several and could be an outside service. Rename to something that does not assume where the checker lives.
2. **Keep the trichotomy and its argument.** Two outcomes conflate REFUTED with UNVERIFIABLE — "the error of concluding falsehood from inability to check" — and would spike conflict over the checker's own blind spots. Three outcomes fix it: the obstruction clears, increments, or is left untouched. A checker crash returns UNVERIFIABLE, never a refutation. **[PROVED]** as an epistemic argument.
3. **Keep the consolidation gate.** $\gamma(\nu)$ reads the verdict and is what stops unverified structure from entering the invariant. This is the link that makes verification load-bearing rather than advisory — state it as such, cross-referencing Chapter 12.
4. **Keep the derivation for unchecked ticks.** "No test ran" is absence of evidence, not a failed test, and the rate for such ticks was *derived* by shrinkage rather than hand-set, degrading exactly to the old constant on day one. Keep the honest caveat: it is conditional on selection bias, since ticks that ran a check may not resemble ticks that did not. Testable, strictly better than a constant, never to be quoted as unbiased.
5. **State the implementation status.** The trichotomy is computed; it has never been connected to a real prover or sandbox, so the verdict is not earned from evidence in any live loop. **[PROPOSED]**.
6. **Hand the open question to Chapter 24** rather than answering it here, and say so explicitly so the two chapters do not drift apart.

Done when. The trichotomy is stated with its epistemic argument, its link to the invariant is explicit, and no text presumes an external service.

Intuition. Coherence governs whether to keep exploring or stop and reconcile. Only an

> external checker touches *truth*. And "I could not check" must never be mistaken for "it is false".

Definition 21.1 (Three-valued verification)**.** VERIFYOP calls a sandboxed symbolic oracle returning one of three verdicts, acting on an obstruction counter Ω:

exit	verdict	effect on state
0	VERIFIED	obstruction cleared; state may crystallise
1	REFUTED	Ω += 1000; forced into RESOLVE
2	UNVERIFIABLE	Ω *untouched*

Remark 21.2 (Absence of proof is not proof of absence)*.* An earlier version treated every nonzero exit as failure, conflating REFUTED with UNVERIFIABLE. That would spike conflict over the oracle's own blind spots and lock the system into RESOLVE over perfectly good mathematics. A checker crash also returns UNVERIFIABLE. This three-valued logic is the promotion gate of Construction 14.6: a composite becomes a generator of $\mathfrak{D}$ only if it survives the oracle, and surviving means VERIFIED, not merely "not refuted".

> **Claims and non-claims.** The asymmetry in the table is deliberate and is the load-bearing design decision. A refutation moves Ω by a large fixed amount; an inability to check moves it not at all. The system is therefore allowed to be ignorant without being punished for it, and is punished immediately for being wrong. The alternative — treating unverifiable as mildly bad — produces a gradient toward only ever asserting things the oracle happens to understand, which is a narrower system, not a better one.

Chapter 22

The Controller

Definition 22.1 (Coherence-gated mode)**.** With threshold ε, the controller selects

$$\mathrm{mode}(x) = \begin{cases} \textsc{Resolve}, & \rho > \varepsilon, \\ \textsc{Explore}, & \rho \le \varepsilon, \\ \textsc{Unknown}, & \rho = \bot. \end{cases} \tag{22.1}$$

Remark 22.2 (Three modes, and an empirical ε)*.* The UNKNOWN branch is essential: when ρ is undefined the controller *refuses* to default to EXPLORE, because "no measurement" is not "all is well" (Remark 9.5). The threshold is calibrated rather than guessed: on real 384-dimensional embeddings, agreement gave $\rho \approx 0.00$, related-but-distinct ≈ 0.06, and off-topic ≈ 0.14, so $\varepsilon \approx 0.10$ is a principled starting gate.

Two cautions were recorded from live runs. First, ρ is *not* monotonic in the number of active organs: adding a well-aligned organ raises the normaliser faster than the discord, so ρ dilutes. Second, comparisons across differently sized complexes must be made with care, because B and $\|X\|_F^2$ both change.

Remark 22.3 (Superseded: gate on $\tilde{\rho}$, not ρ)*.* Both cautions are consequences of a single fact, isolated in Proposition 9.6: $\rho = \alpha \cdot \tilde{\rho}$, and *only* α *has the pathologies.* Organ-count dilution moves α and leaves $\tilde{\rho}$ invariant. Cross-complex comparison is unsafe for ρ, but $\tilde{\rho}$ is comparable across complexes once expressed as its position inside the window (9.4). The controller should therefore gate on $\tilde{\rho}$ — or on its window position — and report α alongside as the answer to a genuinely different question.

The typed reading is what the single number could not give:

- large α, small $\tilde{\rho}$: a clean two-camp split; the coalition wants restructuring (`split`, or a new `bind`);
- small α, large $\tilde{\rho}$: one organ is an outlier; a single edge wants repair.

Worked example. Apply the controller to the running example. With $\varepsilon = 0.10$ and $\rho = 0.2$, mode (22.1) returns RESOLVE, and a diagnostic reading only ρ would report "mild disagreement, reconcile". The typed reading gives more: $\alpha = 0.5$ is large and $\tilde{\rho}$ sits at window position $\frac{1}{3}$, so by Remark 22.3 this is a two-camp split, and the correct remediation is `split` or a new `bind` — *not* repair of the worst edge. Note also that the worst-edge diagnostic was ambiguous here (the argmax tied), which is the same fact appearing in a second instrument. □

Claims and non-claims. The controller is the one place where every earlier honesty decision becomes operational. $\perp$ propagates into a mode rather than being coerced to a number; the split of Proposition 9.6 changes which quantity is gated on; the window of Proposition 9.8 makes the threshold topology-relative. It is also where a mistake is most expensive, because a controller that mis-reads its instrument does not merely report wrongly — it acts.

Part XI

Reproduction

Chapter 23

Reproduction

¡NEW¿ *The weakest chapter in the book, placed last and labelled as such. It proposes a canonical morphism and claims nothing beyond that.*

Work plan. ¡NEW¿

Goal. Give “the kernel reproduces itself” a precise candidate definition, and be explicit that a canonical morphism is not yet a viable offspring.

Raw material. `TATIANA_V1_IDENTITY_MAP.md` §5 (the proposal and its own warning), §10 open decision 4; `MATH_TOOLBOX.md` II.2 (the adjunction, which is standard and already proved).

Blocked by. Chapter 12 (there is no sub-invariant to carry until $\mathrm{K}(\mathbb{K})$ is established) and Chapter 13 (there is nothing for an offspring to *do* until the wake/sleep cycle is specified). Do not write this chapter before those two are finished.

Steps.

1. **Say what reproduction is not.** Not a copy: a bitwise duplicate carries no information about what the kernel learned to be, and duplicating the whole store is not growth but storage. State this before proposing anything.
2. **Propose the morphism.** The adjoint pair is already in the mathematics: ι^* instantiates a downward-closed subcomplex, $\iota_!$ writes back only what was touched, and because ι is a full subposet inclusion the unit is an isomorphism, $\iota^*\iota_! \cong \mathrm{Id}$ — so the round trip provably does not corrupt the fragment. An offspring is a downward-closed subcomplex together with its pullback sheaf.
3. **State the invariant it carries.** The offspring’s K is the restriction of the parent’s dimension vector to the subcomplex — a *sub-invariant.* Establish that this is well-defined and that it is conserved by the offspring’s own consolidation, which follows from Chapter 12 applied to the subcomplex.
4. **Do not claim viability.** Tag the whole chapter **[PROPOSED]**. The honest statement is: the morphism is canonical, and nothing follows from that about whether the offspring can run, learn, or survive contact with input. Say so in those words.
5. **Write the falsifier.** Before any implementation: what observation would show an offspring is *not* viable? Candidates to develop — the offspring’s wake phase never stalls (so it can never grow); or it stalls immediately everywhere (so its address carries no information); or its invariant drifts where the parent’s did not. Specify at least one as a runnable experiment.
6. **Connect to the selection question, and leave it open.** If reproduction is cheap and viability is rare, something must choose which fragments reproduce. Note that V0 explicitly **[REFUTED]** random variation-and-selection as the growth mechanism — variation needs randomness *because it has no address*, and this architecture computes an address — so the same argument constrains

what a selection rule here may look like. Do not propose one yet.

Done when. The morphism is defined, the sub-invariant is proved to be well-defined, the viability claim is explicitly *not* made, and at least one falsifier is specified precisely enough to run.

Part XII

The Encoder and the Verification Lift

Chapter 24

The Encoder and the Verification Lift

¡NEW¿ *Two roles the architecture cannot do without and has not yet derived. This chapter exists to keep them visible as open problems rather than letting them be quietly filled by an import.*

Work plan. ¡NEW¿

Goal. State precisely what the encoder and the verification lift must *do* — as constraints the rest of the mathematics imposes on them — without assuming any particular architecture for either.

Raw material. The scope refusals of §1.4; `MATH_TOOLBOX.md` I.8 (the verification trichotomy, already fixed epistemically), I.5 (what the drive term needs to be for the wake flow to act on it).

Steps.

1. **Derive the encoder's type, do not choose it.** Work backwards from the wake phase: the drive term enters $\dot{s} = -L_{\mathcal{F}}s + \text{drive}$, so o_t must live where the restriction maps can receive it. Establish the minimum: which space, what normalisation, and whether anything more than a vector is required. The answer is a *constraint*, and the chapter's contribution is to state it exactly.
2. **Ask what would make an encoder wrong.** Two failure modes to develop: an encoder whose output geometry already contains the structure the sheaf is supposed to learn (in which case the sheaf is decorative and the architecture is a dressed-up embedding lookup), and an encoder whose output is isotropic noise to the sheaf (in which case nothing can ever glue). Both are checkable. Specify how.
3. **Derive the verification lift's contract.** What must $\mathcal{L}$ guarantee for the trichotomy — VERIFIED, REFUTED, UNVERIFIABLE — to remain meaningful? The third value is the load-bearing one: it exists precisely so that inability to check is never recorded as falsehood, and a checker crash must return it rather than a refutation. State the contract as requirements on the lift, not as a description of a particular checker.
4. **Connect the contract to consolidation.** $\gamma(\nu)$ reads the verdict: γ_0 if verified, $\varepsilon\gamma_0$ if unverifiable, 0 if refuted. So the lift's guarantees are exactly what stops unverified structure from entering the invariant. Make the dependency explicit — it is why this chapter cannot be deferred indefinitely.
5. **Check the surviving toolbox before reaching outward.** Ask whether either role reduces to something already present: is an encoder a restriction map into the complex from a boundary cell? Is a checker a discrete verification map with the same signature as the operators the book already defines? Record the answers even if negative, since a negative answer is what would justify building something new.

6. **State the standing constraint, in the chapter, not in a footnote.** Neither role may be discharged by wrapping a pretrained, externally-hosted model. If a derivation arrives at machinery that also appears in such models, that is permitted — as a conclusion with a derivation attached, never as a default and never as an import.

Done when. Both roles have stated contracts derived from what the rest of the architecture requires of them; at least two failure modes are specified as checkable; and no architecture has been assumed for either.

Part XIII

Limits: What Is and Is Not Claimed

Chapter 25

Limits: What Is and Is Not Claimed

Work plan. ¡RECAST¿

Goal. Keep this chapter's function — the honest boundary — and update its content to V2's boundary, which is in some ways wider and in most ways narrower than V0's.

Raw material. This chapter's existing text; `MATH_TOOLBOX.md` Part VIII (the consolidated dead-end registry), Part VI (four failed cover-detection attempts), Part IX (the methodology findings); `TATIANA_V1_IDENTITY_MAP.md` §2, §11, §12.

Steps.

1. **Replace the scope statement.** The V0 text scopes to three papers and a research-tool track. That track is retired. State V2's scope: one architecture, evaluated by whether its invariant survives its own dynamics.
2. **Import the dead-end registry wholesale** into Appendix C and reference it here. Everything refuted, rejected or dissolved, with the reason, so none of it is re-proposed without knowing why it died. This is one of the project's genuine assets.
3. **Keep the cover-vs-partition saga at full length.** Four instruments, all inconclusive, each diagnosed: an aggregation unit that fixed the answer before any data was seen; a null model with free row margins that manufactured a false positive on identical data ($p = 0.45$ versus $p = 0.02$ purely by changing the null); an instrument blind without dimensionality reduction; a label/score relation mismatch. State the acceptance criterion earned from it: any instrument must pass a synthetic positive, a synthetic negative *with planted structure recovered in the right direction*, and a structure-free geometric control, at one shared parameter setting, *before* touching real data. None of the four met it.
4. **Keep "four failures to reject the partition are not evidence that the partition is true."** Verbatim.
5. **State the live external threat and the defence.** Published work reports that identity maps can replace learned sheaf Laplacians on a benchmark family. It concerns *graphs*. The 2-cell argument of Chapter 5 is the defence and must appear before a reader raises the objection — and it must be stated as a theorem, since as of now it is asserted.
6. **Reproduce the kill-tests as a live checklist**, with current status: routing (unrun), localisation on real data (unrun), invariant drift (**[MEASURED]**, survived, with pruning untested), sheaf-not-decorative (unrun). Anything that fires goes in the logbook and in Appendix C.
7. **Keep the prior-art honesty.** Cover detection has mature machinery in at least four fields under different names, and roughly a week was spent rediscovering it. What remains genuinely original is the sheaf over the structure, the growth law, the accumulating-not-recomputing memory, and the two-complex architecture. Infrastructure should be borrowed, not invented.
8. **Carry the citation warning.** Nearly every citation in the source corpus is flagged as un-

searched. Verify before any bibliography.

Done when. A reader finishing this chapter can state exactly what has been established, what has been refuted, and what has merely not yet failed.

Intuition. The deepest thing the topology buys is not that it makes the system smart. It is that it makes the claim of augmentation *falsifiable*: memory, consistency and growth become operations on one object, with cheap provable guarantees and a live measure of whether augmentation is actually happening. A framework that cannot fail is not doing any work, and most of this chapter is about where this one can.

25.1 The bet and the refusal

Claims and non-claims. The bet (defensible). A persistent, self-checking, self-restructuring *organiser* wrapped around a fixed language model can, over time, produce better-grounded and more internally consistent research artifacts than the raw model, through measured cross-organ error correction and accumulation. The advantage is *architectural*, expressed as operations on the single object $\mathcal{M} = (\mathcal{C}, \mathcal{F}, s)$.

The refusal. No simplex proves a theorem and no Laplacian integrates a differential form. The per-inference reasoning quality is bounded by the underlying model. The mathematics here does not raise that ceiling; it organises, measures and error-corrects *around* it.

The bet is about the *slope*, not the intercept. That makes it an empirical claim, and Open Problem 26.3 records that it has not been tested.

25.2 The scope is three papers, not one

Claims and non-claims. A single document containing the ρ splitting, the operator algebra, the coned concept, the cohomological obstruction and a spectral prediction on $\mathbb{CP}^3$ will produce something no referee can accept, because those claims have incompatible standards of evidence. The audit's recommended split is:

Paper A — a cellular-sheaf coherence measure for multi-agent orchestration. Content: $\mathcal{M} = (\mathcal{C}, \mathcal{F}, s)$; ω and ρ; the $\alpha/\tilde{\rho}$ splitting with the window; weighted Anderson–Morley *with* the $\|\mathcal{F}\| \le 1$ hypothesis stated; worst-edge localisation; the three-valued verification gate. Every claim is a theorem plus a measurement on a running system. This paper is nearly writable today.

Paper B — sheaf-cohomological obstruction as a diagnostic. Content: Proposition 11.3 — the derived cochain is exact, a negative result worth publishing on its own; the architecture needed to produce non-derived 1-cochains; the Hodge trichotomy; and the connection to contextuality in the Abramsky–Brandenburger sense, which is the thread that would make this a quantum-information contribution rather than a software paper. This paper *requires new engineering* and cannot be written before that lands.

Paper C — the spectral work. This is separate mathematics. It has nothing to do with TATIANA and should not be contaminated by it; the architecture's role is to be the tool that helped, mentioned in a methods note, not a co-author of the theorem.

25.3 What must not be claimed

Carrying forward the honesty discipline that is the best thing about the source corpus:

- **Not** "we prove the whole exceeds the sum". A structural novelty statement says a composite is *new*, not that it is true or useful. Verification remains the sole arbiter and runs first.
- **Not** "capacity measures learning". It is gameable by adding independent useless operators. Report it; never optimise it.
- **Not** "the system uses D-module theory". It uses Gaussians, which *are* the simplest holonomic modules — an exact identification that currently buys three invariants, all of them constant. The defensible claim is that the concept type admits a conservative extension to holonomic modules, feasible only because of the existing low-rank design.
- **Not** "Riemann–Hilbert applies". One leg is rigorous (cellular sheaves are constructible sheaves via exit paths); the other requires a commitment to an algebraic stratification that has not been made.
- **Not** "the threshold is derived". It is *normalised* (Proposition 9.8).
- **Not** "b_1 measures conceptual gaps" without Remark 11.2. The justification is the whole content of the claim.
- **Not** "the architecture is an operad". See Remark 16.2 and the three defects following it.

25.4 The findings, restated as boundaries

The audit's fourteen findings are not a list of bugs; each marks a place where a claim outruns its evidence. Restated as standing limits:

1. **The audit itself was against a stale snapshot.** Several findings were repaired before the audit was read. Any use of this list must be re-checked against the current branch — and that instruction applies to this book.
2. **Theorem 7.5 is stated more generally than proved.** The gap between hypothesis and proof has not been located precisely, and until it is, the theorem should be cited in its proved form only.
3. **The $H^1(\mathcal{C};\mathcal{F})$ repair is half circular.** The circularity must be identified before the repair is adopted (Open Problem 26.2).
4. **Proposition 12.4 is false in the deployed case.** A counterexample exists and must be reproduced in Appendix C before that entry is complete.
5. **The K_0 classes are trivial for every object the system produces.** An invariant that separates nothing is not a diagnostic (Open Problem 26.7).
6. **The independence relation is not symmetric**, its supports are placeholders, and its empty-set convention is inverted (Chapter 16).
7. **The restriction maps are the identity**, so the sheaf is a graph Laplacian in disguise (Open Problem 26.1).

Claims and non-claims. The honest summary of the whole corpus: the *measurement* layer is real, proved and running — ω, ρ, the splitting, the window, the three-valued gate, the coned concept and the growth law all survive scrutiny. The *expressive* layer is not yet built: while restrictions are the identity and 1-cochains are derived, the sheaf and its cohomology are carrying no information the graph did not already carry. That is not a fatal criticism; it is a precise statement of what the next piece of work has to be, and the fact that it can be stated this precisely is what the formalism bought.

Part XIV

Open Problems

Chapter 26

Open Problems within MOS

Work plan. ¡RECAST¿

Goal. Re-rank the open problems for V2 and add the ones the pivot created. Retitle: "within MOS" names the retired system.

Raw material. This chapter's existing text; `TATIANA_V1_IDENTITY_MAP.md` §10 (the ranked open decisions); `MATH_TOOLBOX.md` II.1 (Tier C material and why it was inert), III.5 (the contextuality bridge); `TATIANA-V2/MATH/phase0_invariant_dimension.md` §4.

Steps.

1. **Rank, and say what the ranking means:** an open problem here is one where a construction or a measurement is missing, not merely one where work remains.
2. **#1 — the source of a non-exact η for a single kernel.** V0 got its measured 1-cochain from organs disagreeing. V2 has no organs. Without a source, Proposition 8.2 leaves growth with no address and the kernel cannot grow at all. This is the most important open problem in the book. Candidate to develop and test: disagreement between the kernel's own competing candidate transformations.
3. **#2 — co-allocation versus separation.** When a harmonic residual names a cycle, does the kernel cone it (integrate) or keep the memories distinct? The allocation literature says excitability history decides, which in this formalism means it depends on the coupling weights — and those weights are *computed and never read*. That is a live bug, not merely a gap.
4. **#3 — can molding erode the obstruction growth depends on?** Raised by the Phase 0 experiment, which bounds molding to one nudge per tick and so never tests it. If slow molding can dissolve a harmonic class, the layer handoff is a race rather than a handoff. State the experiment that settles it.
5. **#4 — does Ext^2 vanish for the incidence algebra of the 2-complex?** This decides whether "triangles make molding obstructible" is a theorem or an analogy. Quiver categories are hereditary; poset categories with relations should have higher global dimension. Currently unverified — do not build on it either way.
6. **#5 — the MDL exchange rate**, and **#6 — the stall tolerance**: the design's only two admitted free parameters. Keep them listed so that a third would be conspicuous.
7. **#7 — reproduction viability** (Chapter 23), stated as currently carrying no claim.
8. **Revisit the shelved homological material on purpose.** Characteristic cycles, b-functions and Hochschild cohomology were shelved as inert for one stated reason: every concept the V0 engine produced was simple, so the machinery had nothing to act on. A coned concept is a limit of a diagram and is not obviously simple, so the premise may no longer hold. Re-examine, carrying the standing corrections so they are not re-derived wrongly — extension obstructions live in HH^3 and not HH^2; Ore extensions are unobstructed given their data; and a classification

theorem used here needs an algebraically closed field and was once silently applied over $\mathbb{R}$.

9. **Keep the contextuality bridge as a well-posed problem.** Both halves are standard separately; the intersection is not an established area. State the real gap sharply: a sheaf of vector spaces always admits the zero section, so this obstruction is about *dimension*, while the contextuality one is about *existence*.

Done when. The problems are ranked by what blocks what, and #1 is unmistakably identified as the one that gates the architecture.

These are stated as problems, not as directions, because each one has a definite answer that is currently unknown and whose resolution would change what the system can claim. They are ordered by how much else depends on them.

Open Problem 26.1 (Non-identity restriction maps). Find a procedure that learns restriction maps $\mathcal{F}_{v \trianglelefteq e}$ from organ data such that (i) the maps are not all the identity, (ii) functoriality (6.1) holds on every filled simplex, and (iii) the learning is closed-form or otherwise costs no gradient training. Closed-form Procrustes alignment satisfies (i) and (iii) but is not known to satisfy (ii). Everything expressive about the sheaf is downstream of this problem: while the maps are the identity, $\ker L$ is the diagonal and the sheaf is a graph Laplacian in disguise (Proposition 7.8).

Open Problem 26.2 (A non-vacuous $H^1(\mathcal{C}; \mathcal{F})$). Under identity restrictions the cohomological hierarchy carries no information: the 1-cochain the engine emits is δx for a global state x, and a coboundary is exact by definition, so its class in $H^1(\mathcal{C}; \mathcal{F})$ vanishes identically — for every state, every sheaf, and every choice of restriction map. Find a source of 1-cochains that is *not* of the form δx — pairwise judgements formed independently of any global state are the candidate — and determine whether the resulting classes are stable enough to serve as a diagnostic. The audit records that one proposed repair is half circular; the circularity must be identified precisely before a fix is adopted.

Open Problem 26.3 (Does the architecture actually compound?). The central bet is on the *slope*, not the intercept: that a self-checking, self-restructuring organiser improves with use even though its per-inference ceiling is fixed by the underlying language model. This is an empirical claim and it has not been tested. Design an experiment with a pre-registered metric, a control (the same model with no architecture), and a horizon long enough for accumulation to show, and state in advance what result would falsify the bet.

Open Problem 26.4 (The right normaliser). The Anderson–Morley bound B over-bounds $\lambda_{\max}$, which compresses ρ toward zero and contributes to its dead dynamic range. Determine whether a tighter, still eigensolver-free bound exists for the sheaf Laplacian with non-identity restrictions, or prove that no degree-based bound can be tight in this regime.

Open Problem 26.5 (Stability of the diagnostics under growth). The growth operators change the topology, and both the calibration window and the kernel projection are topology-dependent. Give a stability estimate: if `bind` or `collapse` changes the complex by one simplex, how much can $\tilde{\rho}$ move for a fixed state? Without such an estimate, a change in $\tilde{\rho}$ across a growth step cannot be attributed to the state rather than to the rewiring.

Open Problem 26.6 (Functoriality repair by hybridisation). When restriction maps are obtained as projections onto an interface, functoriality fails predictably. The canonical repair is to promote the interface to a cell of its own, which is what hybridisation methods do in the finite-element setting. Determine whether the same repair applies to the cognitive complex, and what it costs: does promoting every offending interface preserve the two-level stratification, or does it force a third stratum?

Open Problem 26.7 (Composition factors as memory atoms). A later development formalises memory atoms as Jordan–Hölder composition factors, with retrieval ordered by stratum, then integer content, then structure. Determine whether the composition series is canonical for the objects the system actually produces — the audit records that the associated K_0 classes are trivial for every such object, which if correct means the invariant distinguishes nothing.

Part XV

Potentials for Quantum Implementation

Chapter 27

Potentials for Quantum Implementation

Work plan. ¡KEEP¿

Goal. Keep as a forward-looking chapter, unchanged in substance. Guard it against the one failure mode it invites: becoming a claim rather than a programme.

Raw material. This chapter's existing text; `MATH_TOOLBOX.md` I.4 (restriction maps as CPTP channels — the structural reason this chapter is not fanciful), III.5 (the proposed toy object).

Steps.

1. **Keep the structural motivation, which is genuine.** Storing a stalk as a rank-k SPD matrix makes it the cone over density-matrix space; congruence by an orthogonal map preserves trace, so restriction maps already *are* unitary channels. The quantum reading is not imported — it is what the objects already are. **[PROVED]**.
2. **Keep the obstruction section prominent.** Input and output remain the hard part; a block-encoding that cannot be loaded or read is not an implementation. This section is the reason the chapter is honest and it must not shrink.
3. **Keep the toy object as a proposal.** A cellular sheaf with quantum-state stalks, whose Laplacian is a Hamiltonian, whose ground space is the globally consistent assignments and whose ground-state energy is a contextuality witness. **[PROPOSED]**, clearly.
4. **Add a status line at the head of the chapter.** Nothing here is on the critical path for V2. It is a direction, retained because the architecture points at it for structural reasons, not because it is being built. Say so explicitly so that a reader in a hurry cannot mistake it for work in progress.

Done when. The chapter's structural motivation is stated as proved, everything else as proposed, and its status relative to V2's actual work is unambiguous.

Claims and non-claims. **[PROPOSED]** Nothing in this part is implemented, and nothing in it is required by any result elsewhere in the book. It is included because the sheaf Laplacian is exactly the kind of object for which quantum linear algebra has a story, and because that story has a specific shape worth recording before it is attempted. Every claim here should be read as a research programme with an identified failure mode, not as a capability.

27.1 Why this object at all

The two quantities TATIANA computes on every tick are a Dirichlet energy $\omega(x) = x^\top Lx$ and a normalised version of it. Both are quadratic forms in a sparse, symmetric, positive semidefinite operator built from local data. This is the standard input to quantum linear-algebra routines, and the standard caution applies: the speedups are in the linear algebra, and they are destroyed by the cost of getting classical data in and scalar answers out.

27.2 The Dirac operator and its block-encoding

Definition 27.1 (Sheaf Dirac operator)**.** Let $d_\mathcal{F} = \bigoplus_k \delta^k$ be the total coboundary of the sheaf. The *sheaf Dirac operator* is

$$D_\mathcal{F} \;=\; d_\mathcal{F} + d_\mathcal{F}^\dagger,$$

a self-adjoint operator on $\bigoplus_k C^k(\mathcal{C};\mathcal{F})$ satisfying $D_\mathcal{F}^2 = \Delta$, the total Hodge Laplacian. Its kernel is the space of harmonic cochains, which by Hodge theory is isomorphic to $\bigoplus_k H^k(\mathcal{C};\mathcal{F})$.

The programme, stated as a pipeline:

1. **Block-encode $D_\mathcal{F}$.** Construct a unitary U whose top-left block is $D_\mathcal{F}/\gamma$ for a subnormalisation $\gamma \geq \|D_\mathcal{F}\|$. For a cellular sheaf with *matrix-valued* restriction maps this is the substantive step, and the subnormalisation γ scales with the heterogeneity of those matrices — which is precisely the quantity that makes the sheaf interesting in the first place.
2. **Filter onto $\ker D_\mathcal{F}$.** Apply a polynomial approximation to a projector via quantum singular value transformation, isolating the harmonic subspace.
3. **Extract a scalar.** Estimate ω, or an overlap with a prepared state, by amplitude estimation.

Remark 27.2 (Where this programme is honest and where it is not)*.* Step 1 is the only step with genuine novelty for cellular sheaves with matrix-valued restrictions; steps 2 and 3 are standard. The honest statement of the advantage is therefore about the *encoding* and its resource scaling, not about an end-to-end speedup for TATIANA. In the deployed regime the restriction maps are the identity (Proposition 7.8), $L = L_K \otimes I_d$, and the whole object reduces to a graph Laplacian on seven vertices — for which classical computation is instantaneous and any quantum treatment is theatre. The quantum question only becomes non-trivial once Open Problem 26.1 is solved and the maps are genuinely matrix-valued and heterogeneous.

27.3 The obstruction: input and output

Claims and non-claims. Two costs dominate and both are classical.

Input. The stalk data is a set of 384-dimensional embeddings produced by a classical encoder. Preparing an amplitude encoding of these costs, in the absence of a QRAM, time linear in the data — which is the same order as computing ω classically. Any claimed speedup that assumes free state preparation is assuming away the problem.

Output. ω is one real number, and amplitude estimation returns it to precision ε in $O(1/\varepsilon)$ queries. For a diagnostic thermometer read once per tick, ε need not be small, but

> neither is the classical cost large. The regime where the quantum route can win is not "compute ρ faster"; it is "compute something about the spectrum of L that is classically infeasible", and no such quantity is currently required by the architecture.

27.4 What would make this real

Open Problem 27.3 (A quantity worth the encoding). Identify a functional of L that (i) the architecture genuinely needs, (ii) is classically infeasible at the scale TATIANA reaches, and (iii) is extractable from the block-encoded $D_{\mathcal{F}}$ by a constant number of scalar measurements. Without (i) the programme is solving an unposed problem; without (ii) it is slower than the classical route; without (iii) the output cost swamps the encoding.

Open Problem 27.4 (Subnormalisation in terms of sheaf data). Give an explicit bound on the block-encoding subnormalisation γ for a cellular sheaf Dirac operator in terms of the restriction-map norms and the degrees of the complex, and determine how it degrades as the maps become heterogeneous. This is the resource-scaling statement on which any advantage claim rests.

Remark 27.5 (Relation to the eddy-current programme). The block-encoding of a cellular sheaf Dirac operator with matrix-valued restriction maps is the technical core of a separate research programme on magnetoquasistatic problems, where the sheaf arises from a discretised PDE rather than from a cognitive architecture and the subnormalisation scales in material contrast. The mathematics is the same; the physical problem supplies what Open Problem 27.3 asks for and TATIANA, at present, does not. That asymmetry is the honest reason this part is short.

Appendices

Appendix A

Notation

Symbol	Meaning
$\mathcal{M} = (\mathcal{C}, \mathcal{F}, s)$	cognitive state space
$\mathcal{C}$	stratified base complex
K	coarse complex (organs as vertices)
$\mathcal{C}_v$	fine complex inside organ v
$\sigma \trianglelefteq \tau$	σ is a face of τ
$\mathcal{F}$	cellular sheaf on $\mathcal{C}$
$\mathcal{F}(\sigma)$	stalk over σ
$\mathcal{F}_{\sigma \trianglelefteq \tau}$	restriction map $\mathcal{F}(\sigma) \to \mathcal{F}(\tau)$
$C^k(\mathcal{C}; \mathcal{F})$	k-cochains $\bigoplus_{\sigma \in \mathcal{C}_k} \mathcal{F}(\sigma)$
δ	coboundary $C^0(\mathcal{C}; \mathcal{F}) \to C^1(\mathcal{C}; \mathcal{F})$
$L = \delta^\top \delta$	degree-zero sheaf Laplacian
$\omega(x) = x^\top L x$	discord (Dirichlet energy)
ρ	coherence score in $[0, 1]$
α	dissent fraction
$\tilde{\rho}$	mode index
B	Anderson–Morley normaliser
π_e	edge precision
$\perp$	undefined / measurement not made
b_0, b_1	Betti numbers of the 1-skeleton
$H^k(\mathcal{C}; \mathcal{F})$	k-th sheaf cohomology
$\mathfrak{D}$	operator algebra of growth operators
$D_{\mathcal{F}}$	sheaf Dirac operator $d_{\mathcal{F}} + d_{\mathcal{F}}^\dagger$
d	stalk dimension (384 in deployment)

Appendix B

Discrepancy Register

Places where the sources disagree with each other, or where the implementation determines a mathematical fact that the prose states differently. Each is a fact about TATIANA, not a typo.

D1 — The two-scale complex, and "concepts inside the stalk". Earlier drafts described the base complex as carrying two nested scales, and described the fine scale as living *inside* a coarse vertex's stalk. Both are withdrawn. A stalk is a vector space and a complex is a complex, so the second description was a category error; and the two-scale structure itself is no longer assumed, for the reasons in §5.3. The base complex has one scale, and the word *stratified* now applies only to the configuration space (§8.2).

D2 — Restriction maps point the wrong way in the poster. `POSTER/sections/05_sheaf.tex` defines $\rho_{\tau\to\sigma} : \mathcal{F}(\tau) \to \mathcal{F}(\sigma)$ and draws the arrows downward, then asserts $s \in H^0(\mathcal{C}; \mathcal{F})$. These are incompatible (Proposition 6.4); the downward convention defines a cosheaf. `TATIANA_Mathematical_Descript` has it correct. **Action: fix the poster before showing it.**

D3 — Two different systems named MOS. The `README` on `main` describes a retrieval pipeline (EvidenceEngine, KnowledgeCurator, Canonicalizer). The poster and the mathematical description define a sheaf-theoretic state space. These are different objects sharing a name, and the README is what a reader sees first.

D4 — `main` is stale. At the time of writing, `main` is four commits and roughly six weeks behind `fix-11-13-and-curvature-decisions`, which carries +33,450 lines including all of Part VI's material and the PRECILLA package. Any reader who clones the default branch gets a snapshot that this book does not describe.

D5 — `hodge.py` is scalar, not sheaf-valued. The Hodge decomposition code operates on a scalar complex — the trivial sheaf $\mathcal{F}(\sigma) = \mathbb{R}$ — with $(\delta^0 f)_{uv} = f(v) - f(u)$. It is therefore a computation about the *shape*, not about the sheaf. This is correct for what it measures and should not be cited as evidence about sheaf-valued cohomology.

D6 — The Hebbian-weight gap appears to be closed. An earlier scrutiny session recorded that plasticity weights were computed but never read by the coherence measurement, which would make ρ a different functional from the one documented. The current `coherence.py` computes a precision-weighted degree $d_\pi(v) = \sum_{e \ni v} \pi_e$ and a weighted Anderson–Morley bound, so the gap looks repaired. **This needs confirming against the precision source** before the finding is retired.

Appendix C

Refuted-Claims Ledger

Claims that were made, tested, and failed. They are kept because the refutations are the expensive part.

R1 — Encoder anisotropy explains the dead dynamic range of ρ. **[REFUTED]** (2026-07-28). The deployed encoder returns unit-normalised vectors, for which $\alpha = (n-1)(1-\bar{c})/n$ exactly, $\bar{c}$ the mean pairwise cosine. At the observed $\bar{c} \approx 0.449$ with $n = 7$ this gives $\alpha \approx 0.46$, which is not small. The compression is produced jointly by α, by $\tilde{\rho}$ sitting mid-window, and by B over-bounding $\lambda_{\max}$.

R2 — The cohomological hierarchy is informative as built. **[REFUTED]**. The 1-cochain the engine emits is δx for a global state x; a coboundary is exact, so its class in $H^1(\mathcal{C}; \mathcal{F})$ vanishes identically, for every state and every sheaf. The growth mechanism has no address in $H^1(\mathcal{C}; \mathcal{F})$. See Open Problem 26.2.

R3 — Proposition 12.4 of the earlier description. **[CONTESTED BY AUDIT]** The audit records it as false in the deployed case. The statement and the counterexample must be reproduced here before this ledger entry is complete.

R4 — Theorem 7.5 as stated. **[CONTESTED BY AUDIT]** The audit records that it is stated more generally than it is proved. The gap between hypothesis and proof must be located precisely.

R5 — K_0 classes as memory invariants. **[CONTESTED BY AUDIT]** The audit records that the K_0 classes are trivial for every object the system produces, which would mean the invariant separates nothing. See Open Problem 26.7.

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